



accredited
program

ACQUIN



MODULE HANDBOOK



70610101 —
Computer Systems and
Their Software

Master's program

Submitted to **ACQUIN**
by Termiz State University
as part of the Accreditation
Process



Termiz city, "Barhayot avlod"
street, house number 43



+998 76 221 74 55



info@terdu.uz



www.terdu.uz



@terdu340



@terdu340



@terdu_university



@Termizdavlatuniversiteti

Contents

<u>Curriculum Structure Overview of 70610101-Computer Systems and Their Software Master's Program.</u>	3
<u>ECTS overview</u>	4
<u>1.01. Methodology of Scientific Research</u>	5
<u>1.02. Methodology for Teaching Specialized Subjects</u>	19
<u>1.03. Design of Information Systems</u>	30
<u>1.04. Administrative Management of Computer Networks</u>	45
<u>1.05. Design and Analysis of Algorithms</u>	63
<u>1.06. Foundations of Parallel Computing</u>	75
<u>2.01. Java Technologies</u>	85
<u>2.01. Distributed Systems</u>	97
<u>2.02. Artificial Intelligence and Neural Networks</u>	109
<u>2.02. Cloud Technologies</u>	119
<u>2.03. Computational Methods and Modelling</u>	128
<u>2.03. Web Application Development Technologies</u>	144
<u>2.04. Prompt Engineering</u>	156
<u>2.04. Machine Learning</u>	166
<u>2.05. Mathematical Modelling of Nonlinear Processes</u>	177
<u>2.05. Mathematical Foundations of Artificial Intelligence</u>	193
<u>2.06. Electronic Government</u>	206
<u>2.06. Fundamentals of the Django Framework</u>	220
<u>3.01. Research Seminar va Research Internship</u>	198
<u>3.02. Research Work, Preparation and Defense of Master's Thesis</u>	207

Curriculum Structure Overview

70610101-Computer Systems and Their Software Master's Program

(2 years / 4 semesters, 120 credits)

1st Semester (1st Year)	Methodology of Scientific Research (6 cr.)	Methodology for Teaching Specialized Subjects (6 cr.)	Design of Information Systems (6 cr.)	Administrative Management of Computer Networks (6 cr.)	Scientific-pedagogical work and qualification practice (4 cr.) Research work, preparation and defense of master's thesis (2 cr.)
2nd Semester (1st Year)	Design and Analysis of Algorithms (6 cr.)	Foundations of Parallel Computing (6 cr.)	Java Technologies / Distributed Systems (4 cr.)	Artificial Intelligence and Neural Networks / Cloud Technologies (4 cr.)	Scientific-pedagogical work and qualification practice (4 cr.) Research work, preparation and defense of master's thesis (6 cr.)
3rd Semester (2nd Year)	Computational Methods and Modelling / Web Application Development Technologies (4 cr.)	Prompt Engineering / Machine Learning (4 cr.)	Mathematical Modelling of Nonlinear Processes / Mathematical Foundations of Artificial Intelligence (4 cr.)	Electronic Government / Fundamentals of the Django Framework (4 cr.)	Scientific-pedagogical work and qualification practice (4 cr.) Research work, preparation and defense of master's thesis (10 cr.)
4th Semester (2nd Year)	Research work, preparation and defense of master's thesis (22 cr.)	Scientific-pedagogical work and qualification practice (8 cr.)	Final State Attestation		
	 Compulsory modules Elective modules Scientific-pedagogical work and qualification practice Final State Attestation Research work, preparation and defense of master's thesis				

ECTS overview

Module	ECTS	Workload			
		Semester I	Semester II	Semester III	Semester IV
Mandatory Courses	36	24	12		
Elective Courses	24		8	16	
Scientific pedagogical work and qualification practice	20	4	4	4	8
Scientific research work, preparation and defense of master's thesis	40	2	6	10	22
Total	120	30	30	30	30

1.01. METHODOLOGY OF SCIENTIFIC RESEARCH

Field	Description
Module Title	METHODOLOGY OF SCIENTIFIC RESEARCH
Module Code	ILTM1106
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 120 hours
Frequency/Duration	I semester
Module objective	The objective of the Module Methodology of Scientific Research is to organise the process of scientific research for postgraduate students. Students learn to identify a scientific problem. The research aims and objectives are defined. Theoretical knowledge in organising the scientific research process is developed. Practical skills in identifying a scientific problem are developed. Practical skills in defining research aims and objectives are also developed.
Module Learning Outcomes	Knowledge: LO1. Can explain the main concepts, principles and contemporary development trends of research methodology. LO2. Can describe the content and objectives of the types, stages, methods and techniques of scientific research. LO3. Can explain the concepts of a scientific problem, the research object and subject, the aim, the task and the hypothesis. LO4. Explain the principles of working with sources of scientific information, bibliographic databases and scholarly literature. LO5. Describe the theoretical foundations of the statistical and mathematical methods used in the

	processing, analysis, and generalisation of scientific results. LO6. Can explain the principles of academic integrity, scientific ethics, copyright, and the prevention of plagiarism. Skills: LO7. Identifies a research problem and formulates the research aims and objectives. LO8. Selects research methods and uses them effectively in scientific research. LO9. Prepares and formalises materials for a scientific article, thesis, report and master's dissertation. LO10. Collects, organises, analyses scientific data and presents the results in the form of tables, graphs and diagrams. LO11. Works with electronic scientific databases (Scopus, Web of Science, Google Scholar, etc.) and conducts bibliographic analysis. LO12. Evaluates research findings, and develops scientific conclusions and practical recommendations. Competencies: LO13. Independently plans and organises research activities. LO14. Critically analyses scientific information and makes scientifically justified decisions. LO15. Utilises modern information and communication technologies and software tools in scientific research. LO16. Is competent in presenting scientific results to academic and professional audiences. LO17. Demonstrates the competence to implement research findings into practice and apply them in innovative activities.
Content	Lectures Science as a Socio-Cultural Phenomenon. The concept of science as a socio-cultural phenomenon is

	studied. The relationship between science, society, and culture, and the influence of scientific development on civilization, are analyzed. The social functions of science and contemporary trends in scientific development are examined. Science as a Special Form of Knowledge. The nature and characteristics of scientific knowledge, and its differences from non-scientific knowledge, are studied. The structure, functions, and methods of acquiring scientific knowledge are analyzed. The role of science in human development is examined. Creative and Scientific Research Are the Products of the Activity of Knowing. The relationship between cognition, creativity, and scientific research is studied. The stages of the cognitive process and creative thinking in scientific inquiry are analyzed. Scientific discovery, innovation, and the role of creativity in research activities are examined. Forms of Manifestation of Scientific Research. The concept and forms of scientific research, including fundamental, applied, and experimental research, are studied. Individual and collaborative research activities and the organization of scientific investigations are analyzed. Practical applications of research outcomes are examined. Innovation and Novations in Scientific Research. The concepts of innovation and novation and their role in scientific progress are studied. Methods of developing innovative research are analyzed. The evaluation of scientific innovations and their impact on social and technological development are examined. The Balance Between Rationality and Scientific Research. The concept of rationality in scientific research, logical reasoning, and scientific methodology are studied. The importance of objectivity and critical thinking is analyzed. Rational decision-making and contemporary approaches to scientific rationality are examined. Harmony of Rationality and Irrationalism. The concepts of rationalism and irrationalism and their role
--	--

	in the development of scientific thought are studied. The interaction between logical and intuitive thinking is analyzed. The significance of creativity and intuition in research, and philosophical perspectives on scientific inquiry, are examined. Key Methodological Approaches and Principles. The concept of scientific methodology and general methodological approaches in research are studied. The principles of objectivity, consistency, and reliability are analyzed. The selection and application of appropriate research methods are examined. Types of Scientific Research. The classification of scientific research, including qualitative and quantitative methods, is studied. Fundamental, applied, descriptive, and experimental research types are analyzed. Comparative analysis of research types and the selection of research design based on objectives are examined. Problem and Problematic Situation Are the Initial Basis of Scientific Research. The concept of a scientific problem and the identification and formulation of research problems are studied. The characteristics of a problematic situation are analyzed. The relationship between research objectives and scientific problems, and methods of problem analysis, are examined. The Role (Importance) of Evidence in Scientific Research. The concept and significance of scientific evidence, and its sources and types, are studied. Data collection and verification methods are analyzed. The reliability and validity of research findings, and evidence-based scientific decision-making, are examined. The Impact of Information and Information Technologies on Scientific Research. The role of information in scientific research and modern information technologies are studied. Electronic databases and scientific information systems are analyzed. Data processing and analysis technologies,
--	--

and the influence of digital transformation on research activities, are examined.

The Importance of Understanding and Explanation in Scientific Research. The concepts of understanding and explanation in science, and the interpretation of scientific facts and phenomena, are studied. Methods of scientific explanation are analyzed. The relationship between theory and evidence, and the role of interpretation in research, are examined.

The Social Responsibility of the Scientist (Researcher). The ethical responsibilities of researchers, academic integrity, and research ethics are studied. Responsible conduct in scientific investigations is analyzed. The impact of research on society and professional standards for scientists are examined.

The Development of Scientific Research in the Republic of Uzbekistan. The development of science and research in the Republic of Uzbekistan, and national policies supporting scientific research, are studied. Scientific institutions and research infrastructure are analyzed. Priority directions of scientific development, and the role of research in national innovation and sustainable development, are examined.

Practical lessons

Collection, Classification and Study of Theoretical Material for Scientific Research. Methods of collecting and classifying scientific literature are practiced. Theoretical sources and academic publications are evaluated, and research materials are organized and analyzed. Literature review skills are developed, and theoretical foundations for research are prepared.

Methodological Foundations of Conducting Scientific Research. Discussion of the Master's Dissertation Topic, Defining Its Aim and the Tasks to Be Carried Out to Achieve It. The principles of scientific research methodology are applied. The

	research topic is selected and justified, and research aims and objectives are formulated. Research methods are identified, and the stages of dissertation research are planned. Determining the Expected Outcomes of the Master's Dissertation and the Scientific Novelty of the Research. Expected research outcomes are determined, and scientific novelty and practical significance are identified. Research hypotheses and expected contributions are formulated. Research originality is evaluated, and outcomes are aligned with research objectives. Scientific and Innovative Activities and Research Directions. Provide Information on the Department's Scientific and Innovative Activities and Research Directions. Scientific and innovative activities and departmental research priorities are reviewed. Current scientific projects and innovations are analyzed. Research collaboration opportunities are explored, and reports on research activities are prepared. Language and Style of the Master's Thesis (Research Paper). The principles of academic writing and scientific language and style are practiced. Requirements for clarity, coherence, and objectivity are applied. Academic texts are formatted, and research papers are prepared according to academic standards. Initial Aim: Clarifying the Tasks and Expected Outcomes and Gathering Information Sources. Research Ethics. Citing Sources and Materials. Research and Bibliography. Research objectives and expected outcomes are clarified, and reliable information sources are collected. The principles of research ethics and academic integrity are applied. Citation methods and bibliography preparation are practiced, with attention to the prevention of plagiarism. Conducting an Experimental Study. Formatting Statistical Materials, Tables and Appendices.
--	--

	Experimental research is planned and implemented, and research data are collected and analyzed. Statistical tables, figures, and appendices are prepared. Experimental results are interpreted and presented according to academic standards. Procedure for Conducting and Formalising an Analytical Review (A Comparative Analysis of Viewpoints on the Topic Concepts) and the Formulation of a Master's Thesis Problem. The principles of analytical review writing are practiced through the comparative analysis of scientific viewpoints. Research gaps are identified, and the master's thesis problem is formulated. Literature analysis is organized, and critical thinking in research is developed. Provide Examples of Analytical Reviews Carried Out by Postgraduate Students. The structure and content of analytical reviews are examined through sample postgraduate research reviews. Strengths and weaknesses of the reviews are identified. Analytical review methods are applied, and academic writing skills are developed. Structure of the Master's Thesis (Research Project). The standard structure of a master's thesis and the organization of chapters and research sections are studied. Requirements for the introduction, literature review, methodology, results, and conclusions are applied. Formatting standards are followed in preparing a complete research project. Approval of the Results of the Master's Dissertation (Scientific Research). Procedures for presenting research results and methods of scientific discussion are practiced. Research reports and presentations are prepared. The publication, dissemination, and academic approval of research findings are examined. Application of Mathematical Methods in Research. The role of mathematical methods in scientific research and quantitative analysis techniques are studied. Appropriate mathematical
--	--

models are selected. Analytical results are interpreted, and practical applications in research activities are examined.

Application of Mathematical Methods in Research. In Accordance with the Nature of the Topic, Discuss Organising and Conducting Experiments, Analyses and Presenting the Results of Applying Mathematical Models.

Mathematical methods are integrated into research design, and experiments are planned and conducted. Experimental data are analyzed using mathematical models. Results are interpreted and presented, and research reliability and validity are evaluated.

Forecasting Scientific Research. The concept and importance of scientific forecasting, and forecasting methods and predictive models, are studied. Research trends and future developments are analyzed. Forecasting techniques are applied in scientific investigations, and forecast reliability is evaluated.

Preliminary and Formal Defence of the Master's Thesis. Procedures for the preliminary and final defence of the master's thesis are practiced. Presentations and supporting materials are prepared, and responses to reviewers' comments and questions are formulated. Academic communication skills and final evaluation requirements are examined.

Self-study

History of the Formation of Scientific Research Methodology and Modern Development Trends.

The historical development of scientific research methodology and the major stages in the evolution of research methods are studied. Contemporary trends in scientific methodology are analyzed. The role of methodology in modern research and future directions of scientific investigation are examined.

The Distinctive Characteristics of Scientific Knowledge and the Types of Scientific Knowledge. The concept and characteristics of scientific knowledge, and its classification, are

	studied. Differences between empirical and theoretical knowledge are analyzed. The role of scientific knowledge in research and its practical applications are examined. Types of Scientific Research and Their Differences (Fundamental, Applied, Innovative). The classification of scientific research and the characteristics of fundamental, applied, and innovative research are studied. A comparative analysis of research types is conducted. The selection of appropriate research approaches and their practical significance are examined. Rules for Defining the Scientific Problem, Aim and Objectives. Analyse the Existing Scientific Problems within the Field of Specialisation and Define the Research Aim and Objectives. The principles of identifying scientific problems and formulating research aims and objectives are studied. Current research issues within the field of specialization are analyzed. Research questions are developed, and problem formulation is practically applied. Classification of Scientific Research Methods (SRM) and Their Application. Provide Examples of SRMs and Prepare a Comparative Table. The classification of scientific research methods, including qualitative and quantitative approaches, is studied. A comparative analysis of research methods is conducted. Appropriate methods are selected for different studies, and a comparative evaluation is prepared. Working with Sources in Scientific Research: Methods of Bibliographic Analysis. Analysis of Five Articles Selected from a Scientific Article Database (Scopus, Web of Science, Google Scholar). The principles of bibliographic analysis and the searching and selection of scientific literature from international databases such as Scopus, Web of Science, and Google Scholar are studied. The quality and relevance of scientific sources are evaluated.
--	---

	Selected articles are comparatively analyzed, and a bibliographic review is prepared. International Standards for Formatting Scholarly Works (APA, MLA, Chicago, etc.). The principles of academic formatting and citation styles are studied. The features of APA, MLA, Chicago, and other international standards are analyzed. Citation and reference formatting techniques are practically applied to avoid plagiarism. Principles of Scientific Ethics and Academic Integrity. The concept of scientific ethics and the principles of academic integrity and responsible research conduct are studied. The prevention of plagiarism and research misconduct is analyzed. Ethical standards in publishing scientific work are practically applied. Developing a Sample Plan for the Master's Dissertation. The structure and organization of a master's dissertation are studied. Research chapters and sections are developed, and the research process is planned. A sample dissertation outline is prepared following the logical sequence of dissertation components. Development and Justification of the Scientific Problem, Hypothesis and Research Concept for the Chosen Research Topic. A scientific problem and hypothesis are formulated, and the research concept is developed. Research significance and novelty are justified. Research objectives are aligned with the hypothesis, and a research framework is practically prepared. Compiling a Bibliography for a Scientific Paper (Based on 10–15 Sources). The principles of preparing a scientific bibliography based on 10 to 15 sources are studied. Academic references are selected and organized, and citation standards are applied. Source credibility is evaluated, and a complete reference list is prepared. Preparing a Sample Structure for a Scientific Article or Thesis. The standard structure of scientific
--	--

	articles and theses is studied. Sections and research content are organized according to formatting requirements for academic writing. A model research manuscript is prepared, and scientific writing quality is evaluated. Present the Results of the Scientific Work in the Form of Charts and Tables. The principles of presenting scientific data are studied. Charts, graphs, and tables are prepared to visualize research findings. Statistical information is interpreted for the effective presentation of scientific results. Processes for Implementing Scientific Results into Practice. The concept of research implementation and methods of applying scientific findings in practice are studied. Technology transfer and innovation processes are analyzed. The impact and practical significance of scientific outcomes are evaluated. Prepare a Short Scientific Lecture and a Presentation with Slides on the Chosen Topic. The principles of preparing scientific presentations are studied. Lecture content and presentation materials are organized, and effective visual aids are developed. Academic communication skills are practiced through the delivery of a scientific presentation.
Teaching Methods	The module is delivered through interactive lectures, seminars, practical classes, case study analysis, problem-based learning, project-based learning, group discussions, academic writing workshops, presentations, independent study, and research-oriented assignments. Students develop competencies in identifying scientific problems, selecting appropriate research methodologies, conducting literature reviews, applying qualitative and quantitative research methods, analyzing research data, and preparing scientific articles, reports, and master's dissertations. Modern digital learning resources, scientific databases such as Scopus, Web of Science, and Google Scholar, reference

	management software, statistical analysis tools, and information and communication technologies are used to enhance students' research competence, critical thinking, academic writing, analytical skills, and ethical responsibility in conducting scientific research.
Prerequisites	Academic Writing Probability and Statistics Programming Database
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, seminar discussions, practical research activities, literature review exercises, academic writing tasks, presentations, case-study analysis, and the successful completion of independent study assignments, including scientific article reviews, bibliography preparation, research proposal development, comparative analysis of research methodologies, and the application of academic writing and citation standards. The midterm assessment consists of written assignments, short in-class quizzes, oral presentations, practical research tasks, and the evaluation of independent study projects, such as developing a research problem and hypothesis, preparing a master's dissertation plan, conducting analytical literature reviews, applying scientific research methods, presenting research findings using charts and tables, and analyzing ethical issues in scientific research. The final assessment is a written examination consisting of theoretical questions and analytical and problem-solving tasks designed to assess students' understanding of the methodology of scientific research, research design, scientific methods, academic writing principles,

	research ethics, literature analysis, data interpretation, and their ability to apply scientific methodologies in conducting independent research and preparing scholarly works. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. N.A. Shermuhamedova. Methodology of Scientific Research. Textbook. T.: Science and Technology, 2014. 464 pages. 2. Sukhoverkov A.V., Katsko I.A. Methodology of Scientific Research: A Textbook. Krasnodar: KubGAU, 2019. 86 pp. 3. N.V. Lipchiu, K.I. Lipchiu. Methodology of Scientific Research. A Textbook. Krasnodar: Kuban State Agrarian University, 2013. 290 pp. 4. Slepko, M.S. Methodology of Scientific Research: Practical Course. Vladimir State University named after A.G. and N.G. Stolyetov. Vladimir: VSU Press, 2015. 67 pp. 5. Ponomarev, A.B., Pikuleva, E.A. Methodology of Scientific Research: Textbook. Perm: Perm National Research Polytechnic University Press, 2014. 186 pp. 6. Mineev, V.V. Methodology and Methods of Scientific Research: A Textbook for Master's Students. Krasnoyarsk State Pedagogical University named after V.P. Astafiev. Krasnoyarsk, 2014. 90 pp. 7. Dmitriyenko, G.V., Mukhin, D.V. Methodology and Methods of Scientific Research: A Textbook. Ulyanovsk: Ulyanovsk State Technical University, 2021. 225 pp.
Responsible Lecturer	Zaripova Mukaddas Djumayozovna

1.02. METHODOLOGY FOR TEACHING SPECIALISED SUBJECTS

Field	Description
Module Title	METHODOLOGY FOR TEACHING SPECIALISED SUBJECTS
Module Code	MFOM1106
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 120 hours
Frequency/Duration	I semester
Module objective	The objective of the Module Methodology for Teaching Specialised Subjects is to provide postgraduate students with a thorough grounding in the methods of teaching specialised subjects. Students are taught to conduct scientific research and scholarly pedagogical work. This prepares them for their pedagogical and scientific activities. Master's students are systematically equipped with the theoretical foundations of teaching methods for specialised subjects. This is achieved in line with the tasks set out in the Law on Education. The module also follows the requirements of the State Education Standard.
Module Learning Outcomes	Knowledge: LO1. Methods for conducting teaching, laboratory, practical, seminar and other activities in accordance with the educational standards. LO2. The teaching process, and the learning activities are pedagogically and organisationally supported. LO3. Introduction to curriculum planning and workload management, the assessment system, and academic and pastoral work. Skills:

	LO4. Skill in designing and organising lessons in specialised subjects - drawing up lesson plans, setting objectives and tasks, and selecting effective teaching methods. LO5. Competence in the practical application of modern pedagogical technologies and methods - the ability to use interactive techniques, information and communication technologies, and innovative approaches in the teaching process. LO6. Skill in assessing student knowledge and analysing the teaching process - organising types of assessment, evaluating results and drawing conclusions to improve the learning process. Competencies: LO7. Competence to teach specialised subjects in a vocational context - a deep knowledge of the content of specialist subjects and the ability to explain them in relation to practical activities. LO8. Competence in the professional organisation of pedagogical activity - planning the teaching process, selecting modern teaching methods and ensuring educational effectiveness. LO9. Competence in the use of innovative and information technologies in professional activity - applying e-learning tools, digital resources and new pedagogical approaches to the teaching process.
Content	Lectures The Subject of Teaching Methodology for Specialised Subjects: Scope, Purpose, Tasks and Content. The scope, objectives, and tasks of teaching methodology for specialised subjects are examined. Its content and theoretical foundations are studied. Its role in organising effective professional education is explained. Informatisation of Education. The concept and objectives of educational informatisation are studied. Its stages and significance are examined. The role of digital technologies in improving teaching, learning,

	management, and access to educational resources is considered. Use of Information and Communication Technologies in Educational Institutions. The application of information and communication technologies in educational institutions is examined. Digital platforms and multimedia resources are considered. Communication tools and learning management systems are also examined. Didactic Foundations of Using Information and Communication Technologies. The didactic principles for integrating information and communication technologies into teaching are studied. Pedagogical requirements for this integration are examined. Their influence on learning motivation, activity, interaction, and knowledge acquisition is analyzed. Requirements for the Creation and Use of Educational Electronic Tools. Pedagogical and technical requirements for developing electronic educational resources are examined. Methodological, ergonomic, and design requirements are also studied. Criteria for their effective and safe use in the educational process are discussed. Application of Information and Communication Technologies in the Educational Process. Methods for integrating digital technologies into lectures and practical sessions are studied. Their integration into independent study, assessment, and educational management is examined. Appropriate ICT tools are selected according to learning objectives and student needs. Information and Methodological Support for the Teaching and Educational Process. The structure and functions of information and methodological support are examined. Electronic textbooks, teaching materials, and databases are considered. Digital libraries, methodological guidelines, and online resources are also examined.
--	---

	Computing and Information and Communication Technologies. The main types and functions of computing and communication technologies are studied. Their educational applications are examined. Their role in processing, storing, presenting, and exchanging educational information is explained. Requirements for Equipping the Information and Communication Technology Room. Technical and sanitary requirements for ICT classrooms are examined. Ergonomic, safety, software, and organisational requirements are also studied. The appropriate arrangement and maintenance of computers, networks, multimedia equipment, and workplaces are discussed. Development of Information and Communication Technology Tools. The stages and principles of designing ICT-based educational tools are studied. Planning and content preparation processes are considered. Interface design, testing, implementation, and evaluation processes are examined. Knowledge Assessment Technologies. Modern methods for assessing students' knowledge, skills, and competencies are examined. Digital assessment tools, including electronic testing and automated assessment, are studied. Formative feedback, learning analytics, and result-monitoring systems are discussed. Methodology for Using Electronic Teaching and Learning Materials in the Teaching Process. Methods for selecting and adapting electronic teaching and learning materials are studied. Their application in the teaching process is examined. Their integration into lectures, practical activities, independent learning, and assessment is analyzed. Methodology for Conducting Sessions in Specialist Subjects. The principles and stages of planning lectures, practical sessions, and laboratory work in specialised subjects are studied. The delivery of seminars and independent activities is also
--	---

	examined. Appropriate teaching methods and assessment tools are selected. Planning of Teaching Loads, Educational-Methodological, Scientific and Educational Work. Procedures for planning teaching loads are examined. The organisation of educational-methodological, scientific, and organisational activities is studied. Documentation, workload distribution, scheduling, and performance monitoring are considered. Planning the Teaching Process in Higher Education Institutions. The principles and stages of planning the educational process in higher education are studied. Curriculum development and academic schedules are examined. Course documentation, learning outcomes, teaching methods, and assessment procedures are also studied. Practical lessons Study of the Law of the Republic of Uzbekistan on Education and Other Regulatory Documents. The principal provisions of education legislation are examined. Relevant regulatory documents governing the organisation and management of the educational process are studied. Quality assurance requirements for the educational process are also examined. Planning and Organising the Teaching Process in the Credit-Module System. The principles of planning curricula within the credit-module system are studied. The allocation of credits and organisation of academic activities are examined. The management of student workloads within the credit-module system is also studied. Organising Students' Independent Work and Monitoring Their Knowledge in the Credit-Module System. Methods for assigning and guiding students' independent work are examined. Monitoring and assessing independent work are practiced. Procedures for evaluating learning outcomes and recording assessment results are applied.
--	--

	Familiarisation with Qualification Requirements, Study Plans and Course Syllabi for Educational Programmes and Specialisations. Qualification requirements, curricula, and academic plans are analysed. Course syllabi are examined to understand their structure and content. Their learning outcomes and role in educational programme design are studied. Determining Study Loads and Developing Curricula for Specialised Subjects, Vocational Training Programmes and Calendar Schedules. Procedures for calculating teaching loads are studied. The preparation of subject curricula and vocational training programmes is practiced. Thematic plans and academic calendars are also developed. Study of Scales and Teaching Methods for Specialised Subjects. Assessment scales appropriate for specialised subjects are examined. Teaching methods for specialised subjects are studied. These methods are compared according to learning objectives, content, student characteristics, and expected competencies. Organising Lectures, Structuring a Lecture Script and Delivering It. The stages of preparing a lecture are practiced. Lecture plans and scripts are developed. Visual materials and methods for presenting subject content clearly and systematically are prepared. Methodology for Designing Practical and Laboratory Sessions. Practical and laboratory sessions are planned by defining objectives and tasks. Equipment and procedures are specified. Safety requirements, assessment criteria, and expected learning outcomes are also defined. Planning, Organising and Conducting Student Assessment. Methods for designing formative, continuous, midterm, and final assessments are studied. The administration of these assessments is practiced. Assessment tasks, criteria, rubrics, and feedback procedures are prepared.
--	--

	Distribution of Independent Work Hours and Their Assessment. The principles of allocating independent study hours according to course topics are examined. Workload requirements for independent study are analyzed. Appropriate assignment types, submission schedules, and assessment criteria are developed. Organising and Assessing Demand-Driven Coursework and Projects in Specialised Subjects. Methods for selecting relevant project topics are studied. Requirements for coursework and projects are defined, and project work is supervised. Outcomes are assessed based on practical needs and established criteria. Calculation of Technical Specifications for the Manuscript of Teaching Materials. Procedures for determining the volume and structure of teaching-material manuscripts are practiced. Formatting, illustrations, and references are prepared according to established requirements. Technical requirements of teaching-material manuscripts are examined. Development of Teaching-Methodological Complexes for a Specific Subject. The structure of a teaching-methodological complex is examined. The preparation of the syllabus, lecture materials, and practical tasks is studied. Independent work, assessment tools, and learning resources are also prepared. Designing and Planning Pedagogical Technologies for Specialised Subjects. Teaching models are developed based on appropriate pedagogical technologies. Learning objectives and instructional stages are planned. Methods, resources, interaction formats, and assessment procedures are also planned. Planning Electronic Teaching and Learning Materials for the Subject. The design of electronic textbooks and presentations is studied. Multimedia resources, tests, and interactive tasks are developed.
--	--

	Digital learning modules are designed according to pedagogical, technical, and usability requirements. Self-study Study of the Law of the Republic of Uzbekistan on Education and Other Regulatory Documents. The principal provisions of education legislation and regulatory documents are examined. These documents govern educational organisation, management, quality assurance, and the rights and responsibilities of participants in the educational process. The assignment is completed through the preparation and defence of a presentation in Week 1. Planning and Organising the Teaching Process in the Credit Module System. The principles of credit allocation and academic workload planning within the credit-module system are studied. Learning outcome assessment and the organisation of the teaching process are examined. An assessment criteria table is developed during Weeks 2 to 3. Organising Students' Independent Work and Monitoring Their Knowledge in the Credit-Module System. Methods for planning and supervising students' independent learning are examined. Monitoring and assessment procedures for independent learning are studied. A literature review is prepared during Weeks 4 to 5 as Teacher-Guided Independent Work. Familiarisation with the Qualification Requirements, Curricula and Course Programmes for Educational Fields and Specialisations. The structure, objectives, and learning outcomes of qualification standards, curricula, and course programmes are analysed. The subject content and workload requirements of these programmes are examined. A comparative table and presentation are prepared during Weeks 6 to 7. Determining the Workload, and Developing the Curricula and Working Programmes for Specialised Subjects, and Their Calendar Plans. Procedures for calculating teaching loads are studied.
--	--

	The preparation of curricula, working programmes, thematic plans, and academic schedules is examined. An analytical summary supported by tables and charts is prepared during Weeks 8 to 9. Study of Teaching Scales and Methods for Specialised Subjects. Teaching methods and assessment scales appropriate for specialised subjects are examined. These are compared according to learning objectives, content, student needs, and expected competencies. An analytical commentary is prepared during Weeks 10 to 11 as Teacher-Guided Independent Work. Organising Lectures, Structuring Lecture Texts and Delivering Them. The principles of planning and conducting lectures are studied. This includes defining objectives, structuring content, preparing lecture texts, and selecting visual materials. Effective presentation methods are applied, and an analytical report is prepared during Weeks 12 to 13. Study of the Methodology for Designing Practical and Laboratory Sessions. The design of practical and laboratory sessions is studied, including defining objectives, tasks, and equipment. Procedures, safety requirements, assessment criteria, and expected learning outcomes are examined. A mini-research project is prepared during Weeks 14 to 15.
Teaching Methods	The module is delivered through lectures, practical sessions, and interactive and problem-based learning. Group discussions, presentations, analytical assignments, and project-based activities are also used. Students study regulatory documents and plan teaching processes within the credit-module system. They develop lesson plans, lecture materials, practical and laboratory sessions, assessment criteria, and electronic teaching resources. Teacher-guided and student independent learning, literature reviews, comparative analyses, reports, and methodological recommendations are incorporated. Mini-research projects further develop students'

	pedagogical, methodological, analytical, digital, and professional teaching competencies.
Prerequisites	Scientific Research Methodology Administrative Management of Computer Networks Information Systems Design
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and final assessment (50%). The continuous assessment evaluates participation in 15 practical sessions and completion of two Teacher-Guided Independent Work assignments. A maximum of 8 points is awarded for practical participation, while 12 points are allocated to teacher-guided assignments, with each assignment worth 6 points. The midterm assessment consists of a theoretical component worth 12 points and independent work worth 18 points. The theoretical component may be conducted as a test or a written examination. The written examination includes three theoretical and two practical questions, while the independent component consists of reports, analytical papers, projects, or presentations submitted through the HEMIS system. The final assessment may be conducted as a written examination or a test. The written examination consists of five questions worth 10 points each, while the test consists of 25 questions worth 2 points each. All required independent study assignments must be submitted before the final examination.
Recommended Literature	1. H. Khonboboyev. Methodology of Teaching Specialised Subjects [Text]: A Textbook. Tashkent: Ilm va fan Publishing House, 2023. 144 pp. 2. Zakirova F., Muxamedxanov U., Sharipov Sh., Isyanov R., Esanboboyev F., Dottoyev S. Methodology for Creating Electronic Teaching and Methodological Complexes and Educational Resources. Methodological Guide. Tashkent: Ministry

	of Education and Science of the Republic of Uzbekistan, 2010. 64 pp. 3. A.A. Abduqodirov, A.X. Pardaev. Theory and Practice of Distance Learning. Monograph. Tashkent: Fan, 2009. 146 pp. 4. Yuldashev U.Yu., Boqiev R.R., Zokirova F.M. Methodology of Teaching Informatics. Methodical Guide. Tashkent, 2004. 5. Nazarov U.A., Karimov A.A., Elmuradov B.E. Lecture Notes in the Subject 'Information Technologies in Education'. Samarkand: SamDAQI Publication, 2014. 6. M. Aripov et al. Information Technologies. Textbook. Tashkent: Noshir, 2009. 368 pp.
Responsible Lecturer	Xatamov Ochildi Qurbanovich, Doctor of Economic Sciences, Professor, Department of Computer and Software Engineering, Termiz State University

1.03. DESIGN OF INFORMATION SYSTEMS

Field	Description
Module Title	INFORMATION SYSTEMS DESIGN
Module Code	ATL1106
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 90 hours (44 hours Lecture, 46 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	I semester
Module objective	The objective of the Module Information Systems Design is to develop in postgraduate students knowledge of the theoretical foundations and modern technologies for the analysis, modelling, design, and implementation of information systems. The management of the information system life cycle is also addressed. Theoretical knowledge and practical skills in applying international standards are developed. Systematic approaches to information systems design are studied. Students learn to create complex information systems based on UML and CASE tools. These systems are implemented in practice.
Module Learning Outcomes	Knowledge: LO1. Can explain the fundamental concepts of information systems, systems theory, and the design of information systems, as well as their contemporary development trends. LO2. Can describe the working principles and functions of the technologies, methods and tools used in information systems design. LO3. Can explain the stages of the information systems life cycle, the design processes and the contents of technical documentation.

	LO4. Can distinguish the components of modelling methods for information systems, including DFD, ERD, IDEF and UML diagrams. LO5. Can explain the theoretical foundations of formulating system requirements, developing a technical specification, and making design decisions. LO6. Describe international standards, quality requirements, information security and professional ethical norms in information systems design. Skills: LO7. Performs the processes of analysing, designing and modelling information systems. LO8. Develops and applies UML, DFD, ERD and IDEF diagrams in practical activities. LO9. Formulates the functional and non-functional requirements for information systems and develops technical specifications. LO10. Creates system models using PlantUML, PlantText UML Editor, Lucidchart and other CASE tools. LO11. Develops conceptual models of data flows, business processes and databases. LO12. Analyses, tests, evaluates information system projects and develops proposals for their improvement. Competencies: LO13. Responsibly utilises modern technologies and international standards in the design of information systems. LO14. Critically evaluates information systems architecture, models and design solutions, and analyses their effectiveness. LO15. Designs and implements information systems in accordance with the activities of various sectors and organisations.
--	--

	LO16. Applies research and innovative approaches in the development and modernisation of systems. LO17. They will independently study new software platforms, CASE tools and information system design technologies, and will be qualified to apply them in practice.
Content	Lectures The Concept of a System and Its Various Definitions. Classification of Systems. Systems are introduced as integrated sets of interacting elements forming a coherent whole with a defined purpose and structure. Different definitions from system theory are compared to highlight properties such as boundaries, environment, hierarchy, and feedback. Systems are classified into simple and complex, open and closed, and static and dynamic types for analytical use. Information Systems. Information systems are defined as socio-technical systems that collect, process, store, and distribute information to support decision-making, coordination, and control in organisations. Their main components, including data, procedures, hardware, software, and users, are outlined. Typical types such as transaction processing, management information, decision support, and enterprise systems are examined. Automated Information Systems. Automated information systems are described as information systems where key processes of data acquisition, processing, and dissemination are carried out automatically using computer technologies. The integration of databases and application software is examined. Communication networks are studied for automating routine operations, reporting, and control functions in organisational settings. Information, Software and Technological Support for Information Systems. Information support, covering the structures, formats, and flows of data required for the operation of information systems, is

	studied. Software support, comprising system and application software that implement information processing and control logic, is examined. Technological support, including hardware platforms, communication networks, and technical infrastructure, is also analyzed for its role in reliability and scalability. Design of Information Systems (IS). Definitions of System Theory Concepts. Information system design is presented as a structured process that moves from requirements analysis through conceptual and logical modelling to implementation and deployment. Fundamental system theory concepts, including elements, structure, environment, inputs and outputs, states, and feedback, are defined. These concepts are linked to design decisions about architecture, interfaces, and data models in information system projects. International Standards for Information Systems. International standards for information systems are reviewed as normative frameworks that specify requirements for quality, security, interoperability, and lifecycle management. Examples include standards for software and systems engineering, information security management, and IT service management. These standards guide organisations in designing, documenting, and operating information systems according to recognised best practices. Modelling of Information Systems. Modelling of information systems is introduced as the use of abstract representations to describe the structure, behaviour, and data flows of systems. Conceptual, logical, and physical models are distinguished. Common modelling techniques, such as data-flow diagrams, entity-relationship models, and UML notations, are presented as tools for analysis, design, and communication among stakeholders. Modelling Data Flows. Modelling data flows between processes, data stores, and external entities within an information system is studied. Data-flow diagrams
--	---

and related notations are used to show how information moves, is transformed, and accumulated. These techniques support the identification of bottlenecks, redundancies, and control points.

Stages of Information Systems Design. The stages of information systems design are described, from the initial feasibility study and requirements elicitation through analysis, conceptual and logical design, physical design, and implementation. Each stage is associated with specific deliverables, such as models, specifications, and prototypes. Decision points that affect cost, risk, and system quality are examined at each stage.

Technical Specification. Technical specification is defined as a formal document that describes the functional and non-functional requirements, interfaces, performance criteria, and constraints of an information system. It serves as a basis for design, implementation, and testing. It ensures that stakeholders share a clear and precise understanding of system expectations.

Requirements for Information Systems. The identification, classification, and documentation of requirements for information systems are studied. Functional, non-functional, user, organisational, and regulatory requirements are collected using interviews, workshops, and analysis of existing processes. These requirements are structured into requirement sets that guide design and validation.

Technical Design. Technical design is presented as the phase in which the logical solution is translated into detailed descriptions of system components, data structures, and algorithms. User interfaces and technical configurations are also specified. This phase bridges the gap between abstract models and implementable artefacts, providing the blueprint for developers and administrators.

Main Products. The main products of the information system design process are listed as models, specifications, prototypes, configurations, and

documentation. These products support communication among stakeholders and implementation activities. They also support testing, deployment, and the subsequent maintenance and evolution of the system.

Sample Information Systems Project. A sample information systems project is examined to illustrate the complete lifecycle from initial idea to deployment. The integration of requirements, models, specifications, design decisions, and implementation artefacts is shown. Typical challenges and success factors in information system projects are highlighted.

The UML Language. The UML language is introduced as a standard modelling language for specifying, visualising, and documenting software-intensive systems. Its main diagram types, including use case, class, sequence, and activity diagrams, are presented. State and deployment diagrams are also examined as tools to capture different views of system structure and behaviour.

CASE Systems. CASE (Computer-Aided Software Engineering) systems are described as software tools that support analysis, design, coding, testing, and documentation in information system development. Their functions include model editing and consistency checking. Code generation and reverse engineering functions are also examined for improving productivity and quality in information system projects.

Commercial Object Model. The commercial object model addresses the representation of business entities, services, and processes as objects in an information system. Object classes, attributes, methods, and relationships are defined to reflect commercial logic. The model is used as a basis for implementation in object-oriented or component-based architectures.

Conceptual Model. Conceptual modelling is presented as the creation of technology-independent models that capture the essential structure and meanings of data and processes in a domain. Entity-

	relationship diagrams and high-level process descriptions are used to represent concepts and their relationships. This approach provides a stable foundation for later logical and physical models. Interrelationships and System Requirements. How different system components, processes, and data objects are interrelated is examined. How these interrelationships influence system requirements is analyzed. Dependencies, constraints, and interactions are analysed to refine and validate requirements and to ensure coherent and consistent system behaviour. Practical lessons System Design Using the UML Modelling Language. The UML modelling language is used to describe the structure and behaviour of a system. Key diagram types are applied to capture requirements. High-level architecture is represented in a formal, visual form. Modelling Commercial Premises. Models of commercial premises are built as an example domain. Entities, processes, and relationships within this domain are identified. These elements later appear in UML diagrams used for system analysis and design. Improving Commercial-Like Models. Creating Requirements for the System. Initial commercial-style models are refined. Functional and non-functional requirements are derived from these models. The requirements are organised as structured input for subsequent system design activities. PlantText UML Editor Online UML Diagramming Environment and Its Features. The PlantText UML Editor is explored as an online environment for editing and rendering UML diagrams from textual descriptions. Its interface is examined. Supported diagram types and basic features are also examined. Registration on the PlantText UML Editor Platform. Logging In. User Interface. Creating
--	--

UML Diagrams. Students register and log in to the PlantText platform. The main user interface elements are learned. A set of UML diagrams, including class, sequence, collaboration, activity, and component diagrams, is created using text-based definitions.

Lucidchart Online UML Diagramming Environment and Its Features. Lucidchart is used as an online diagramming tool. Its templates and collaboration features are reviewed. Export options for producing UML-based models in team and individual work are also examined.

Registration in the Lucidchart Online UML Diagramming Environment. Logging In. User Interface. Creating UML Diagrams. Accounts are created in Lucidchart. Students work with the user interface. Various UML diagrams are built, focusing on consistent notation, layout, and organisation of model elements within a shared workspace.

PlantUML Online UML Diagram Creator and Its Features. PlantUML is studied as a text-based UML diagram generator. How textual specifications are converted into graphical diagrams is learned. The diagram types and options available for modelling systems are examined.

Registering with PlantUML. Logging In. User Interface. Access to the PlantUML online environment is configured. Login procedures are performed where required. The editing and preview interfaces used to enter descriptions and view generated diagrams are examined.

Creating UML Sequence Diagrams and Other Helper Commands in the PlantUML Online Environment. Sequence diagrams in PlantUML are studied. Syntax and helper commands are used to represent time-ordered interactions between objects. Layout, lifelines, and messages are controlled using these commands.

Creating UML Use Case and Class Diagrams in the PlantUML Online Environment. Actors, use cases, classes, and relationships are defined in PlantUML

	syntax. Use case diagrams describing system functionality are generated. Class diagrams describing structural design are also generated. Create UML Object and Activity Diagrams in the PlantUML Online Environment. Object diagrams are produced to show instance-level structures. Activity diagrams are created to model workflows and business processes. States, actions, and transitions are specified in textual form. Create UML Component and Deployment Diagrams in the PlantUML Online Environment. Component diagrams representing software components and interfaces are created. Deployment diagrams representing physical nodes are also created. Logical architecture is mapped to implementation and deployment views of the system. Create UML Statechart and Timing Diagrams in the PlantUML Online Environment. Statechart diagrams are generated to model object lifecycles. Timing diagrams are created to depict time-based interactions. PlantUML syntax is applied for states, events, and temporal constraints. Creating UML Diagrams in the Canva Online Environment. Canva is used as a general-purpose design tool to construct UML-style diagrams. Shapes and connectors are adapted to represent system elements. This approach is applied when specialised UML tools are not available. The ARGOUML Environment and Its Features. ArgoUML is examined as a desktop UML modelling environment. Its supported diagram types are identified. Navigation features and model management capabilities for software-intensive systems are also examined. User Interfaces in the ARGOUML Environment. Creating Diagrams and Other Auxiliary Commands. ArgoUML's user interface is navigated. Basic diagrams are created. Auxiliary commands for
--	--

	editing, layout, and consistency checks within UML-based projects are used. Creating Similar Diagrams in the ARGOUML Environment: Use Case, Class, Sequence and Collaboration Diagrams. Use case, class, sequence, and collaboration diagrams are built in ArgoUML. Consistency between different views of the same system is maintained. Behavioural and structural models are linked. Create Statechart, Activity, Component and Deployment Diagrams in the ARGOUML Environment. ArgoUML projects are extended with statechart and activity diagrams. Component and deployment diagrams are also added. These diagrams cover the behavioural and implementation perspectives of system architecture. Generating Program Code in the ARGOUML Environment. Project Formalisation. ArgoUML's code generation features are used to produce skeleton program code from UML models. The project structure is formalised. Diagrams are linked to implementation artefacts and documentation. Discussion of Projects Created on Selected Topics. Projects developed on selected topics are presented and discussed. Modelling choices and tool usage are evaluated. The alignment between requirements, UML models, and proposed implementations is assessed. Discussion of Projects Created by Postgraduate Students. Projects prepared by postgraduate students are reviewed and analysed. Exposure to more advanced modelling and design examples is gained. Methodological and technical approaches used in these projects are reflected upon. Self-study Key Concepts of Information Systems Design Technologies, the Purpose and Content of Its Methodology, and Its Stages of Development. The core concepts of information systems design
--	--

technologies, including goals and principles, are studied. Typical approaches to information systems design are examined. The methodology is analysed in terms of its structure, main activities, and historical stages of development in organisational practice.

Subsystems in Information Systems and the Stages and Tasks of Information Systems Design.

Subsystems within information systems, such as data management, user interaction, and control components, are analysed. Design stages associated with each subsystem are identified. These stages link analysis, modelling, specification, and implementation activities.

The IDEF Functional Methodology, Data Flow Modelling and the Object-Oriented Modelling Methodology.

The IDEF family of functional modelling methods is reviewed, focusing on process and function diagrams. Data flow modelling is examined. Object-oriented modelling is compared with data flow modelling as a complementary approach for representing system behaviour and structure.

Comparative Analysis of Modelling Methodologies.

A comparative analysis of different modelling methodologies used in information systems design is performed. Criteria such as expressiveness and tool support are applied. Learning effort and suitability for specific project types are also evaluated to determine the advantages and limitations of each methodology.

Modelling Business Processes.

Business processes are modelled by identifying activities, participants, inputs, and outputs. Process maps and activity diagrams are used to represent workflows. Opportunities for optimisation and automation are revealed through this analysis.

Structured Modelling Software for RAMUS Processes.

Structured modelling software for RAMUS-based processes is explored. The tool is used to create hierarchical models of functions and

	processes. Attention is given to notation rules, consistency checks, and documentation features within the software. Context, Decomposition, Secondary Decomposition and Creating DFD Diagrams. System context is defined. Primary and secondary decomposition of processes into sub-processes is performed. Data Flow Diagrams are created to visualise flows between processes, data stores, and external entities. Formulating Functional Requirements for Information Systems. Functional requirements for information systems are formulated based on the analysis of processes and stakeholder needs. Requirements are structured into categories and prioritised. Documentation of requirements is prepared in a way that supports later design and validation. UML Modelling Language: Online and Offline Tools, and a Comparative Analysis. UML modelling using both online and offline tools is examined. Features such as diagram support, collaboration options, and integration with other development tools are compared. A comparative analysis highlights which tools are most suitable for different project environments. Designing a System and Creating Models of Various Domain Objects Using the UML Modelling Language. A system is designed, and UML models of domain objects are built. Classes, relationships, and interactions are represented in these models. Different diagram types are used to represent structural and behavioural aspects of the domain. Creating Class, Sequence, Collaboration, Activity and Component Diagrams Using PlantText UML Editor. The PlantText UML Editor is used to create class, sequence, collaboration, activity, and component diagrams from textual descriptions. Consistency across diagrams is maintained. The
--	--

	editor's automation capabilities are exploited to streamline diagram creation. Create Class, Sequence, Collaboration, Activity and Component Diagrams Using Lucidchart. Class, sequence, collaboration, activity, and component diagrams are constructed in Lucidchart. The graphical interface is utilised for diagram construction. Collaboration features are used to develop and refine shared UML models. Create Use Case, Class, Object, Activity, Component, Deployment, Statechart and Timing Diagrams Using PlantUML. PlantUML is employed to generate a wide range of UML diagrams, including use case, class, and object diagrams. Activity, component, deployment, statechart, and timing diagrams are also generated. Text-based specifications are used to maintain and update the full set of models efficiently. Creating Models of Commercial Objects in the ArgoUML Environment, Code Generation and Project Documentation. Commercial objects are modelled in ArgoUML, defining classes and relationships that reflect business logic. Code generation functions are applied to produce implementation skeletons. Project documentation functions are used to produce formal project artefacts. Capabilities for Creating UML Diagrams in the Canva Online Environment. The capabilities of Canva for creating UML-style diagrams are investigated. General-purpose shapes and connectors are adapted to represent system elements. Canva can be used when specialised UML tools are unavailable or when visual presentation is the main focus.
Teaching Methods	The module is delivered through lectures, practical sessions, and problem-based and case-based learning. Analytical activities, presentations, independent research, and mini-projects are also incorporated. Students use system theory, information-systems and automated-systems

	concepts, modelling methodologies such as DFD, IDEF, and UML, and requirements engineering to analyse and design information systems. Practical work focuses on modelling business processes and data flows. Specifications and designs are prepared, and UML and CASE-tool diagrams are created for sample information-systems projects, thereby developing students' analytical, modelling, documentation, and design skills.
Prerequisites	Programming 1,2 Database Data Structures and Algorithms
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, practical and laboratory performance in information systems modelling and design, seminar presentations, and the successful completion of independent study assignments, including modelling business processes and data flows, preparing subsystem descriptions, analysing modelling methodologies (DFD, IDEF, UML), and producing sample UML diagrams with online and offline CASE tools. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as an information systems design concept paper, a context and DFD diagram set, a requirements specification for an information system, a comparative review of modelling tools, and a small design report for a commercial-object information system. The final assessment is a written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of systems and information systems, modelling methodologies, requirements engineering, UML and CASE-based design, analytical skills, and their ability

	to apply information-systems design principles. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Arthur M. Langer. Analysis and Design of Information Systems. 3rd Edition. Springer, 2007. 432 p. 2. V.I. Gerekul, N.L. Korovkina, Yu.V. Kupryanov. Design of Information Systems. Practical Workbook: A Textbook with Illustrations. Moscow: National Open University INTUIT, 2012. 187 pp. 3. D.V. Chistov (ed.). Design of Information Systems: Textbook and Practical Work for Academic Undergraduate Studies. Moscow: Yurayt Publishing House, 2016. 258 pp. 4. I.Yu. Kotsyuba, A.V. Chunaev, A.N. Shikov. Fundamentals of Information Systems Design: A Textbook. St. Petersburg: ITMO University, 2015. 206 pp. 5. D.Yu. Ivanov, F.A. Novikov. Unified Modeling Language (UML): A Textbook. St. Petersburg: Polytechnic University Press, 2011. 229 pp.
Responsible Lecturer	Juraev Olim Albaevich, Head of Department of Computer and Software Engineering, Termiz State University

1.04. ADMINISTRATIVE MANAGEMENT OF COMPUTER NETWORKS

Field	Description
Module Title	ADMINISTRATIVE MANAGEMENT OF COMPUTER NETWORKS
Module Code	KTMB1106
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 90 hours (46 hours Lecture, 44 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	I semester
Module objective	The objective of the Module Administrative Management of Computer Networks is to develop theoretical knowledge and practical skills in the administrative management of modern computer networks. Students learn to configure network services. Network monitoring techniques are studied. Methods for ensuring network security are developed. Backing up and restoring network infrastructure are practiced. Students learn to effectively manage corporate networks.
Module Learning Outcomes	Knowledge: LO1. Can explain the operating principles, architecture and main components of computer networks. LO2. Can explain the functions of IP addressing, IPv4/IPv6, subnetting, VLSM, DHCP and DNS services. LO3. Can distinguish between the OSI and TCP/IP models, and the protocols at the link, network, transport and application layers. LO4. Describe routing, switching, Ethernet, WLAN, FDDI, Frame Relay, ATM, ISDN and DSL technologies.

	LO5. Has theoretical knowledge of network security, authentication, firewalls, VPNs, NAC, cryptographic protection and monitoring systems. LO6. Explain the fundamentals of server services, web hosting, HTTP, web server configuration and client-server technology. Skills: LO7. Performs practical tasks on configuring router, switch, LAN/WAN, VLAN, VPN, DHCP, DNS, SNMP and NetFlow services. LO8. Analyses and optimises network traffic, identifies and diagnoses faults. LO9. Manages users and groups, Active Directory, resource access rights and account policy. LO10. Implements measures to ensure the continuous operation of backup, archiving, recovery and network services. LO11. Configure web server, virtual hosts, proxy server, cache and authentication settings. LO12. Demonstrates skills in developing network security policies, detecting attacks and using protection tools. Competencies: LO13. Will have competence in planning, managing and documenting corporate network infrastructure. LO14. Applies FCAPS, NMS, SNMP, QoS, MPLS, NGN and Softswitch-based management approaches in practical projects. LO15. Can make independent decisions regarding the reliable, secure and efficient management of network resources. LO16. Independently studies and implements new standards, services and security solutions in network technologies.
Content	Lectures

	Principles of Internet Operation. Internet Networks and Their General Structure. The Internet is presented as a global network of networks based on the TCP/IP model and packet switching. The general structure of Internet networks is described in terms of backbone networks, routers, providers, and local access networks arranged as a connected chain. Client-server architecture, IP addressing, domain names, and the DNS system form the basis for the path of data packets from source to destination. IP Address Standards and Types. IP Address Classes and Special Addresses. The concept of an IP address is given as the fundamental format for identifying devices in a network. IPv4 and IPv6 standards are compared in terms of numerical structure, address space size, and areas of application. IP address classes, the separation of network and host parts, and special addresses, including private, loopback, and multicast addresses, are distinguished. Creating Subnets Using the VLSM Method. The DHCP Protocol, Message Format and Operating Algorithm. Network subdivision into smaller segments based on variable-length subnet masks is studied. Stepwise determination of network and host ranges is used to show efficient allocation of address resources. The role of the DHCP protocol, its main message types and formats, and the sequence of its operation are described. Domain Hierarchy, DNS Service and Protocol, NetBIOS Naming System. The Domain Name System is defined as a mechanism for managing IP addresses through hierarchical textual identifiers. The DNS hierarchy is described as a chain of root servers, top-level domains, and lower-level zones. The main functions of the DNS service, the query-response process, and protocol-level resource records are specified, and the NetBIOS naming system is
--	--

introduced as a mechanism for identifying computers in local networks.

The OSI Model and the TCP/IP Protocol Stack. The OSI reference model structures communication into seven layers with clearly separated functions. The TCP/IP protocol stack is outlined as a practical four-layer architecture used in real networks. Relationships between OSI layers and TCP/IP layers are noted to show how both models serve as frameworks for analysing services, interfaces, and data flow in networked systems.

Link Layer Protocols. The IP Protocol, the IP Datagram Format and the Packet Transmission Process. Link layer protocols are presented as mechanisms for framing, addressing, and error control on a local physical medium between directly connected nodes. The IP protocol is introduced as the core network-layer protocol that defines the IP datagram structure and supports forwarding of packets across interconnected networks. The packet transmission process is outlined from encapsulation at the source host through intermediate routers to decapsulation at the destination host.

ARP and RARP Address-Mapping Protocols. ICMP Message Types and Caching of Query Results. ARP and RARP are described as address-mapping protocols that translate between IP addresses and hardware addresses in local networks. ICMP message types are listed as control and error-reporting messages exchanged between network devices for diagnostics and signalling. Caching of query results is linked to reduced control traffic and faster resolution of repeated address and status requests.

Ping and Traceroute Utilities. PPP Protocol, Network Access via PPP and Security Issues. Ping and traceroute utilities are introduced as basic diagnostic tools that generate test traffic to measure reachability, delay, and path characteristics. The PPP protocol is described as a data link protocol for point-

to-point connections, with encapsulation, negotiation, and optional authentication features. Network access via PPP is associated with configuration procedures and specific security issues related to authentication and line integrity.

Application-Level Protocols. Fundamentals of Switching and Routing in IP Networks. Application-level protocols are grouped as services that define message formats and interaction rules for web access, file transfer, email, and remote management. Fundamentals of switching and routing in IP networks cover the separation of frame forwarding in local segments from packet forwarding between networks. The use of addressing, forwarding tables, and path selection criteria is indicated as central to data delivery.

Static and Dynamic Routing. Routers and the Distance-Vector Routing Algorithm. Static routing is characterised by manually configured entries that remain fixed unless changed by an administrator. Dynamic routing is based on automatic exchange of routing information that adapts paths to topology changes. Routers are described as devices that maintain routing tables and apply the distance-vector algorithm, which propagates cost values between neighbours to compute preferred routes.

Routing Algorithm Taking Channel State into Account. UDP Protocol and Header Fields. Routing algorithms that take channel state into account use metrics such as delay, bandwidth, and load to select paths that balance performance and resource use. The UDP protocol is introduced as a connectionless transport protocol with minimal header information for ports, length, and checksum. UDP header fields are linked to simple multiplexing of applications and low-overhead delivery without built-in reliability.

TCP Protocol, SNMP, MIB Database, SMTP, POP3, IMAP and FTP Protocols. The TCP protocol is presented as a connection-oriented transport protocol that provides reliable, ordered delivery of a byte

	stream between endpoints. SNMP is described as a management protocol that uses the MIB database to organise information about network devices for monitoring and control. SMTP, POP3, and IMAP are grouped as core email protocols for message submission, retrieval, and synchronisation, while FTP is identified as a standard protocol for file transfer between hosts. Local and Distributed Network Technologies. Ethernet Technology and Topologies. Local and distributed network technologies are outlined in terms of how computing resources are interconnected within a site and across multiple locations. Ethernet is introduced as the dominant LAN technology, with physical and logical characteristics defined by standard speeds and media. Ethernet topologies, such as bus, star, and switched structures, are described through their wiring patterns and collision domains. Ethernet Addressing, Access Control, Wireless Local Area Networks, FDDI and Frame Relay Technologies. Ethernet addressing is presented using MAC addresses as unique hardware identifiers for devices on the link. Access control mechanisms and frame handling are related to media arbitration and switch-based forwarding. Wireless local area networks extend Ethernet concepts to radio-based communication, while FDDI and Frame Relay are given as legacy high-speed and wide-area technologies with their typical usage contexts. ATM, ISDN and DSL Technologies: Basic Concepts and Areas of Application. ATM technology is characterised by fixed-size cells and virtual circuits for integrated voice, video, and data transmission. ISDN is described as a digital telephony service that provides separate channels for signalling and user data over traditional lines. DSL technologies are outlined as broadband access methods that use copper pairs to deliver higher-speed Internet connections to homes and offices.
--	---

	User and Group Management. Domain, Workgroup, Access Rights, Account Policy and Backup Principles. User and group management is introduced as the basis for organising identities and permissions in networked systems. Domains and workgroups are distinguished as different models for structuring authentication and resource sharing. Access rights and account policies define what operations are permitted, while backup principles ensure that user data and configuration information can be restored in case of failure. Archiving and Restoring System Data. Local Document Storage, Monitoring and Management Methodology. Archiving and restoring system data are introduced as core procedures for preserving and recovering configuration and user information. Local document storage is described in terms of directory structures, access policies, and versioning practices on workstation and server systems. Monitoring and management methodology covers regular checks, logging, and the use of administrative tools to track system state and plan maintenance. NAT System, Address Translation, Network Services, File Sharing and the Interaction Between Operating Systems. The NAT system is presented as a mechanism that translates private internal addresses to public external addresses at network boundaries. Address translation is related to access to network services such as web, email, and file transfer from multiple hosts behind a single public IP. File sharing and interaction between operating systems highlight common protocols and methods that allow heterogeneous systems to exchange data in a local or wide-area network. Organising Email and Managing Basic Network Services. Email organisation is outlined through mailboxes, user identifiers, and domain-based addressing. Managing basic network services includes the configuration and supervision of DNS, DHCP, web, mail, and directory services that support everyday communication and resource access.
--	--

	Service dependencies and startup, shutdown, and logging procedures are indicated as part of the operational environment. Network Security. Authentication, Instrumental Protection Tools, Cryptographic Protection Systems and Firewalls. Network security is described as a set of measures that protect data, services, and devices from unauthorised access and misuse. Authentication identifies users and systems before granting access to resources. Instrumental protection tools, cryptographic protection systems, and firewalls are grouped as technical means that enforce security policies, filter traffic, and secure communication channels. Web Hosting, the HTTP Protocol, Client-Server Technology, Web Server Installation and Configuration, Virtual Hosts, Caching and Proxy Servers. Web hosting is introduced as the provision of storage and processing resources for websites on server platforms. The HTTP protocol and client-server technology form the basis of web communication between browsers and servers. Web server installation and configuration, the use of virtual hosts, caching mechanisms, and proxy servers are presented as practical elements of deploying and optimising web services. Server Application Development Technologies. CGI Interface, the PHP Language, Dynamic Content, HTML Forms and Loops. Server application development technologies are outlined as tools and interfaces for generating responses to client requests on the web. The CGI interface is described as a standard for connecting web servers with external programs, while the PHP language represents a widely used scripting environment for dynamic pages. Dynamic content, HTML forms, and control structures such as loops are combined to implement interactive web functionality. Integrating Web Database Architecture and Network Services with Web Applications.
--	---

	Integration of web applications with database architecture is presented as a way to store, retrieve, and manipulate structured data behind web interfaces. Network services such as authentication, mail, and directory services are related to web-based front-ends that use them as back-end components. The overall architecture links web servers, application logic, and databases to support transactional and information-oriented web systems. Practical lessons Study the Principles of Network Management. Analyse the Role and Importance of Various Stakeholders. Basic principles of network management, including monitoring, configuration, accounting, performance, and security functions in managed networks, are studied. The roles of administrators, service providers, users, and regulatory bodies are identified. These roles are compared in terms of responsibilities and influence on network operation. Installation and Configuration of Network Equipment in Telecommunications Networks. Resource Identification. Installation and initial configuration of network devices, such as switches, routers, and access points, are performed in a telecommunications environment. Network resources, including interfaces, addresses, services, and device identifiers, are discovered. These resources are documented using command-line tools and management utilities. Exploring Network Management Features in the Windows Operating System. Shared Access to Resources and Setting Up an Access List. Built-in network management features of the Windows operating system, including management consoles and administrative tools, are explored. Shared access to files, printers, and other resources is configured. Access lists are defined to control permissions for users and groups in a local or domain environment.
--	--

	Network Management Protocols. SNMP Protocol. Network management protocols are examined, with a focus on the structure and operation of the SNMP protocol. SNMP components such as managers, agents, and MIB objects are reviewed. Message types are examined using examples from real or simulated network devices. Configuring SNMP on Network Equipment Using a Cisco Router as an Example. SNMP parameters are configured on network equipment, using a Cisco router as a representative platform. Community strings, SNMP versions, access restrictions, and trap destinations are set and tested. The resulting configuration is verified through SNMP queries from a management station. Configuring the VLAN System and VPN. VLANs are configured on switches to segment network traffic by department or service. Basic VPN connections are set up to provide secure remote access. Tunnels, authentication parameters, and encryption settings are defined for the VPN connections. Learning Management Tools and Solutions within Cisco Networking Services. Performing Tasks within the FCAPS Framework. Cisco networking management tools are used to perform monitoring, configuration, and troubleshooting tasks. Activities are mapped to the FCAPS framework, covering fault detection and configuration changes. Accounting data collection, performance checks, and security controls are also examined within this framework. Study of Standard Tools and Corporate Solutions for Network Management. Management Platforms and Tools. Standard network management tools and commercial management platforms used in enterprise environments are reviewed. Laboratory exercises compare features for device discovery and topology visualisation. Alerting, reporting, and automated configuration features are compared across multiple vendors.
--	---

	Explore the IMC QoS Manager, IMC MPLS VPN Manager and IMC Network Traffic Analysis Tools. IMC modules, including QoS Manager, MPLS VPN Manager, and Network Traffic Analysis, are used to observe and adjust quality of service settings. VPN configurations and traffic patterns are examined using these tools. Tasks include viewing policies, modifying parameters, and interpreting graphs and reports produced by these tools. Network Backup and Recovery Strategies. Backup procedures for configuration files, critical services, and key data in a networked environment are designed and implemented. Recovery strategies are tested by restoring saved configurations and data. These strategies are tested after simulated failures or misconfigurations. Analysis and Optimisation of Network Traffic. Network traffic is captured and analysed using monitoring and packet inspection tools. Utilisation, delays, and bottlenecks are observed. Optimisation measures, such as policy adjustments, QoS rules, or topology changes, are proposed and evaluated. Active Directory Integration and Management. Computers and users are integrated into an Active Directory domain. Organisational units, users, and groups are managed within the domain. Practical tasks include configuring Group Policy, applying policies to specific containers, and verifying domain-based authentication and authorisation. Service Virtualisation and Network Integration. Virtualised services, such as virtual machines or containers that provide networked applications, are deployed. Network integration covers assigning addresses and configuring virtual switches and bridges. Virtual services are connected to existing LAN and WAN segments. Configuration and Management of LAN and WAN Networks. LAN segments are configured using switches, VLANs, and basic security features. WAN links are set up using routers and appropriate
--	---

encapsulation methods. Management activities include documenting topology, applying configuration changes, and verifying connectivity between sites.

Managing IPv4 and IPv6 Addresses. IPv4 and IPv6 address ranges are planned and assigned for subnets and devices. Static and dynamic addressing configurations are practiced. Dual-stack connectivity is tested, and address allocation tables are maintained for administrative control.

Network Monitoring and Analysis: SNMP and NetFlow. SNMP-based tools and NetFlow exporters are used to collect performance and traffic data from network devices. Collected statistics are analysed to identify utilisation patterns. Potential bottlenecks and anomalies in flows between key endpoints are identified.

Analysis of Network Errors. Network errors, such as connectivity loss, routing loops, address conflicts, and protocol misconfigurations, are investigated. Diagnostic tools and logs are used to trace error causes. Corrective actions are applied to restore stable operation.

Router and Switch Configuration. Essential configuration tasks on routers and switches, including setting hostnames, interfaces, and routing parameters, are performed. VLANs and trunk links are also configured. Verification steps ensure that forwarding, segmentation, and access between subnets work as intended.

Configuration and Management of Wireless Networks. Wireless access points and controllers are configured, defining SSIDs, security modes, and access policies. Signal coverage, client associations, and throughput are monitored. Parameters are adjusted to improve reliability and performance.

Network Access Control (NAC) Systems. Network Access Control systems are examined as mechanisms that regulate which users and devices may connect to a network in accordance with defined security policies. NAC components are identified, and

the enrolment of endpoints is practiced. Access decisions applied based on authentication and device compliance are observed.

MPLS-Based Network Management. NGN Network Management, Softswitch Architecture and Control Protocols. Network management aspects of MPLS-based infrastructures, including label-switched paths and traffic engineering, are reviewed. Quality-of-service policies are examined. NGN management is considered together with softswitch architecture, focusing on the separation of control and media planes and the use of signalling and control protocols in converged voice-data networks.

Methods for Implementing Control Systems and Fundamentals of Ensuring Security in Control. Methods for implementing industrial and networked control systems, from architecture definition to deployment and monitoring, are studied. Security fundamentals in control environments are addressed through the identification of threats. Authentication, segmentation, and protective measures that reduce the risk of unauthorised or unsafe operations are applied.

Self-study

Concept of Network Management and Functions of Network Management Systems (NMS). The basic concept of network management as a structured set of activities that ensure reliable and efficient operation of computer networks is studied. Network Management Systems are reviewed in terms of their core functions, including monitoring, configuration, and performance analysis. Fault handling and security support in single-vendor and multi-vendor environments are also examined.

Network Design Based on IP Addressing, Subnetting, and VLSM. Network design methods that rely on hierarchical IP addressing and subdivision of address space into subnets are analysed. Fixed and variable-length subnet masks are used to create subnets of different sizes. Design tasks focus on

allocating address ranges, planning routing, and documenting addressing schemes for corporate networks.

Configuration and Analysis of DHCP and DNS Services. The roles of DHCP and DNS as fundamental services in IP networks are reviewed. Configuration examples show how DHCP distributes addresses and parameters to hosts, while DNS resolves names to addresses. Analysis tasks concentrate on service dependencies, typical errors, and the impact of these services on user access and application performance.

Role of the OSI and TCP/IP Models in Network Management. The OSI and TCP/IP reference models are examined as frameworks for understanding layered communication in networks. The relation of these models to management functions is explored by mapping protocols, services, and monitoring tasks to specific layers. This mapping supports structured analysis and troubleshooting of network behaviour.

Network Monitoring Using SNMP, MIB, and NetFlow. SNMP and MIB are studied as standard mechanisms for collecting status and performance information from network devices. NetFlow is introduced as a flow-based approach to traffic monitoring. The combined use of these tools is considered for detecting anomalies, planning capacity, and documenting utilisation in managed networks.

Cisco Router and Switch Configuration and Documentation. Configuration practices for Cisco routers and switches, including interfaces, routing parameters, VLANs, and basic security features, are studied. Emphasis is placed on systematic documentation of configurations and topology diagrams. Change records are maintained to support administration, audit, and future modifications.

Implementation of VLAN and VPN Technologies. VLANs are explored as a means of logical segmentation of local networks, and VPNs are

	examined as mechanisms for secure remote connectivity over public infrastructures. Design and configuration topics include assigning VLANs to ports and defining VPN tunnels. These technologies are related to organisational structure, performance, and security requirements. Active Directory and User Management in Corporate Networks. Active Directory is examined as a centralised directory and authentication system in corporate environments. User and group management, organisational units, and policy-based administration are studied. The focus is on controlling access to resources, supporting role-based management, and integrating services within the domain. Network Traffic Analysis and Optimisation Using Wireshark and NetFlow. Wireshark and NetFlow-based tools are used to capture, classify, and analyse network traffic at different layers. Findings from packet-level and flow-level observations are used to identify bottlenecks and misconfigurations. Inefficient patterns are identified, and optimisation measures such as policy adjustments or architectural changes are proposed. Network Backup and Recovery Strategies. Strategies for backing up critical network-related data, including configuration files, service databases, and documentation, are studied. Recovery procedures are considered in the context of failures, migrations, and upgrades. The importance of tested backup plans, storage policies, and version control in network administration is highlighted. Firewall, IDS/IPS, and Network Access Control (NAC) Systems. Firewalls, intrusion detection and prevention systems, and Network Access Control systems are reviewed as complementary components of network defence. Topics cover the placement and configuration of these systems. Their roles in enforcing security policies and limiting unauthorised or risky access are examined.
--	--

	Web Server, HTTP, Virtual Host, and Proxy Server Configuration. Web server architecture and the HTTP protocol are examined as the basis of web services. Configuration tasks include setting up sites and using virtual hosts for multiple domains. Proxy and caching functions are deployed, with emphasis on their relationship to performance, security, and resource management. Approaches to Managing MPLS and Next Generation Networks (NGN). Management approaches for MPLS-based networks, including label distribution, traffic engineering, and service differentiation, are studied. NGN management is considered through softswitch and control-plane concepts. Attention is given to converged voice-data services, quality of service, and coordination of signalling and media resources. Establishing Security Policies and Monitoring in Corporate Networks. The definition and implementation of security policies that govern access, usage, and protection of network resources in organisations are studied. Monitoring activities, logs, alerts, and reporting mechanisms are examined. These mechanisms are used to check policy compliance, detect incidents, and support continuous improvement of security posture. Final Mini-Project: Corporate Network Management Model. A mini-project is developed to design and document a corporate network management model integrating concepts from the module. The project typically includes network architecture, addressing plan, key services, management tools, and security measures. Operational procedures are also documented, resulting in a coherent blueprint for administrative management of a computer network.
Teaching Methods	The module is delivered through interactive lectures, case-based and problem-based learning, and laboratory exercises with real and virtual network equipment. Computer simulations, seminars, group

	discussions, and project-oriented activities are also used. Students perform configuration and management tasks on routers, switches, servers, and directory services. Monitoring tools are used to collect and interpret network performance data, and administrative scenarios related to users, groups, and access rights are analysed. Independent learning and the use of digital educational resources support the development of analytical thinking, documentation, and reporting skills, and the ability to apply modern concepts of administrative management of computer networks in organisational environments.
Prerequisites	Computer Science Operating Systems Databases
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance with routers, switches and servers, practical configuration tasks (IP addressing, VLAN/VPN, DHCP/DNS, Active Directory), seminar presentations, and the successful completion of independent study assignments, including network design exercises, management system (NMS) case studies, documentation of corporate network architectures, and short technical reports on network security and monitoring. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as a network management plan for a small enterprise, a topology and addressing diagram for a corporate network, a video presentation on best practices in network administration, a mini-project on SNMP/NetFlow-based monitoring, and an essay on policy-based management and access control in IP networks. The final assessment is a written examination consisting of theoretical questions and

	practical problem-solving tasks designed to assess students' understanding of administrative management of computer networks, analytical skills, and their ability to apply management, configuration, monitoring and security principles in realistic scenarios. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Jo'rayev N.M. Technical Maintenance of Telecommunications Networks. Textbook. Tashkent: Alohachi, 2020. 416 pp. 2. Oriental Renaissance: Innovative, Educational, Natural and Social Sciences. Scientific Journal, Vol. 2, Issue 11, November 2022. ISSN 2181-1784. 3. Cisco Systems. Network Management Configuration Guide, Cisco IOS Release 15.0S. Cisco Systems, Inc., 2010. 4. Forouzan B.A. Data Communications and Networking. 5th Edition. New York: McGraw-Hill Education, 2012. 1264 p. 5. Kiritchenko A.A. Administration of Computer Networks. Network-based electronic textbook. 2014. 189 pp.
Responsible Lecturer	Xatamov Ochildi Kurbonovich, Doctor of Economics Sciences, Associate Professor

1.05. DESIGN AND ANALYSIS OF ALGORITHMS

Field	Description
Module Title	DESIGN AND ANALYSIS OF ALGORITHMS
Module Code	ALTQ1206
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	II semester
Module objective	The objective of the Module Design and Analysis of Algorithms is to equip master's students with fundamental knowledge and practical skills in designing, analysing and optimising efficient algorithms. The Module establishes a solid foundation for algorithmic thinking, problem decomposition and the application of algorithmic methods to programming practice. It introduces methods for evaluating time and space complexity using Big-O, Big-Ω and Big-Θ notation, and applies these methods to recursive, sorting, searching and graph algorithms. Divide-and-conquer, greedy and dynamic programming paradigms are examined together with their application to optimization problems such as scheduling, minimum spanning trees and the knapsack problem. The Module further covers combinatorial optimization, computational geometry, randomized algorithms and the theory of NP-completeness, including reduction techniques and approximation methods for computationally difficult problems. Students also study the fundamental principles of designing and executing parallel algorithms. Through this combination of theoretical analysis and practical implementation, students develop the competence to select, design and optimise algorithmic solutions for complex computational problems.

Module Learning Outcomes	Knowledge: LO1. Learns to apply Divide and Conquer, Greedy Algorithms, Dynamic Programming and other methods. LO2. Is able to assess the time and space complexity of algorithms and to use the Big-O, Big-Ω, and Big-Θ notations. LO3. Learns to select and design an appropriate algorithmic approach to create an optimal solution for a given problem. LO4. Learns the practical application of algorithmic analysis in a programming environment, alongside a theoretical understanding of it. LO5. Understands how to apply techniques for improving and optimising algorithms. LO6. Has experience in analysing real-world problems and developing effective algorithmic solutions for them. LO7. Learns the fundamental principles for designing and efficiently executing parallel algorithms. LO8. Breaks down complex problems into smaller parts and analyses them. LO9. Finds the optimal solution for search-based problems, among others. Skills: LO10. Possesses systematic thinking skills for understanding and solving algorithmic problems. LO11. Acquires skills in working with stacks, queues, graphs, trees and other data structures to ensure the efficient operation of algorithms. LO12. Is able to solve a given problem using an algorithmic approach. LO13. Logically analyses problems and selects optimised solutions. LO14. Breaks down the solution into stages and develops an effective strategy for the problem.
---------------------------------	--

	LO15. Evaluates the time and space complexity of an algorithm using Big-O notation. LO16. Reduces execution time and conserves resources through efficient algorithms. LO17. Tunes the algorithm to run faster by optimising the code. LO18. Verifies that the algorithm works correctly for all possible states. LO19. Identifies and fixes bugs and performance issues. LO20. Verifies the correctness of algorithms through automated tests. Competencies: LO21. Is able to solve complex problems algorithmically and to implement them in software tools. LO22. Analyses problems, breaks them down into parts and finds the most effective solution. LO23. Is able to improve existing algorithms and to develop new algorithmic approaches.
Content	Lectures Introduction to Algorithm Design. This topic introduces the fundamental concepts of algorithm design, including algorithmic thinking and problem-solving strategies. The characteristics of efficient algorithms are also examined. Students learn the importance of algorithms in computer science. Analysis of Algorithms. Students study methods for evaluating algorithm performance using time complexity. Methods for evaluating algorithm performance using space complexity are also studied. The topic introduces asymptotic notations such as Big-O, Big-Ω, and Big-Θ. Recursive Algorithms. This lecture explains the concept of recursion. Recursive problem-solving techniques are also explained. Students learn how

	recursive functions simplify complex computational tasks. Sorting Algorithms. Students examine common sorting algorithms such as Bubble Sort, Selection Sort, and Insertion Sort. Merge Sort, Quick Sort, and Heap Sort are also examined. Their efficiency and practical applications are compared. Search Algorithms (Binary Search Algorithms). This topic introduces searching techniques with a focus on binary search. Students analyze its efficiency. Students also compare it with linear search methods. Algorithms on Graphs. Students study graph representations. Graph traversal algorithms such as Depth-First Search (DFS) and Breadth-First Search (BFS) are also studied. Applications in networking and pathfinding are discussed. Greedy Algorithms. This lecture introduces greedy algorithm design techniques where locally optimal choices lead to global solutions. Practical examples include scheduling problems. Minimum spanning tree problems are also included as practical examples. The Divide and Conquer Method. Students learn how complex problems can be divided into smaller subproblems. Students also learn how these subproblems are solved recursively. Applications include Merge Sort, Quick Sort, and binary search. Design and Analysis of Dynamic Programming Algorithms. This topic explains dynamic programming techniques for solving optimization problems. These techniques rely on storing intermediate results. Students study classic examples such as the Knapsack and Fibonacci problems. The Dynamic Programming Problem. Students apply dynamic programming methods to solve complex computational problems efficiently. The topic emphasizes overlapping subproblems. Optimal substructure is also emphasized.
--	---

	Combinatorial Problems. This lecture introduces combinatorial optimization and counting problems. These problems involve permutations, combinations, subsets, and graph structures. Efficient solution techniques are discussed. Computational Geometry Algorithms. Students study algorithms for solving geometric problems such as convex hull construction and line intersection. The closest pair of points problem is also studied. Applications in computer graphics and GIS are explored. Random Algorithms. This topic introduces randomized algorithms that use random choices to improve efficiency or simplify computation. Their advantages and limitations are analyzed. Their practical applications are also analyzed. NP – Estimating the Difficulty of Algorithms for Solving Complete Problems. Students learn the concepts of P, NP, NP-Complete, and NP-Hard problems. The topic explains computational complexity. The challenges of solving difficult optimization problems are also explained. Selecting and Colouring Vertices in Graphs. Identifying Subsets of Sets and Merging Them. This lecture examines graph coloring algorithms and independent set problems. Disjoint-set (Union-Find) data structures are also examined. Students learn their applications in scheduling, network optimization, and graph analysis. Practical lessons Complete Steps for Constructing Algorithms and Methods of Representation. This seminar introduces the stages of algorithm development, including problem analysis and algorithm design. Representation using pseudocode and flowcharts is also introduced. Students learn how to document algorithms clearly and systematically. Assessing and Analysing the Complexity of an Algorithm. Students evaluate algorithm efficiency by
--	--

analyzing time complexity. Space complexity is also analyzed. The seminar emphasizes the use of Big-O notation to compare algorithm performance.

Designing and Analysing Programs for Recursive Algorithms. This topic focuses on implementing recursive algorithms. Their correctness and efficiency are analyzed. Practical examples include factorial, Fibonacci, and recursive tree traversal.

Programming and Analysis of Sorting Algorithm Programs. Students develop and compare sorting algorithms such as Bubble Sort, Merge Sort, Quick Sort, and Heap Sort. The seminar examines their computational complexity. Their practical applications are also examined.

Programming and Analysis of Search Algorithms. This seminar covers the implementation of linear search and binary search. Graph search algorithms such as DFS and BFS are also covered. Students analyze their performance in different data structures.

Design and Analysis of Graph Algorithms. Students study graph algorithms for traversal and shortest path. Minimum spanning tree construction and cycle detection are also studied. Applications in networking and optimization are discussed.

Working on Graph-Related Problems. A Program for Constructing a Graph Using the Kruskal Method. This topic focuses on solving graph optimization problems by implementing Kruskal's algorithm. Students construct minimum spanning trees. Students also analyze algorithm efficiency.

Greedy Algorithms and Their Analysis. Students examine the greedy algorithm design strategy through practical examples such as shortest path and Huffman coding. Scheduling problems are also examined. The seminar evaluates their advantages and limitations.

An Algorithm Operating on the "Divide and Conquer" Principle. An Algorithm for Sorting a Set of Elements According to a Given Key. This

	seminar explores divide-and-conquer techniques through sorting algorithms such as Merge Sort and Quick Sort. Students analyze recursive decomposition. Computational efficiency is also analyzed. Programming Problems Based on Dynamic Programming. Students solve optimization problems using dynamic programming techniques. Practical examples include the knapsack problem and the longest common subsequence problem. The coin change problem is also included. Combinatorial Problems. This topic introduces algorithms for solving combinatorial problems involving permutations and combinations. Subsets and partitions are also addressed. Applications in optimization and scheduling are discussed. Elementary Problems in Computational Geometry. Students study geometric algorithms for convex hull construction and line intersection. Point location algorithms are also studied. Practical applications in computer graphics and geographic information systems are explored. Kruskal's Algorithm, Prim's Algorithm, and Huffman Trees. This seminar focuses on minimum spanning tree algorithms and Huffman coding. Students compare their implementation and efficiency. Their practical applications in network design and data compression are also compared. NP-Complete Problems. Methods of Reduction to NP-Complete Problems. Students examine the theory of NP-complete problems. Students also learn reduction techniques used to prove computational complexity. The seminar discusses the importance of these concepts in algorithm design. Graph Cut Problem. Travelling Salesman Problem. This topic explores graph cut optimization and the Traveling Salesman Problem (TSP). Students discuss exact and heuristic algorithms for solving
--	--

	computationally difficult problems. Approximation algorithms are also discussed. Self-study Designing Algorithms. Students learn practical techniques for designing efficient algorithms using systematic problem-solving approaches. Emphasis is placed on selecting appropriate algorithm design strategies. These strategies are applied to different computational problems. Evaluating the Correctness and Efficiency of Algorithms. Analysis of Algorithms. This topic focuses on verifying algorithm correctness. Time and space complexity are also analyzed. Students apply Big-O notation to compare algorithm performance. Recursive Algorithms: Factorial and Fibonacci Numbers. Students implement recursive solutions for computing factorials and Fibonacci numbers. The topic demonstrates recursion principles. Recursive and iterative approaches are also compared. Sorting Algorithms: Problems of Arranging Data in a List or Array. This practical introduces common sorting algorithms such as Bubble Sort, Selection Sort, and Merge Sort. Quick Sort and Heap Sort are also introduced. Students compare their efficiency through implementation and testing. Search Algorithms: Element Search, Trees, and Graphs. Students implement linear search and binary search. Depth-First Search (DFS), Breadth-First Search (BFS), and shortest path algorithms are also implemented. Applications in data retrieval and graph traversal are emphasized. Algorithms on Graphs. This topic covers graph algorithms including Dijkstra, Bellman-Ford, and Floyd-Warshall. Prim, Kruskal, Ford-Fulkerson, Edmonds-Karp, topological sorting, and cycle detection are also covered. Students solve practical graph optimization problems. Greedy Algorithms. Students implement greedy algorithms for shortest path finding and minimum
--	--

	spanning trees. Huffman coding, resource allocation, and coin change problems are also implemented. The effectiveness of greedy strategies is analyzed. The Divide and Conquer Method. This practical explores divide-and-conquer algorithms such as Merge Sort, Quick Sort, and Binary Search. Strassen's Matrix Multiplication is also explored. Students evaluate their recursive structure and efficiency. Design and Analysis of Dynamic Programming Algorithms. Students develop dynamic programming solutions for optimization problems. These solutions rely on storing intermediate results. Practical examples demonstrate improved computational efficiency. Dynamic Programming Problems. This topic includes practical implementations of the Longest Common Subsequence (LCS) and Edit Distance problems. Coin Change and Matrix Chain Multiplication problems are also implemented. Students analyze optimal substructure and overlapping subproblems. Combinatorial Problems. Students solve combinatorial optimization problems involving permutations and combinations. Partitions and arrangements are also addressed. Applications in scheduling and optimization are discussed. Computational Geometry Algorithms. This practical introduces algorithms for Convex Hull construction and polygon triangulation. Voronoi diagrams, Delaunay triangulation, and geometric intersection detection are also introduced. Applications in computer graphics and GIS are explored. Randomized Algorithms. Students implement randomized algorithms including Monte Carlo methods and Las Vegas algorithms. Random pivot selection, Miller-Rabin primality testing, and Karger's
--	--

	minimum cut algorithm are also implemented. Their probabilistic behavior is analyzed. NP – Complexity of Algorithms for Solving Complete Problems. This topic examines computational complexity, including the classes P and NP. NP-Complete and NP-Hard classes are also examined. Students analyze the computational difficulty of optimization and decision problems. NP-Complete Problems: 3-SAT, Hamiltonian Cycle, and Traveling Salesman Problem. Students study well-known NP-complete problems, including 3-SAT, Hamiltonian Cycle, and the Traveling Salesman Problem. Students explore approximation algorithms for obtaining practical solutions to computationally difficult problems. Heuristic methods for obtaining practical solutions are also explored.
Teaching Methods	The Module is delivered through lectures combined with seminar-based practical sessions in which students design, implement and analyse algorithms in a programming environment. Practical sessions employ problem-based learning, requiring students to document algorithms using pseudocode and flowcharts and to implement and test sorting, searching, recursive, greedy, divide-and-conquer and dynamic programming algorithms. Group work on graph-related problems, including the construction of minimum spanning trees with Kruskal's and Prim's algorithms, develops collaborative problem-solving skills. Independent study assignments require students to implement and analyse combinatorial, computational-geometry, randomized and NP-complete algorithms, reinforcing the link between theoretical complexity analysis and practical programming. Automated testing is used to verify the correctness of implemented algorithms and to identify performance issues. This combination of lecture-based instruction, guided practical work and independent implementation develops both the analytical and programming competencies required for algorithm design.

Prerequisites	Artificial Intelligence Operating Systems Algorithms and Data Structures Programming
Exams and Assessment	Type of examination (written, oral, project) and its weight Master's student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, programming exercises, seminar presentations, and the successful completion of independent study assignments, including implementing sorting and searching algorithms, designing recursive solutions, analyzing algorithm complexity, solving graph-related problems, and preparing reports on algorithm design techniques and their practical applications. The midterm assessment consists of written assignments, short in-class quizzes, practical programming tasks, and the evaluation of independent study projects, such as implementing graph algorithms including Dijkstra, Kruskal, and Prim, developing dynamic programming solutions, designing divide-and-conquer algorithms, comparing algorithm efficiency using complexity analysis, and preparing presentations on modern algorithmic approaches used in software engineering, artificial intelligence, and optimization. The final assessment is a written examination consisting of theoretical questions, algorithm design tasks, complexity analysis problems, and programming-oriented problem-solving exercises designed to assess students' understanding of algorithm design paradigms, computational complexity, graph algorithms, greedy methods, divide-and-conquer techniques, dynamic programming, randomized algorithms, and NP-complete problems. Successful completion of all laboratory work, programming assignments, and independent study projects is

	required as an essential component of the module assessment.
Recommended Literature	1. Introduction to Algorithms (3rd Edition), Cormen, Leiserson, Rivest, Stein. MIT Press, 2002. 2. Erickson J. Algorithms. Urbana-Champaign: University of Illinois, 2019. 472 p. https://jeffe.cs.illinois.edu/teaching/algorithms/book/Algorithms-JeffE.pdf 3. Massachusetts Institute of Technology. Introduction to Algorithms. MIT OpenCourseWare, 6.006, Spring 2020. https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/download/ 4. Stanford University. CS161: Design and Analysis of Algorithms. Stanford University, 2026. https://opendsa-server.cs.vt.edu/ODSA/Books/Everything/html/5. 5. CP-Algorithms. Algorithms for Competitive Programming. https://cp-algorithms.com/index.html. 6. V. D. Koldaev. Structures and Algorithms for Data Processing: A Textbook. Moscow: IORR Publishing House: NIIR INFRA-M, 2014. 296 pp. 7. A.N. Mirzaev, H.A. Mamadaliyev, Y.K. Alikulov. Design of Algorithms. Textbook. – Tashkent: Muhammad al-Khwarizmi IT University. 2022. -142 pp.
Responsible Lecturer	Zaripova Mukaddas Djumayozovna, Doctor of Philosophy (PhD) in Economics, Associate Professor ad interim.

1.06. FOUNDATIONS OF PARALLEL COMPUTING

Field	Description
Module Title	FOUNDATIONS OF PARALLEL COMPUTING
Module Code	PHA1206
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 90 hours (44 hours Lecture, 46 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	I semester
Module objective	The objective of the Module Foundations of Parallel Computing is to develop theoretical knowledge and practical skills in parallel computing architectures. Programming models for multicore processors, clusters, and GPU environments are studied. Algorithm design for these environments is developed. Students learn parallel algorithm design, analysis, optimization, and implementation. OpenMP, MPI, and GPU/CUDA are used for these purposes. These skills are applied to scientific and industrial applications.
Module Learning Outcomes	Knowledge: LO1. Explain fundamental concepts, theoretical foundations and development directions in parallel computing. LO2. Describe architecture of parallel systems, memory models and Flynn classification. LO3. Explain issues of designing, decomposing and synchronizing parallel algorithms. LO4. Differentiate parallel programming models: OpenMP, MPI, CUDA/OpenCL. LO5. Explain Amdahl's and Gustafson's laws and metrics of speedup, efficiency, scalability.

	LO6. Describe security, resource efficiency and academic integrity requirements in parallel computing. Skills: LO7. Create and test parallel programs using OpenMP for shared-memory systems. LO8. Develop message-passing parallel programs using MPI for distributed systems. LO9. Analyze parallel algorithm performance using profiling and benchmarking. LO10. Optimize parallel solutions based on speedup, efficiency and scalability analysis. LO11. Implement parallel algorithms for matrix operations and linear algebra problems. LO12. Develop parallel sorting algorithms and optimize their performance. Competencies: LO13. Design parallel solutions for scientific computing and data-intensive problems. LO14. Evaluate and select appropriate parallel programming models for given problems. LO15. Deploy parallel applications on clusters and GPU environments responsibly. LO16. Apply parallel computing technologies in research and professional practice with academic integrity.
Content	Lectures Introduction to Parallel Computing. Parallelism concepts and their importance in modern computing are introduced, including pipeline instruction processing. Vector and matrix processors are examined. The evolution from sequential to parallel computing and its applications in scientific computing, data analysis, and high-performance systems are discussed.

Parallel Algorithms and Computing Systems.

Shared-memory systems and distributed-memory systems are studied. Hybrid parallel architectures and communication schemes in multiprocessor systems are examined. Parallel versus sequential algorithm design and the trade-offs in parallel computing are analyzed.

Levels of Parallelism. Bit-level parallelism and instruction-level parallelism are studied. Data parallelism and task parallelism are examined. Methods for achieving parallelism at different levels, combining multiple parallelism types, and granularity considerations are discussed.

Fundamentals of Parallel Algorithm Construction.

Parallelization approaches, including coarse-grain, fine-grain, and algorithmic strategies, are studied. Global parallelization strategies and the identification of parallelizable components are examined. Problem transformation for parallelism and design patterns for parallel algorithms are discussed.

Stages of Parallel Algorithm Development.

Problem partitioning into independent tasks is studied. Data dependencies and communication patterns are identified. Scaling task sets across processors, load balancing, workload distribution, and performance prediction are examined.

Key Challenges in Parallel Computing.

Synchronization mechanisms and problems, including data race conditions and their prevention, are studied. Deadlock, livelock, and starvation are examined. Load imbalance issues and scalability limitations related to Amdahl's law are analyzed.

Data Dependency Analysis. A computational model using operation-operand graphs is studied. Dependency types and detection methods are examined, including loop-carried and inter-procedural dependencies. Parallelization feasibility analysis is discussed.

Algorithms for Array Reductions. Summation and reduction patterns are studied. Parallel reduction tree

	structures, scan operations, and prefix sums are examined. Logarithmic complexity reductions and work-efficient algorithms are discussed. Matrix-Vector Multiplication Parallelization. The algorithm structure and data dependencies of matrix-vector multiplication are studied. Matrix partitioning strategies, including row-wise, column-wise, and block-based decomposition, are examined. Communication patterns and load balancing approaches are discussed. Matrix Multiplication Algorithms. Sequential algorithm analysis and strip-wise matrix partitioning for distributed memory are studied. Block decomposition and data distribution methods are examined. The Fox algorithm for block-striped data and the Cannon algorithm for load balancing are discussed. Gaussian Elimination for Linear Systems. The sequential Gaussian algorithm structure is studied. Forward elimination and backward substitution parallelization are examined. Data distribution schemes and pivot selection in parallel are discussed. Gradient Algorithms for Linear Systems. The conjugate gradient method and the sequential gradient algorithm are studied. Parallel vector operations, reduction, and broadcast operations are examined. Convergence monitoring in parallel is discussed. Parallel Sorting Fundamentals. Comparison-based sorting analysis and parallelization strategies are studied. Bubble sort, even-odd sort, sorting networks, and bitonic sort are examined. Performance analysis and complexity of these methods are discussed. Advanced Sorting Methods. Shellsort parallelization approaches are studied. Quicksort variants, partition strategies, radix sort, and counting sort are examined. The selection of algorithms based on problem size and hardware-software co-design are discussed.
--	---

	Parallel Systems and Performance Analysis. The speedup definition and the implications of Amdahl's law are studied. Gustafson's law and scaled speedup are examined. Efficiency metrics, scalability, isoefficiency analysis, and profiling for bottleneck identification are discussed. Practical lessons Array Element Summation Algorithms. Sequential sum calculation and parallel reduction tree implementation are practiced. Performance measurement techniques are applied. Sequential versus parallel execution times are compared. Exact Integral Calculation in Parallel. Numerical integration methods, including the rectangle rule and trapezoid rule, are practiced. Parallel domain decomposition is implemented. Result aggregation and accuracy verification are performed. Parallelizing Iteration Processes. Sum and series calculations are parallelized. Trigonometric function computation is implemented. Convergence monitoring and iteration count optimization are practiced. Matrix-Vector Multiplication with Strip Decomposition. Matrix row-wise partitioning is implemented. Distributed vector distribution is practiced. Local computation, result collection, and communication pattern analysis are performed. Matrix-Vector Multiplication with Block Decomposition. Block-based data distribution is implemented. A 2D processor grid organization is practiced. Local operations, allgather communication, and efficiency calculation are performed. Parallel Matrix Multiplication Fundamentals. Algorithm steps and dependencies are studied. Test cases are created, and correctness is verified. Performance profiling is practiced. Matrix Multiplication with Strip Partitioning (Algorithm 1). The data distribution scheme and communication patterns are implemented.
--	---

	Implementation in OpenMP or MPI is practiced. Load balancing analysis is performed. Matrix Multiplication with Strip Partitioning (Algorithm 2). An alternative communication strategy is implemented to reduce communication overhead. Pipeline-based communication is practiced. Scalability is compared with the previous algorithm. Fox Algorithm for Block-Striped Multiplication. The algorithm steps and data movement of the Fox algorithm are implemented. A distributed implementation is practiced. Synchronization patterns and performance characteristics are analyzed. Cannon Algorithm for Block Multiplication. The block rotation scheme of the Cannon algorithm is implemented. Its systolic array analogy and optimized communication are practiced. The Cannon algorithm is compared with the Fox algorithm. Gaussian Elimination Implementation. Forward elimination is implemented in parallel. Pivot row broadcasting and backward substitution parallelization are practiced. Testing with known solutions is performed. Gradient Algorithm for Linear Systems. The conjugate gradient method is implemented. Parallel vector operations and reduction operations are practiced. Convergence testing is performed. Bubble Sort and Even-Odd Sort Implementation. The algorithm structure and dependencies of bubble sort and even-odd sort are studied. Comparison-exchange operations are implemented. Parallel implementation steps and performance measurement are practiced. Odd-Even Sort Performance Analysis. Sorting network organization is analyzed. Processor assignment strategies and communication patterns are examined. Scalability limits are evaluated. Bitonic Sort and XOR Sort Methods. Bitonic network construction is implemented. Sorting with regular patterns and logarithmic depth networks is
--	---

	practiced. Hardware implementation considerations are examined. Self-study Essence and Development History of Parallel Computing. The evolution from vector to parallel computing is studied. Technological drivers and market trends are examined. Current research directions and future prospects are analyzed. Parallel System Architectures and Flynn Classification. SISD, SIMD, MISD, and MIMD architectures are studied. Shared versus distributed memory and hybrid systems are examined. Architecture selection criteria and interconnection networks are analyzed. Decomposition, Granularity and Dependency Analysis. Problem decomposition strategies are studied. Fine-grain versus coarse-grain parallelism is examined. Dependency detection and analysis, data flow graphs, and algorithms for dependency identification are analyzed. Synchronization Problems in Parallel Programs. Data race condition mechanisms are studied. Deadlock and resource allocation, starvation, and fairness are examined. Synchronization primitives and verification techniques are analyzed. OpenMP Technology and Directives. The shared-memory parallel programming model is studied. Compiler directives and clauses are examined. Thread creation and management, variable scoping rules, and work distribution are analyzed. OpenMP Parallelization Techniques. Parallel for loops and reductions are studied. Static, dynamic, and guided scheduling, and nested parallelism are examined. Synchronization constructs, critical sections, and locks are analyzed. MPI Model and Point-to-Point Communication. The message-passing model and processes are studied. MPI communicators and ranks are examined. Send and receive operations, blocking and non-
--	--

	blocking communication, and deadlock avoidance are analyzed. MPI Collective Operations. Broadcast and scatter operations are studied. Gather and all-gather operations, and reduce and all-reduce patterns are examined. Barrier synchronization and performance optimization are analyzed. Parallel Algorithm Efficiency Evaluation. Amdahl's law and speedup limits are studied. Gustafson's law and scaled speedup are examined. Parallel overhead analysis, the communication-to-computation ratio, and efficiency metrics are analyzed. Parallel Matrix Operations Analysis. Complexity analysis of parallel algorithms is studied. Communication cost analysis and asymptotic efficiency are examined. Best-case and worst-case scenarios are compared through algorithm analysis. Load Balancing and Scheduling. Static and dynamic load balancing are studied. Task scheduling strategies and data distribution for load balance are examined. Heterogeneous system considerations are analyzed. Profiling and Performance Bottlenecks. Profiling tools and methodologies are studied. Communication bottlenecks and memory bandwidth limitations are identified. Compute-to-memory access ratios and optimization strategies are analyzed. GPU Computing and CUDA/OpenCL. Graphics processing unit architecture is studied. The CUDA programming model, kernel functions, and blocks are examined. Memory hierarchy optimization and massive parallelism exploitation are analyzed. Cluster and Cloud Parallel Computing. Cluster architecture and programming are studied. Cloud computing platforms and grid computing concepts are examined. Distributed file systems and resource management are analyzed. Mini-Project: Parallel Algorithm Development. A problem is selected and analysed for the mini-project.
--	---

	Algorithm design, implementation, performance evaluation, and optimization are practiced. Documentation, presentation, code review, and demonstration are completed.
Teaching Methods	The module is delivered through interactive lectures with theoretical foundations and hands-on practical sessions with algorithm implementation. Computational experiments and benchmarking are conducted. Code review and optimization practice are included. Research-based learning with HPC systems and independent mini-projects with final presentations are incorporated. Profiling and analysis tools are used, along with peer discussion and collaborative problem-solving.
Prerequisites	Programming Algorithms and Data Structures Computer Architecture
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation in practical sessions, algorithm implementation ability, code quality, and completion of teacher-guided independent work assignments, including OpenMP/MPI implementations, performance analysis reports, optimization studies, and mini-projects on parallel algorithms. The midterm assessment consists of a written test covering theoretical knowledge of parallel computing fundamentals, architecture, algorithm design, and Amdahl's and Gustafson's laws, together with independent work projects evaluating practical skills in algorithm parallelization and performance analysis using profiling tools. The final examination is a written examination with problems requiring students to design and implement parallel algorithms, analyze performance, propose optimizations, and explain architectural decisions.

	The assessment measures theoretical understanding, algorithmic thinking, and the ability to apply parallel computing to complex problems.
Recommended Literature	1. Pacheco P. An Introduction to Parallel Programming. Morgan Kaufmann, 2011. 2. Grama A., Gupta A., Karypis G., Kumar V. Introduction to Parallel Computing. 2nd Edition. Addison-Wesley, 2003. 3. Quinn M.J. Parallel Programming in C with MPI and OpenMP. McGraw-Hill, 2004. 4. Kirk D.B., Hwu W.W. Programming Massively Parallel Processors. 3rd Edition. Morgan Kaufmann, 2016.
Responsible Lecturer	Zaripova Mukaddas Djumayozovna, PhD, Associate Professor, Department of Computer and Software Engineering, Termez State University

2.01. JAVA TECHNOLOGIES

Field	Description
Module Title	JAVA TECHNOLOGIES
Module Code	JVT104
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	II semester
Module objective	The objective of the Module Java Technologies is to develop in postgraduate students the practical skills to create software solutions using the Java programming language and Java technologies. The principles of object-oriented programming, classes, and objects are applied. Theoretical knowledge in creating applications is developed. Practical skills based on inheritance, polymorphism, and interfaces are developed. Exception handling and graphical interfaces are studied. Design patterns are also applied in software development.
Module Learning Outcomes	Knowledge: LO1. Can explain the structure, syntax, data types, operators and control structures of the Java programming language. LO2. Can explain the principles of object-oriented programming: class, object, encapsulation, inheritance and polymorphism. LO3. Can describe the functions of packages, exceptions, interfaces, graphics programming and Swing components in Java. LO4. Can explain the essence of design patterns, their place in software architecture and the situations in which they are applied. Skills:

	LO5. Creates console applications in Java, performing data input/output and type conversion operations. LO6. Develops algorithmic solutions for practical problems using conditional, selection and iteration operators. LO7. Build programs using classes, objects, methods, constructors, recursion, overloading and nested classes. LO8. Has practical skills in exception handling, creating packages, and using interfaces and abstract classes. LO9. Develops simple graphical user interface applications using Swing components. LO10. Designs software solutions using design patterns such as Adapter, Facade, Composite, Bridge, Singleton, Observer, Mediator, Proxy and Builder. Competencies: LO11. Uses Java technologies to solve professional and scientific-practical problems. LO12. Analyses program code, identifies errors, tests and refines it. LO13. Plans the software project structure, separates components and organises work on a modular basis. LO14. Demonstrates teamwork, project presentation and technical documentation skills. LO15. Will be competent in creating an independent software project using Java technologies, analysing program code, identifying errors, testing, and developing a software solution appropriate to practical problems.
Content	Lectures Structure of a Java Program. Input and Output. Type Conversion (Upcasting, Downcasting – Cast). The basic structure of Java programs and program execution is studied. Input/output operations,

variables, and data types are analyzed. Implicit and explicit type conversion techniques in Java are investigated.

Operators: Logical, Arithmetic, Character, Relational and Bitwise Operators. The String Type. Working with the Math Class and Its Fields and Methods. Java operators and their applications in programming are studied. String manipulation methods and mathematical computations using the Math class are analyzed. Operator precedence and expression processing are evaluated.

Control Statements: Branching, Selection, Loops, Break and Continue Statements. Conditional statements and iterative programming constructs are studied. Program flow control using if, switch, for, while, and do-while statements is analyzed. Break and continue statements for loop control are evaluated.

Complex Types: Arrays and Their Use in the Java Programming Language. One-dimensional and multi-dimensional arrays in Java are studied. Array declaration, initialization, and manipulation techniques are analyzed. Arrays are evaluated for efficient data storage and processing.

Creating a Class and an Object. Creating and Using an Existing Class Object. Object-oriented programming concepts in Java are introduced. Classes, objects, and object instantiation are studied. Object interaction and class-based program design are analyzed.

Creating a New Class. Methods. Constructors. Accessing Class Members and Encapsulation. Methods, constructors, and encapsulation principles are studied. Access modifiers and data hiding techniques are analyzed. Class design for secure and reusable software development is evaluated.

Static and Constructor Methods. The Main Method. Recursion. Parameter Passing Mechanisms. Static members, constructors, recursion, and parameter passing methods are

studied. Recursive algorithms and object initialization are analyzed. Program design using static and instance methods is evaluated.

Overloading. Default and Explicit Field Initialisation. Default Constructor and Hidden Parameter. Method and constructor overloading in Java are studied. Field initialization mechanisms and object construction are analyzed. Constructor behavior and object lifecycle are evaluated.

Nested Classes. The Concept of a Subclass. Inheritance. Method Overriding. Polymorphism. Abstract Methods and Classes. The Final Specifier. Advanced object-oriented programming concepts are studied. Inheritance, polymorphism, abstract classes, and nested classes are analyzed. Code reusability and extensibility using inheritance mechanisms are evaluated.

Packages. Using Packages. Importing a Class into a Package and Working with Packages. Package organization and modular programming in Java are studied. Package creation, class importing, and namespace management are analyzed. Modular software development techniques are evaluated.

Exceptions: Try-Catch, Throw, Finally. Creating Custom Exceptions by the Developer. Exception handling mechanisms in Java are studied. Error detection, exception propagation, and custom exception classes are analyzed. Robust program development through exception management is evaluated.

Interfaces. Fundamentals of Graphics Programming. Handling Events. Interfaces and event-driven programming principles are studied. Graphical programming concepts and event handling mechanisms are analyzed. Interface-based software design is evaluated.

Components in the Swing Package for the User Interface and Their Application. Java Swing components for graphical user interface development are studied. Windows, panels, buttons, text fields, and

	event-driven applications are analyzed. User interface design principles are evaluated. Introduction to Java Design Patterns. Adapter and Facade Design Patterns. Software design patterns in Java are introduced. The Adapter and Facade patterns for improving software flexibility and maintainability are studied. Practical implementation scenarios are analyzed. Aggregated Design Patterns: Composite, Bridge, Singleton, Observer, Mediator, Proxy and Builder Patterns. Advanced software design patterns used in object-oriented systems are studied. Structural, behavioral, and creational design patterns are analyzed. Their application in scalable and maintainable Java applications is evaluated. Practical lessons Structure of a Java Program, Creating Practical Applications for Data Input-Output and Type Conversion. Java programs are developed using standard input/output operations. Data type conversion techniques are applied. Program execution and variable management are analyzed. Programming Practical Problems Using Operators, String Types and Methods of the Math Class. Arithmetic, logical, relational, and bitwise operations are implemented. String manipulation methods are applied practically. Mathematical functions from the Math class are used in Java applications. Creating Programs Based on Branching, Selection and Loop Operators. Java programs are developed using conditional statements and loops. Program control flow is analyzed. Iterative algorithms are implemented and evaluated. Working with Arrays: Writing Programs for One-Dimensional and Multi-Dimensional Arrays. Array processing algorithms are implemented. Searching and sorting techniques using arrays are analyzed.
--	---

	Data manipulation techniques using arrays are practiced. Practical Exercises on Creating Classes and Objects and Using Existing Class Objects. Object-oriented Java applications are developed using classes and objects. Object creation mechanisms are practiced. Object interaction mechanisms are analyzed. Programming Based on Methods, Constructors and the Principles of Encapsulation. Classes with methods, constructors, and encapsulated data are implemented. Object-oriented programming principles are evaluated. Encapsulation techniques are applied in practice. Practical Programs on Static Methods, the Main Method, Recursion and Parameter Passing. Java applications are developed using recursion, static members, and parameter passing techniques. Recursive algorithm performance is analyzed. Parameter passing mechanisms are practiced. Overloading, Field Initialisation and Working with Constructors. Overloaded methods and constructors are implemented. Object initialization is analyzed. Constructor execution is examined in practice. Practical Work on Inheritance, Method Overriding, Polymorphism, Abstract Classes and the Final Specifier. Object-oriented applications are developed using inheritance and polymorphism. Code reuse mechanisms are analyzed. Abstraction mechanisms are practiced through abstract classes and the final specifier. Creating Packages, Placing Classes into Packages and Using Them. Modular Java applications are developed using packages. Package organization techniques are analyzed. Code management techniques are practiced. Handling Exceptions: Try-Catch, Throw, Finally and Creating Custom Exception Classes. Exception handling techniques in Java are
--	--

	implemented. Custom exception classes are developed. Reliable applications are built using these exception handling mechanisms. Developing Practical Applications Using Interfaces and the Fundamentals of Graphical Programming. Interface-based applications are implemented. Event-driven programming techniques are applied. Graphical application development is analyzed. Creating User Interface Applications Using Swing Components. Desktop applications are developed using Java Swing. Graphical interface components are analyzed. Event handling mechanisms are practiced. Developing Software Solutions Based on the Adapter and Facade Design Patterns. The Adapter and Facade design patterns are implemented in Java projects. Software modularity is analyzed. Software flexibility is evaluated through these design patterns. Practical Application of the Composite, Bridge, Singleton, Observer, Mediator, Proxy and Builder Design Patterns. Java applications are developed using advanced design patterns, including Composite, Bridge, Singleton, Observer, Mediator, Proxy, and Builder. Software architecture is analyzed. Maintainability and scalability of these applications are evaluated. Self-study The Java Programming Language and Program Structure. Java language fundamentals, program structure, and the execution model are studied. Java platform architecture is analyzed. Application development principles are examined. Operators, Data Types and the Math Class. Java data types, operators, and mathematical functions are investigated. Expression evaluation is analyzed. Numerical computations are examined. Control Operators and Loops. Conditional statements and iterative programming constructs are
--	---

	studied. Decision-making mechanisms are analyzed. Repetitive algorithm implementation is examined. Working with Arrays and Strings. Array structures and String processing methods are investigated. Data manipulation techniques are analyzed. Text processing techniques are examined. Object-Oriented Programming. Object-oriented programming concepts, including classes, objects, inheritance, polymorphism, and encapsulation, are studied. Object-oriented software design principles are analyzed. Practical applications of these concepts are examined. Methods, Constructors and Encapsulation. Methods, constructors, access modifiers, and encapsulation mechanisms are investigated. Reusable class design is analyzed. Secure class design principles are examined. Recursion, Parameter Passing and Overloading. Recursive programming techniques, method overloading, and parameter passing mechanisms are studied. Recursive algorithm implementation is analyzed. Practical examples of these techniques are examined. Inheritance and Polymorphism. Inheritance hierarchies and polymorphic behavior in Java are investigated. Abstraction techniques are analyzed. Software extensibility through inheritance is examined. Modularising Packages and Classes. Java package management and modular application development are studied. Namespace organization is analyzed. Code reuse techniques are examined. Handling Exceptions. Java exception handling mechanisms and custom exceptions are investigated. Reliable application development is analyzed. Fault-tolerant application development techniques are examined. Interfaces and Abstract Classes. Interfaces and abstract classes in object-oriented programming are
--	---

	studied. Interface-based software architecture is analyzed. Abstraction techniques are examined. Working with Swing Components. Graphical user interface development using Java Swing is investigated. Component interaction is analyzed. Event-driven programming techniques are examined. Text Editor or a Small User Application. A small desktop application is developed using Java technologies. Object-oriented programming principles are applied. Graphical interface design and event handling techniques are used in the application. Adapter, Facade and Other Design Patterns. Java design patterns, including Adapter, Facade, Singleton, Observer, and Builder, are studied. Their practical application in software engineering is analyzed. Additional related patterns are examined. Final Mini-Project on Java Technologies. A complete Java application is designed and implemented, integrating object-oriented programming, exception handling, and graphical user interface components. Packages and software design patterns are incorporated into the project. Software quality, maintainability, and functionality are evaluated.
Teaching Methods	To teach master's students the syntax of the Java programming language, data types, operators, control structures, and program structure. Explain the fundamental principles of object-oriented programming, such as class, object, method, constructor, encapsulation, inheritance, and polymorphism. Develop skills in using arrays, strings, recursion, parameter passing, packages, and exception and error handling mechanisms in Java. Develop practical competencies in interfaces, graphical programming, Swing components and creating user interfaces. Teach methods for designing software solutions based on Java design patterns, reusing code and improving application architecture. To develop in postgraduate students the competence

	to create independent software, analyse code, test, optimise, and present the results of a software project.
Prerequisites	Algorithms and Data Structures Object-oriented Programming Fundamentals Operating Systems Database Fundamentals Computer Networks Fundamentals
Exams and Assessment	Type of examination (written, oral, project) and its weight Master's degree candidates are assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates candidates' active participation in lectures and practical sessions, analytical discussions, research presentations, programming and software development activities, and the successful completion of independent study assignments. Assessment focuses on the design and implementation of object-oriented Java applications, analysis of Java program structures, advanced use of classes, inheritance, polymorphism, exception handling, graphical user interface development using Swing, software modularization with packages, and the application of object-oriented design patterns in software engineering. The midterm assessment consists of written assignments, analytical reports, programming projects, presentations, practical software development tasks, and the evaluation of independent research projects. Candidates are assessed on their ability to design object-oriented software solutions, implement advanced Java programming concepts, develop graphical desktop applications, apply exception handling and interface-based programming, evaluate object-oriented software architectures, compare software design patterns, and justify software engineering decisions using modern Java technologies and programming methodologies. The final assessment is conducted as a written examination consisting of theoretical

	questions, analytical problems, software design scenarios, and programming-oriented case studies covering all topics of the module. The examination evaluates candidates' advanced understanding of Java programming fundamentals, object-oriented programming principles, classes and objects, methods and constructors, inheritance, polymorphism, abstract classes, interfaces, exception handling, packages, Swing-based graphical user interface development, event-driven programming, software design patterns, and modern Java software development practices. Candidates are expected to demonstrate critical thinking, software architecture design skills, and the ability to apply advanced Java programming techniques to the development of scalable, maintainable, and high-quality software systems. Successful completion of all practical programming assignments, software development projects, and independent research tasks is required as an essential component of the overall module assessment.
Recommended Literature	1. Baqoyev O., Boltayev T.B. Programming in Java. Bukhara, 2009. 64 pp. 2. Vyazovik N.A. Programming in Java. 2003. 592 pp. 3. Downey A.B., Mayfield C. Think Java: How to Think Like a Computer Scientist. 2nd Edition, Version 7.1.0. Needham: Green Tea Press, 2020. 4. Eck D.J. Introduction to Programming Using Java. Version 9.0, JavaFX Edition. Hobart and William Smith Colleges, 2022. 5. Downey A.B. Think Data Structures: Algorithms and Information Retrieval in Java. Version 1.0.0. Needham: Green Tea Press, 2016. 6. Oracle. The Java Tutorials Bundle. Java SE Tutorial, 2022-03-04. Oracle, 2022. 7. Lindholm T., Yellin F., Bracha G., Buckley A., Smith D. The Java Virtual Machine Specification. Java SE 21 Edition. Oracle, 2023.

	8. GoalKicker.com. Java Notes for Professionals. GoalKicker.com, 2018. 982 pp. 9. Wikibooks contributors. Java Programming. Wikibooks, open books for an open world.
Responsible Lecturer	Norqobilov Sobir Hamza o'g'li

2.01. DISTRIBUTED SYSTEMS

Field	Description
Module Title	DISTRIBUTED SYSTEMS
Module Code	TQT104
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	II semester
Module objective	The objective of the Module Distributed Systems is to provide postgraduate students with a deep understanding of the theoretical foundations, architectural models, and operating principles of distributed systems. Client-server, peer-to-peer, and service-oriented architectures are studied. Professional competencies in the design of modern distributed systems are developed. Management and analysis skills for distributed systems are also developed. Students learn the implementation of modern distributed systems based on distributed file systems, Grid, and cloud computing technologies.
Module Learning Outcomes	Knowledge: LO1. Explain the fundamental concepts, principles, characteristics, and contemporary development trends of distributed systems. LO2. Describe the operating principles and application domains of distributed system architecture models, including client-server, peer-to-peer (P2P), service-oriented architecture (SOA), and microservices architecture. LO3. Explain the functions and components of middleware technologies, distributed file systems, and data transmission mechanisms.

	LO4. Explain the theoretical foundations of synchronization, coordination, replication, and consensus algorithms in distributed systems. LO5. Describe the operating principles of grid computing, cloud computing, virtualization, and containerization technologies. LO6. Explain the principles of security, authentication, authorization, and data protection methods in distributed systems. Skills: LO7. Possesses practical skills in designing distributed systems architectures and integrating their components. LO8. Utilises mechanisms for data exchange, resource management and process coordination in distributed systems. LO9. Performs tasks involving the operation, configuration and management of distributed file systems and databases. LO10. Creates and manages distributed services on cloud platforms and in virtual environments. LO11. Applies methods for system performance evaluation, load balancing and fault detection and remediation. LO12. They will have the skills to develop, test and implement software solutions for distributed systems. Competencies: LO13. Effectively utilises distributed systems technologies to create corporate information systems and digital platforms. LO14. Analyses the performance of distributed systems, monitors them, and makes decisions to optimise them. LO15. Has the competence to design and manage modern cloud, Grid and service-oriented infrastructures.
--	--

	LO16. Applies modern technologies in research and innovation projects in the field of distributed systems. LO17. They will independently study new distributed computing platforms, cloud services and software tools, and will be qualified to implement them in practice.
Content	Lectures Introduction to the Field of Distributed Systems. Distributed Systems and Networks: Definition and Composition. Architectural Characteristics of Distributed Systems. The concepts, objectives, and characteristics of distributed systems are introduced. The components and architecture of distributed computing environments are studied. The role of distributed systems in modern information technologies is analyzed. The Role and Importance of Middleware in Distributed Systems. Middleware technologies and their functions in distributed environments are studied. Communication, interoperability, and resource-sharing mechanisms are analyzed. Middleware platforms used in modern distributed applications are evaluated. Interconnection Model for Distributed Systems Components Based on an Ethernet Network. Interconnection Model for Distributed Systems Components Based on a Software-Configurable Network. Ethernet-based and software-defined networking models for distributed systems are studied. Communication protocols and network architectures are analyzed. Efficient methods for connecting distributed components are evaluated. Indicators Determining the Effective Operation of Distributed Systems. Performance metrics used to evaluate distributed systems are studied. Scalability, reliability, availability, throughput, and latency are analyzed. System efficiency under different workloads is evaluated.

	Types and Capabilities of Distributed Systems. Client-Server Technology. R2R Technology. The main types of distributed systems and their applications are studied. Client-server and peer-to-peer architectures are analyzed. Their advantages, limitations, and practical use cases are compared. Object-Oriented Distributed Systems. Object-oriented approaches in distributed computing are introduced. Distributed objects, remote method invocation, and component interaction are studied. Object-oriented middleware technologies are analyzed. Distributed Systems Based on Web Technologies. Web-based distributed systems and Internet technologies are studied. Web services, RESTful APIs, and distributed web applications are analyzed. Modern web technologies for distributed computing are evaluated. Principles of Organising Distributed File Systems. The Logical Structure of File Systems. Distributed file system architecture and logical organization are studied. File storage, replication, consistency, and access mechanisms are analyzed. Distributed storage technologies are evaluated. Distributed Systems Based on the Service-Oriented Architecture Concept. The principles and architecture of Service-Oriented Architecture are introduced. Services, interoperability, and distributed service integration are analyzed. SOA applications in enterprise systems are evaluated. Fundamentals of Shaping Distributed Systems Based on the Service-Oriented Architecture Concept. Methods for designing and implementing distributed systems based on SOA are studied. Service composition, orchestration, and communication are analyzed. Scalable service-oriented solutions are evaluated. Model for Delivering Information and Communication Network Resources and Services on the Basis of an E-, Service-Oriented Distributed
--	---

	System. Service delivery models in distributed information and communication networks are studied. Service management, resource sharing, and quality of service are analyzed. Enterprise service infrastructures are evaluated. Distributed Systems Based on Grid and Cloud Technologies. Principles of Applying Virtualisation Technology in TTs. Foundations for Building a Distributed System Infrastructure Based on the Capabilities of Grid Technologies. Grid computing and Cloud computing architectures are studied. Virtualization technologies and distributed infrastructure are analyzed. Grid technologies for high-performance computing are evaluated. Principles of Providing TT Resources Based on Grid Technologies on a Service-Oriented Architecture (SOA) Basis. Resource allocation and service delivery using Grid technologies are studied. Integration between Grid computing and Service-Oriented Architecture is analyzed. Distributed resource management techniques are evaluated. The Content and Essence of Cloud Technology. A Model for Creating Distributed Systems Based on the Advantages of Cloud Technology. A Comparative Analysis of Distributed Systems Based on Grid and Cloud Technologies. Cloud computing concepts and service models are introduced. Distributed systems built on cloud technologies are analyzed and compared with Grid computing. Cloud-based infrastructure for modern applications is evaluated. Development Prospects for Distributed Systems and Networks. Recommendations for Providing Information and Communication Network Resources and Services Based on Distributed Systems. Emerging trends in distributed computing and communication networks are studied. Future technologies, intelligent distributed systems, and large-scale infrastructures are analyzed.
--	---

	Recommendations for improving distributed services are evaluated. Practical lessons Design Models for Distributed Systems. Architectural models for distributed systems are developed and analyzed. System components are evaluated. Communication methods and scalability are assessed. Design of Distributed Systems with Shared Memory. Distributed systems are implemented using shared memory techniques. Synchronization mechanisms are analyzed. Communication and resource management mechanisms are examined. Security of Distributed Systems. Security threats in distributed systems are investigated. Authentication methods, access control, and encryption techniques are examined. Mechanisms for protecting distributed environments are evaluated. Designing a Workflow System. Workflow models for distributed applications are developed. Task coordination mechanisms are analyzed. Process automation and service integration are examined. Stability of Self-Stabilising Distributed Systems. Fault tolerance and self-stabilization mechanisms are investigated. System recovery techniques are analyzed. Reliability under failure conditions is examined. Operating Principles of Distributed File Systems (NFS, AFS and GFS). Distributed file systems, including NFS, AFS, and GFS, and their storage architectures are practically studied. File sharing and replication mechanisms are analyzed. Consistency and access performance are examined. Peer-to-Peer (P2P) System. Peer-to-peer network architectures are designed and analyzed. Decentralized communication mechanisms are investigated. Resource sharing and distributed applications are examined.
--	---

	Study of Data Processing Centres Based on Distributed Systems. Distributed data centers and cloud infrastructures are investigated. Resource allocation and virtualization techniques are analyzed. Large-scale data processing technologies are examined. Self-study Architectures and Models of Distributed Systems. Distributed system architectures, design models, and communication structures are studied. Centralized architectures are analyzed. Decentralized and hybrid architectures are examined. Network and Communication Protocols in Distributed Systems. Communication protocols used in distributed environments are investigated. TCP/IP and RPC protocols are analyzed. HTTP, gRPC, and other distributed communication mechanisms are examined. Synchronous and Asynchronous Systems: Differences and Applications. Synchronous and asynchronous distributed systems are studied. Their characteristics are analyzed. Their advantages, limitations, and practical applications are examined. Data Exchange and Memory Management in Distributed Systems. Distributed memory management and data communication techniques are studied. Message passing mechanisms are analyzed. Shared memory and synchronization mechanisms are examined. Fault-Tolerant Systems and Their Design. Fault tolerance techniques, redundancy, and recovery mechanisms are investigated. Highly available distributed system architectures are analyzed. Practical fault tolerance strategies are examined. Security and Cryptography in Distributed Systems. Authentication and encryption methods used in distributed environments are studied. Digital certificates and secure communication protocols are
--	--

	examined. Cryptographic techniques for distributed systems are analyzed. Cloud Computing and Service Models (IaaS, PaaS, SaaS). Cloud computing concepts and cloud service models are studied. Infrastructure-as-a-Service and Platform-as-a-Service are analyzed. Software-as-a-Service in cloud environments is examined. Microservices Architecture and Container Technologies (Docker, Kubernetes). Microservices architecture and container-based deployment are investigated. Docker and Kubernetes are analyzed. Modern orchestration technologies are examined. Resource Management and Load Balancing in Distributed Systems. Resource allocation and scheduling techniques are studied. Load balancing techniques are analyzed. Methods for improving system scalability and performance are examined. Distributed Databases and Replication Technologies. Distributed database architectures and replication strategies are investigated. Consistency and availability in distributed data management are analyzed. Fault tolerance mechanisms are examined. Coordination of Time and Events in Distributed Systems. Clock synchronization and logical clocks are studied. Event ordering and distributed coordination techniques are analyzed. Time consistency in distributed applications is examined. Consensus Algorithms in Distributed Systems (Paxos, Raft). Consensus mechanisms used in distributed computing are investigated. The Paxos and Raft protocols are analyzed. Distributed agreement protocols for fault-tolerant systems are examined. Peer-to-Peer (P2P) Systems and Their Operating Mechanism. Decentralized peer-to-peer architectures and communication models are studied. Resource sharing mechanisms are analyzed.
--	--

	Distributed storage and P2P applications are examined. Modern Trends and Future Development Directions in Distributed Systems. Emerging technologies in distributed computing, including edge computing and fog computing, are investigated. Serverless computing is analyzed. AI-enabled distributed systems are examined. Monitoring and Performance Analysis in Distributed Systems. Monitoring tools, performance metrics, logging, and system diagnostics are studied. Techniques for evaluating distributed system performance are analyzed. Methods for optimizing distributed system performance are examined.
Teaching Methods	To provide postgraduate students with theoretical knowledge of the fundamental concepts, principles, characteristics, and architectural models of distributed systems. Teach the structure and operating mechanisms of client-server, peer-to-peer, object-oriented, web-based and service-oriented architectures. To develop practical skills in data exchange, process coordination, resource management, and establishing inter-component interactions in distributed systems. Developing competencies in creating and managing distributed file systems, middleware technologies, virtualisation tools, and network services. Designing distributed infrastructures based on Grid and cloud computing technologies, evaluating their performance, and teaching methods for applying them in modern information and communication networks. To develop skills in applying the principles of security, reliability, load balancing and service management in distributed systems, and in utilising modern distributed platforms for scientific research and practical applications.
Prerequisites	Programming 1,2 Data Structures and Algorithms Computer Networks Computer Architecture

	Operating Systems Cloud Technologies Cryptography and Network Security
Exams and Assessment	Type of examination (written, oral, project) and its weight Master's degree candidates are assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates candidates' active participation in lectures and practical sessions, analytical discussions, research-oriented presentations, case study analyses, and the successful completion of independent study assignments. Assessment focuses on the ability to analyze distributed system architectures, evaluate middleware technologies, compare client-server, peer-to-peer (P2P), service-oriented architecture (SOA), Grid and Cloud computing models, design distributed system solutions, and critically assess current trends and challenges in distributed computing. The midterm assessment consists of written assignments, analytical reports, research presentations, case analyses, practical design tasks, and the evaluation of independent research projects. Candidates are assessed on their ability to design distributed system architectures, evaluate communication protocols, analyze distributed file systems, investigate virtualization technologies, apply cloud and Grid computing concepts, compare microservices and container technologies, assess security and fault-tolerance mechanisms, and justify architectural decisions using modern distributed computing principles. The final assessment is conducted as a written examination consisting of theoretical, analytical, and problem-solving questions covering all topics of the module. The examination evaluates candidates' advanced understanding of distributed system architectures, middleware, communication models, distributed file systems, service-oriented architecture, Grid and Cloud

	computing, virtualization, distributed databases, consensus algorithms, fault tolerance, distributed security, microservices, container technologies, resource management, performance optimization, and emerging trends in distributed computing. Candidates are expected to demonstrate critical analysis, system design skills, and the ability to apply advanced distributed computing concepts to research and real-world enterprise applications. Successful completion of all practical assignments, analytical case studies, and independent research tasks is required as an essential component of the overall module assessment.
Recommended Literature	1. Tanenbaum, Andrew, and Maarten van Steen. Distributed Systems: Principles and Paradigms. Upper Saddle River, NJ: Prentice Hall, 2002. ISBN: 9780130888938. 2. T.N. Nishonboev, N.B. Usmanova. Distributed Systems. Textbook. Tashkent: TATU, 2018. 276 pp. 3. E. Tanenbaum, M. van Steen. Distributed Systems: Principles and Paradigms. St. Petersburg: Piter, 2003. 877 pp. 4. M.S. Kosyakov. Introduction to Distributed Computing. St. Petersburg: ITMO University, 2014. 155 pp. 5. Stein van M., Tanenbaum A.S. Distributed Systems. Trans. from English by V.A. Yarotsky. Moscow: DMK Press, 2021. 584 pp. 6. D.A. Gradusov, A.V. Shutov. Distributed Information Systems: A Practical Textbook. Vladimir State University named after A.G. and N.G. Stolyetov. Vladimir: VSU Publishing House, 2021. 386 pp. 7. S.A. Shelokov, E.N. Chernoprudova. Design of Distributed Information Systems: Lecture Course. Orenburg State University. Orenburg: OGU, 2012. 195 pp.

	8. V.V. Bratishchenko. Design of Information Systems: A Textbook. Irkutsk: BSUEP Publishing House, 2004. 84 pp.
Responsible Lecturer	Zaripova Mukaddas Djumayozovna, Doctor of Philosophy (PhD) in Economics, Associate Professor ad interim

2.02. ARTIFICIAL INTELLIGENCE AND NEURAL NETWORKS

Field	Description
Module Title	ARTIFICIAL INTELLIGENCE AND NEURAL NETWORKS
Module Code	SINT104
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	II semester
Module objective	The aim of the Module Artificial Intelligence and Neural Networks is to provide postgraduate students with a deep understanding of the theoretical foundations and operating principles of artificial intelligence, machine learning, and neural networks. Intelligent agents, expert systems, and machine learning algorithms are also studied. Students learn to design and develop modern artificial intelligence systems based on deep learning. Convolutional and recurrent neural networks are addressed in this design and development process. Natural language processing technologies are also addressed. Professional competencies in teaching, assessment, and implementation into practice are developed.
Module Learning Outcomes	Knowledge: LO1. Can explain the main concepts, principles, stages of development and modern directions of artificial intelligence. LO2. Describe the structure and operating principles of intelligent agents, knowledge bases and expert systems. LO3. Can explain the theoretical foundations of supervised and unsupervised machine learning algorithms.

	LO4. Can describe the operating principles and application areas of artificial neural networks, deep learning, CNN and RNN architectures. LO5. Explain the capabilities and limitations of Natural Language Processing (NLP), generative artificial intelligence and modern neural network technologies. Skills: LO6. Has practical skills in data collection, preparation and analysis for artificial intelligence tasks. LO7. Applies machine learning algorithms, model creation and methods for evaluating results. LO8. Performs practical tasks on creating, training and testing artificial neural networks. LO9. Develops artificial intelligence models using modern tools such as Python, TensorFlow, Keras and PyTorch. LO10. They will have the skills to evaluate, optimise and interpret the results of artificial intelligence models. LO11. Effectively utilises artificial intelligence and neural network technologies to solve practical problems. LO12. Analyses the effectiveness of intelligent systems and machine learning models and makes decisions on improvements. Competencies: LO13. Develops innovative AI-based solutions for medicine, economics, transport, education and other sectors. LO14. Applies modern technologies in research and innovation projects in the field of artificial intelligence and neural networks. LO15. Adheres to the principles of information security, ethical standards, responsible AI and academic integrity when developing artificial intelligence systems.
--	--

Content	Lectures Introduction to the Field of Artificial Intelligence. Stages of Artificial Intelligence Development. The concept, objectives, and importance of artificial intelligence are studied. The historical development of AI from its origins to modern intelligent systems is explored. Major achievements, challenges, and future trends in artificial intelligence technologies are analyzed. Capabilities and Types of Artificial Intelligence. The capabilities and limitations of artificial intelligence systems are studied. The classification of AI into narrow AI, general AI, and superintelligent AI is studied. Practical applications of different types of AI in various fields are examined. Intellectual Agent. The concept of intelligent agents and their components is studied. How agents perceive their environment, make decisions, and perform actions is learned. Different types of intelligent agents and their applications in modern systems are analyzed. Intellectual Analysis of Data. Methods of extracting useful knowledge from large datasets are explored. The principles of data mining, pattern recognition, and predictive analytics are studied. The role of intelligent data analysis in decision-making processes is examined. Knowledge Base. The structure and organization of knowledge bases are studied. Methods of knowledge representation, storage, and retrieval are studied. The role of knowledge bases in intelligent systems and expert applications is examined. Expert Systems. The principles and architecture of expert systems are studied. Knowledge acquisition, inference engines, and decision-making mechanisms are studied. Practical applications of expert systems in medicine, education, business, and engineering are analyzed.
-----------------------	--

Introduction to Machine Learning. The fundamentals of machine learning and its relationship with artificial intelligence are studied. Data-driven learning processes, model training, and prediction techniques are studied. The importance of machine learning in modern AI applications is examined.

Supervised Learning Algorithms. Supervised learning concepts, datasets, and training methods are studied. Classification and regression algorithms are studied. Practical applications of supervised learning in prediction and decision-making tasks are examined.

Unsupervised Learning Algorithms. Unsupervised learning techniques and data clustering methods are explored. How patterns and structures are discovered in unlabeled data is studied. Practical applications of clustering and dimensionality reduction algorithms are analyzed.

Introduction to Artificial Neural Networks. The structure and operation of artificial neural networks are studied. Neurons, layers, weights, activation functions, and learning processes are studied. The role of neural networks in intelligent systems is examined.

Deep Learning Algorithms. The principles of deep learning and multi-layer neural networks are studied. Training processes, optimization techniques, and practical applications of deep learning are studied. The advantages of deep learning in complex problem-solving are explored.

CNN Architecture and Operational Structure. Convolutional Neural Networks (CNNs) and their architecture are studied. How CNNs process images and visual information is studied. The applications of CNNs in image recognition, object detection, and computer vision are examined.

RNN Architecture and Operating Structure. The structure and operation of Recurrent Neural Networks (RNNs) are studied. Sequence processing and time-dependent data analysis are studied. The applications

	of RNNs in speech recognition, text processing, and forecasting are examined. Natural Language Processing. Methods of processing and understanding human language using artificial intelligence are studied. Text analysis, language models, translation systems, and conversational AI are studied. Practical applications of NLP technologies are explored. Applying Neural Networks to the Learning Process. How neural networks can be applied in education and intelligent learning systems is studied. Adaptive learning technologies, intelligent tutoring systems, and educational analytics are studied. The benefits of AI-driven learning environments are analyzed. Practical lessons Introduction to Artificial Intelligence. Familiarisation with Python Programming Libraries. The Python environment is installed and configured. The basic Python libraries used in artificial intelligence are learned. Practical applications of Python in machine learning and intelligent systems are explored. Working with Variables and Data Types in the Python Programming Language. Python variables and data types are studied. Data manipulation techniques are also studied. Basic operations are performed, and data representation in Python programs is understood. Working with Arithmetic, Logical and Assignment Operators in Python. Arithmetic calculations are applied. Logical expressions and assignment operations are also applied. Problem-solving skills are developed through programming exercises. Working with Strings and String Operations in Python. String manipulation techniques are studied. Text processing methods and string functions are also studied. Practical exercises involving text data are performed.
--	---

	Working with Conditional Statements in Python. If, elif, and else statements are used to create decision-making programs. These statements are applied in practical programming exercises. Logical thinking is developed through conditional programming tasks. Working with Looping Operators in Python. For and while loops are studied. Repetitive processes are created using loop structures. Calculations are automated using these loop structures. Working with Functions in Python. Functions are defined and used in Python programs. Parameters and return values are studied. Modular programming principles are also studied. Working with Lists in the Python Programming Language. Lists are created and managed. Data storage and retrieval operations are performed using Python lists. Data manipulation operations are also performed using Python lists. Working with Tuples in the Python Programming Language. Tuple structures and their applications are studied. Tuples are compared with lists. Practical exercises involving tuples are performed. Working with Sets in the Python Programming Language. Sets are used to manage unique data elements. Set operations are performed. Practical data-processing tasks are solved using sets. Working with Dictionaries in the Python Programming Language. Key-value data structures are created and managed. Dictionaries are applied in practical programming tasks. Dictionaries are also applied in machine learning tasks. Working with Files in the Python Programming Language. Reading from files is practiced. Writing to files is also practiced. Data storage and file processing operations are managed using Python. Working with the Main Built-in Modules (random, math) in Python. Built-in Python modules are used for mathematical calculations. These modules are
--	--

	also used for random number generation. Practical computational problems are solved using these modules. Working with Modules in the Python Programming Language. External modules are imported and used. Code reusability principles are studied. Modular programming techniques are also studied. Learning the Basics of OOP in the Python Programming Language. Classes and objects are studied. Inheritance, encapsulation, and polymorphism are also studied. Object-oriented solutions to programming problems are developed. Exploring a Simple Neural Network Program that Determines the Ratio Between Magnitudes. A simple neural network model is built and analyzed. Neural network inputs and outputs are studied. Learning processes are studied through practical examples. Programming a Neural Network in Python to Find the Square of a Number. A simple neural network application is developed in Python. The model is trained and tested to approximate mathematical relationships. Machine learning fundamentals are understood through this process. Self-study Preparing for Practical Sessions and Completing Assignments in Python Programming and Artificial Intelligence. Theoretical materials are reviewed. Programming tasks are practiced. Laboratory assignments related to artificial intelligence and Python are completed. Independent Study of Literature, Scientific Articles and Educational Materials on Artificial Intelligence, Machine Learning and Neural Networks. Textbooks and scientific publications are studied. Online resources are also studied. Current developments and research in artificial intelligence are analyzed.
--	---

	Performing Practical Tasks in Python Programming, Machine Learning and Neural Network Modelling. Practical programming solutions are developed. Machine learning models are also developed. Theoretical concepts are applied to real-world problems. Studying Scientific Sources and Analytical Materials on Artificial Intelligence, ChatGPT, Deep Learning and Neural Networks. Modern AI technologies are explored. Their applications are also explored. Current trends and innovations in artificial intelligence are evaluated. Developing Mini-Projects and Practical Models Based on Artificial Intelligence and Neural Networks. Small AI projects are designed. These projects are also implemented. Machine learning and neural network techniques are applied to practical tasks. Preparation of Presentations, Reports, Software Developments and Independent Works. Technical reports and presentations are prepared. Software prototypes and research projects are also prepared. Acquired knowledge and practical skills in artificial intelligence and intelligent systems are demonstrated.
Teaching Methods	The module is delivered through a combination of lectures, laboratory classes, practical programming sessions, project-based learning, demonstrations, and independent study activities. Teaching methods include classroom instruction, interactive discussions, hands-on Python programming exercises, artificial intelligence modeling, machine learning experiments, and neural network implementation. They also include algorithm design, case studies, problem-solving activities, group projects, seminars, presentations, and the use of modern AI development tools and frameworks. Practical activities involve developing Python programs, working with data structures and algorithms, implementing supervised and unsupervised learning models, and building simple neural networks. They also involve processing and

	analyzing datasets, experimenting with deep learning techniques, and creating artificial intelligence applications for real-world problems.
Prerequisites	Programming 1,2 Data Structures and Algorithms Probability and Statistics Database Foundations of Artificial Intelligence Big Data and Data Analysis
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, Python programming exercises, practical artificial intelligence tasks, machine learning experiments, seminar presentations, and the successful completion of independent study assignments, including data preprocessing, algorithm implementation, neural network modeling, AI-based problem-solving activities, and mini-project development. The midterm assessment consists of written assignments, short in-class quizzes, laboratory evaluations, and the assessment of independent study projects, such as Python-based artificial intelligence applications, machine learning models, neural network implementations, data analysis reports, intelligent agent simulations, AI case studies, and presentations on current developments in artificial intelligence and deep learning. The final assessment is a written examination consisting of theoretical questions and practical problem-solving tasks designed to assess students' understanding of artificial intelligence concepts, intelligent systems, machine learning algorithms, neural networks, deep learning architectures, natural language processing, expert systems, knowledge representation, and their ability

	to apply artificial intelligence techniques to real-world computational problems.
Recommended Literature	1. Murphy, K. P. Probabilistic Machine Learning: A Primer. Trans. by A. A. Slinskykina. Moscow: DMK Press, 2022. 940 pp. 2. Bekmuratov Q.A. Artificial Intelligence and Neural Networks. Textbook. "Mahalla and Family Publishing House", 2021. 373 pp. 3. Bekmuratov Q.A. Artificial Intelligence. Textbook. Tashkent: "Aloqachi", 2019. 312 pp. 4. Khojaqulov T.A., Malikova N.T. Artificial Intelligence. Textbook. Tashkent: "Aloqachi", 2019. 192 pp. 5. Sadullayeva Sh.A., Yusupov D.F., Yusupov F. Artificial Intelligence and Neural Network Technologies. Textbook. Tashkent: "Mahalla and Family Publishing House", 2022. 192 pp. 6. Abdumanonov A.A., Quchqarov A.A. Fundamentals of Artificial Intelligence. Textbook. Farg'ona: "Sunrise-pro" Publishing House, 2025. 344 pp. 7. Maxkamov B.Sh., Zaynidinov X.N., Nurmurodov J.N. Fundamentals of Artificial Intelligence. Textbook. Tashkent University of Information Technologies, 2024. 264 pp.
Responsible Lecturer	Zaripova Mukaddas Djumayozovna

2.02. CLOUD TECHNOLOGIES

Field	Description
Module Title	CLOUD TECHNOLOGIES
Module Code	BTH104
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	II semester
Module objective	The aim of the Module Cloud Technologies is to develop in master's students a deep understanding of the theoretical foundations of cloud technologies. Its architecture, service models, and operating principles are also studied. Virtualisation and cloud infrastructure are addressed. Professional competencies in data storage and processing are developed. The creation, management, and analysis of cloud services are also developed. The implementation of cloud services is addressed as well.
Module Learning Outcomes	Knowledge: LO1. Can explain the fundamental concepts, principles, advantages and current development trends of cloud technologies. LO2. Can describe the service models of cloud technologies (IaaS, PaaS, SaaS) as well as the operating principles and application areas of cloud infrastructure. LO3. Can explain the functions and components of virtualisation technologies, cloud computing resources, data storage and network services. LO4. Can explain the principles of resource management, load distribution, backup and data recovery in cloud technologies.

	LO5. Describe the operating principles of cloud platforms, containerisation technologies and cloud services. Skills: LO6. Explain the methods for information security, authentication, authorisation, and data protection in cloud technologies. LO7. Has practical skills in designing a cloud infrastructure and integrating its main components. LO8. Utilises tools for managing computing resources, data storage and network services on cloud platforms. LO9. Performs tasks of creating, configuring and managing virtual machines, cloud storage systems and cloud services. LO10. Deploys, configures and utilises applications on cloud platforms effectively. LO11. Applies methods for evaluating the efficiency of cloud systems, monitoring them, and using resources wisely. LO12. They will have the skills to develop, test and implement practical solutions based on cloud technologies. Competencies: LO13. Effectively utilises cloud technologies to develop corporate information systems and digital services. LO14. Analyses the performance of the cloud infrastructure, monitors it, and makes decisions on optimisation. LO15. Has the competence to design, manage and use cloud platforms and services effectively. LO16. Applies modern technologies in research and innovation projects in the field of cloud technologies.
--	---

	LO17. They will independently explore new cloud platforms, services and software tools and will be qualified to implement them in practice.
Content	Lectures Introduction to Cloud Technologies. The Concept of Cloud Computing, Its Stages of Development and Its Advantages. The concept of cloud computing, its historical development, and its role in modern information technologies are studied. The advantages of cloud technologies, including scalability, flexibility, cost efficiency, and accessibility, are examined. The impact of cloud computing on businesses, education, and industry is analyzed. Architecture and Key Components of Cloud Technologies. The architecture of cloud systems and their fundamental components are studied. Servers, storage systems, networking infrastructure, and management platforms are also studied. How cloud environments are organized and operated is examined. Service Models of Cloud Technologies (IaaS, PaaS, SaaS) and the Operating Principles of Cloud Infrastructure. Infrastructure as a Service, Platform as a Service, and Software as a Service models are explored. Their characteristics, advantages, and practical applications are examined. The operational principles of cloud infrastructure and service delivery are analyzed. Virtualisation and Hypervisors in Cloud Technologies. Virtualization concepts and technologies are studied. The role of hypervisors in creating and managing virtual machines is studied. The importance of virtualization in resource optimization and cloud environments is examined. Deployment Models of Cloud Technologies: Public Cloud, Private Cloud, Hybrid Cloud and Community Cloud. Different cloud deployment models and their characteristics are studied. Public, private, hybrid, and community clouds are compared.

	Their applications, benefits, and limitations in various organizational environments are examined. Cloud Data Storage Technologies and Data Centres. Cloud storage solutions and data center infrastructure are studied. Methods of data storage, synchronization, and management are studied. The role of data centers in cloud service delivery and reliability is examined. Cloud Computing Resource Management and Monitoring. Techniques for managing and monitoring cloud resources are studied. Resource allocation, scalability, performance monitoring, and cost optimization are also studied. Cloud resource management strategies are examined. Deploying and Managing Applications on Cloud Platforms. Methods for deploying applications in cloud environments are studied. Application lifecycle management, scaling techniques, and maintenance procedures are studied. Practical skills in cloud-based application management are developed. Containerisation Technologies and Docker Fundamentals. The principles of containerization and its advantages over traditional virtualization are studied. Docker architecture, images, containers, and container management are studied. Modern software deployment approaches are examined. Use of Cloud Services in Education, Business and Industry. Practical applications of cloud technologies in different sectors are explored. Cloud-based solutions for education and business operations are analyzed. Cloud-based solutions for industrial automation and digital transformation are also analyzed. Information Security and Data Protection in Cloud Technologies. Security challenges and protection mechanisms in cloud environments are studied. Authentication, authorization, encryption, and access control methods are studied. Cloud security best practices are examined.
--	--

	Backup and Disaster Recovery Technologies in the Cloud Environment. Backup strategies and disaster recovery solutions are studied. Methods of protecting critical data and ensuring business continuity are studied. Cloud-based recovery mechanisms and fault tolerance are examined. Continuous Service and Effective Use of Resources in Cloud Technologies. High availability and load balancing concepts are studied. Fault tolerance and service continuity concepts are also studied. Techniques for optimizing cloud resources and improving system efficiency are learned. Modern Cloud Technologies and Their Development Trends. Current innovations and emerging technologies in cloud computing are explored. Edge computing, serverless computing, multi-cloud environments, and cloud-native applications are studied. Future trends in cloud technology are analyzed. Prospects and Practical Applications of Cloud Technologies. The future potential of cloud computing is examined. Its applications across industries are also examined. How cloud technologies support innovation, digital transformation, and business growth is understood. Practical lessons Creating and Configuring Virtual Machines. Virtual machines are created and configured. They are also managed. Virtual machine settings, resource allocation, and system administration tasks are understood. Managing the Virtualisation Environment. Virtualization platforms and management tools are used. Virtual resources are monitored. Virtual machine performance is optimized. Working with Cloud Storage Services. Cloud storage platforms are used for data management and sharing. They are also used for synchronization and
--	--

	backup. Practical cloud storage operations are understood. Analysis of IaaS, PaaS and SaaS Services. Cloud service models are compared through practical examples. Their advantages and limitations are evaluated. Their real-world applications are also evaluated. Fundamentals of Working with Docker Containers. Docker containers are created and managed. They are also deployed. Container lifecycle management and basic container orchestration concepts are studied. Organising Information Security in a Cloud Environment. Security measures in cloud platforms are implemented. Access controls and authentication systems are configured. Data protection mechanisms are also configured. Deploying Applications on Cloud Platforms. Applications are published and managed in cloud environments. Cloud services are configured. Application performance is monitored, and availability is ensured. Developing a Mini-Project Based on Cloud Technologies. A practical cloud-based solution is designed and implemented. Virtualization, cloud services, and storage concepts are applied in a real-world project. Security and deployment concepts are also applied in this project. Self-study Preparing for Practical Sessions and Completing Assignments. Theoretical materials are reviewed. Laboratory instructions are studied. Assigned practical tasks related to cloud technologies are completed. Study of Scientific Literature and Articles on Cloud Technologies. Scientific publications related to cloud computing are analyzed. Technical documentation related to cloud computing is also
--	---

	analyzed. Research papers related to virtualization technologies are analyzed as well. Complete Practical Tasks on Virtualisation and Cloud Services. Independent practical exercises involving virtual machines are performed. Exercises involving cloud storage and cloud platforms are also performed. Exercises involving service management are performed as well. Analyse Scientific Sources on Cloud Platforms and Modern Technologies. Modern cloud platforms and emerging technologies are studied. Industry trends are also studied. Analytical reviews and summaries are prepared. Preparation of Mini-Projects Based on Cloud Technologies. Small cloud-based applications are designed. Infrastructure solutions are also designed and developed. Modern cloud tools and technologies are used in this process. Preparation of Presentations, Reports and Independent Work. Presentations and technical reports are prepared. Research papers and project documentation are also prepared. These materials cover topics related to cloud computing, virtualization, cloud security, and modern cloud services.
Teaching Methods	The module is delivered through a combination of lectures, laboratory classes, practical demonstrations, project-based learning, workshops, and independent study activities. Teaching methods include classroom instruction, interactive discussions, case studies, hands-on virtualization exercises, cloud platform demonstrations, cloud service configuration activities, problem-solving tasks, group projects, seminars, presentations, and the use of modern cloud computing tools and technologies. Practical activities involve creating and configuring virtual machines, managing virtualization environments, working with cloud storage services, analyzing IaaS, PaaS, and SaaS models, and deploying and managing applications on cloud platforms. They also involve using Docker containers, implementing cloud security

	measures, performing backup and disaster recovery tasks, and developing cloud-based solutions.
Prerequisites	Computer Networks Computer Architecture Operating Systems Cryptography and Network Security
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, practical cloud computing exercises, virtualization tasks, cloud storage management activities, Docker container implementation, seminar presentations, and the successful completion of independent study assignments related to cloud technologies. The midterm assessment consists of written assignments, short in-class quizzes, laboratory evaluations, and the assessment of independent study projects, such as virtual machine deployment, cloud infrastructure design, cloud service comparison reports (IaaS, PaaS, SaaS), Docker-based application deployment, cloud security analysis, backup and disaster recovery planning, and presentations on modern cloud technologies. The final assessment is a written examination consisting of theoretical questions and practical problem-solving tasks designed to assess students' understanding of cloud computing concepts, cloud architectures, virtualization technologies, cloud deployment models, cloud service platforms, containerization technologies, cloud security mechanisms, resource management strategies, and their ability to apply cloud solutions to real-world scenarios.
Recommended Literature	1. Delov T.E. Cloud Technologies. Textbook. Tashkent, 2020. 217 pp.

	2. Balanoff A. N. Cloud Technologies. A Textbook for Vocational Education. Saint Petersburg: Lan', 2024. 104 pp. 3. Buyya, R., Vecchiola, C., & Selvi, S.T. Mastering Cloud Computing: Foundations and Applications Programming. Morgan Kaufmann. ISBN 978-0124114548. Elsevier Inc., 2013. 468 pp. 4. Burns, B., Beda, J., Hightower, K., & Evenson, L. Kubernetes: Up and Running. 3rd Edition, 328 pp. O'Reilly Media, 2022. ISBN 978-1098110208. 5. Arundel, J., & Domingus, J. Cloud Native DevOps with Kubernetes. 2nd Edition. O'Reilly Media, 2022. ISBN 978-1098116828. 6. V.V. Bratishchenko. Design of Information Systems: A Textbook. Irkutsk: BSUEP Publishing House, 2004. 84 pp. 7. Maxkamov B.Sh., Zaynidinov X.N., Nurmurodov J.N. Fundamentals of Artificial Intelligence. Textbook. Tashkent University of Information Technologies, 2024. 264 pp.
Responsible Lecturer	Zaripova Mukaddas Djumayozovna

2.03. COMPUTATIONAL METHODS AND MODELLING

Field	Description
Module Title	COMPUTATIONAL METHODS AND MODELLING
Module Code	HUM204
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	III semester
Module objective	This course equips master's students with both the theoretical foundation and hands-on skills needed to work confidently with numerical methods and algorithms. Students explore error theory and function approximation. Numerical integration is also studied, and students learn to solve algebraic and transcendental equations, individually and as systems. The course also covers system conditioning and numerical differentiation. Practical approaches to Cauchy and boundary-value problems for ordinary differential equations are covered. Throughout, students gain experience designing algorithms and writing software to tackle real computational problems.
Module Learning Outcomes	Knowledge: LO1. Understands the sources and types of computational errors - including rounding, truncation, and absolute and relative error - and how errors accumulate across arithmetic operations. LO2. Understands direct and iterative numerical methods for solving systems of linear algebraic equations, including the conditions that ensure the convergence of iterative approaches. LO3. Understands techniques for isolating and numerically computing roots of nonlinear equations

	and systems, including the theoretical basis behind each method. LO4. Understands numerical methods for computing eigenvalues and eigenvectors of matrices and recognises their significance in applied mathematics and engineering. LO5. Understands the purpose and principles of interpolation, including how interpolating polynomials are constructed and used to approximate function values. LO6. Understands numerical integration methods, including key quadrature formulas, and can reason about their accuracy and associated error bounds. LO7. Understands the foundations of numerical methods for initial-value problems (Cauchy problems) for ordinary differential equations, including approximation error, finite differences, and method classification. LO8. Understands numerical and approximate approaches to boundary-value problems for second-order ordinary differential equations, including finite difference and variational methods. Skills: LO9. Can accurately estimate the absolute and relative errors of computed results and assess how those errors propagate through a sequence of arithmetic operations. LO10. Can apply direct and iterative numerical methods to solve systems of linear algebraic equations and verify results using computational tools. LO11. Can locate and compute roots of nonlinear equations and systems using appropriate iterative methods and verify results programmatically. LO12. Can numerically compute eigenvalues and eigenvectors of matrices and interpret their practical significance in context.
--	---

	LO13. Can construct interpolating polynomials for a given dataset and use them to estimate function values at intermediate points. LO14. Can evaluate definite integrals numerically using quadrature formulas, estimate integration errors, and implement the procedure using software tools. LO15. Can apply single-step and multi-step numerical methods to solve initial-value problems for ordinary differential equations and analyse the accuracy of the solution. LO16. Can solve boundary-value problems for ordinary differential equations numerically and implement the solution using a programming environment. LO17. Can assess the conditioning of a linear system and recognise when ill-conditioning may compromise the reliability of numerical results. LO18. Can design step-by-step algorithms for numerical problems and implement them as working program code. Competencies: LO19. Independently selects and applies the most appropriate numerical method for a given problem, justifying the choice based on the problem's structure and accuracy requirements. LO20. Critically assesses the accuracy and reliability of computational results, identifies potential sources of error, and draws well-founded conclusions from numerical data. LO21. Designs and implements complete software solutions for numerical problems, demonstrating competence in both algorithm development and the use of modern computational tools.
Content	Lectures Introduction. Basic Concepts of Differential Equations. The fundamental concepts of differential equations and their role in mathematical modeling are

introduced. Ordinary and partial differential equations and their classification are examined. Order, initial and boundary conditions, and applications to real-world processes are also examined.

Engineering Problems Reducible to Ordinary Differential Equations. Engineering and physical problems that can be mathematically represented by ordinary differential equations are examined. Mathematical models of mechanical and thermal processes are formulated. Models of electrical and other continuous processes are also formulated and analyzed.

Numerical Solution of Boundary-Value Problems for Ordinary Differential Equations. Selection of Basis Functions. Collocation Method. Numerical methods for solving boundary-value problems for ordinary differential equations are studied. The selection of appropriate basis functions is examined. The implementation of the collocation method is also studied.

Numerical Solution of Boundary-Value Problems for Ordinary Differential Equations: Galerkin Method. The Galerkin method for the numerical solution of boundary-value problems is examined. The construction of approximate solutions and the selection of trial and weighting functions are studied. The formulation of the corresponding algebraic system is also studied.

Solving Boundary-Value Problems Using the Finite Difference Method. The application of the finite difference method to boundary-value problems for ordinary differential equations is studied. Discretization techniques and difference schemes are examined. Numerical solution procedures are also examined.

Partial Differential Equations and Methods for Solving Them. Partial differential equations and their classification into elliptic, parabolic, and hyperbolic types are introduced. Analytical methods for solving mathematical models based on partial differential

equations are studied. Numerical methods for these models are also studied.

Finite Difference Method. General Concepts. Approximation, Stability and Convergence. The fundamental principles of finite difference methods are examined. The concepts of approximation and consistency are studied. Stability and convergence are studied as essential criteria for evaluating numerical schemes.

Methods for Solving First-Order Partial Differential Equations. Analytical methods for solving first-order partial differential equations are studied. Numerical methods for these equations are also studied. Characteristics, initial and boundary conditions, and mathematical models of transport processes are examined.

Numerical Methods for Solving the Heat Conduction Equation. Numerical methods for modeling heat conduction processes are examined. Explicit, implicit, and combined finite difference schemes are studied. Their stability and convergence properties are also studied.

Different Schemes for the Quasilinear Heat Conduction Equation. Numerical schemes for solving quasilinear heat conduction equations are studied. The influence of nonlinear coefficients on discretization and stability is analyzed. Their influence on convergence and computational accuracy is also analyzed.

Numerical Solution of the Two-Dimensional Heat Conduction Equation. Numerical methods for modeling two-dimensional heat transfer are examined. Finite difference schemes and boundary conditions are studied. Computational grids and numerical analysis of temperature distributions are also studied.

Numerical Methods for Solving Laplace and Poisson Equations. Numerical methods for solving elliptic equations, particularly the Laplace and Poisson equations, are studied. Difference schemes

	for obtaining numerical solutions are examined. Iterative methods for obtaining numerical solutions are also examined. Numerical Methods for Solving Second-Order Hyperbolic Partial Differential Equations. Numerical methods for second-order hyperbolic equations describing wave and propagation processes are examined. Difference schemes and stability conditions are studied. Numerical simulation of wave phenomena is also studied. Convection-Diffusion Problems and Their Numerical Solution. Mathematical models describing the simultaneous effects of convection and diffusion are studied. Numerical schemes for convection-diffusion equations are examined. Stability issues and the analysis of transport processes are also examined. Computer Modelling of Differential Equations Using Python. The implementation of mathematical models and numerical methods using Python is studied. Students learn to construct computational algorithms and solve differential equations numerically. Results are visualized, and modeled processes are analyzed. Practical lessons Introduction. Basic Concepts of Differential Equations. The basic concepts and classification of differential equations are introduced. Students solve elementary examples of differential equations. The order, type, and initial or boundary conditions of differential equations are identified. Engineering Problems Reducible to Ordinary Differential Equations. Mathematical models of engineering and physical processes are formulated using ordinary differential equations. Practical examples from mechanical and thermal engineering systems are considered. Examples from electrical and other engineering systems are also considered.
--	---

Numerical Solution of Boundary-Value Problems for Ordinary Differential Equations. Selection of Basis Functions. Collocation Method. Approximate solutions to boundary-value problems are constructed using selected basis functions. The collocation method is implemented. The obtained numerical results are analyzed.

Numerical Solution of Boundary-Value Problems for Ordinary Differential Equations: Galerkin Method. The Galerkin method is applied to boundary-value problems. Approximate solutions are constructed by students. The corresponding system of algebraic equations is formulated.

Solving Boundary-Value Problems Using the Finite Difference Method. Boundary-value problems are discretized using finite difference schemes. The resulting systems of algebraic equations are solved numerically. The accuracy of the obtained solutions is evaluated.

Partial Differential Equations and Methods for Solving Them. Partial differential equations are classified. Appropriate analytical and numerical solution methods are analyzed. Basic mathematical models based on partial differential equations are formulated and investigated.

Finite Difference Method. General Concepts. Approximation, Stability and Convergence. Finite difference schemes are constructed. Their approximation, stability, and convergence properties are evaluated. The reliability and accuracy of numerical solutions are analyzed.

Methods for Solving First-Order Partial Differential Equations. Analytical methods for solving first-order partial differential equations are applied. Numerical methods for these equations are also applied. Problems involving characteristics, initial conditions, and transport processes are considered.

Numerical Methods for Solving the Heat Conduction Equation. Numerical schemes for the heat conduction equation are implemented.

	Temperature distributions are computed. The influence of discretization parameters and stability conditions is analyzed. Different Schemes for the Quasilinear Heat Conduction Equation. Numerical methods for quasilinear heat conduction equations are studied. Different schemes are implemented and compared. The effects of nonlinear coefficients on numerical solutions are analyzed. Numerical Solution of the Two-Dimensional Heat Conduction Equation. Computational grids are constructed for two-dimensional heat conduction problems. Numerical schemes are implemented for these problems. The obtained temperature distributions are visualized and analyzed. Numerical Methods for Solving Laplace and Poisson Equations. The numerical solution of Laplace and Poisson equations is studied. Finite difference and iterative methods are implemented. The accuracy and convergence of the solutions are analyzed. Numerical Methods for Solving Second-Order Hyperbolic Partial Differential Equations. Numerical schemes for second-order hyperbolic equations are implemented. The propagation of waves is investigated. The stability of numerical solutions is investigated through computational experiments. Convection-Diffusion Problems and Their Numerical Solution. The numerical solution of convection-diffusion problems is studied. The influence of convection and diffusion components is investigated. The stability and accuracy of numerical schemes are analyzed. Computer Modelling of Differential Equations Using Python. Numerical methods for solving differential equations are implemented using Python. Computational models are developed. Numerical
--	--

	results are visualized, and the behavior of modeled processes is analyzed. Self-study Subject and Tasks of Computational Methods and Modelling. The subject and objectives of computational methods and mathematical modelling are examined. The fundamental tasks of this field are studied. The role of numerical methods in solving scientific, engineering, and technical problems is analyzed. Basic Concepts of Differential Equations and Their Applications. The fundamental concepts of differential equations are studied. Their classification and applications are examined. Their role in describing and modeling continuous processes is analyzed. Boundary-Value Problems for Ordinary Differential Equations. The formulation of boundary-value problems for ordinary differential equations is studied. Solution methods for these problems are examined. Boundary conditions and their influence on the solution are analyzed. Collocation Method and Selection of Basis Functions. The principles of the collocation method are studied. Methods for selecting appropriate basis functions are examined. The construction of approximate numerical solutions is also examined. Galerkin Method and Its Applications. The theoretical foundations of the Galerkin method are examined. Its application to boundary-value problems is studied. Approximate solutions and corresponding algebraic systems are analyzed. Solving Boundary-Value Problems Using the Shooting Method. The shooting method for solving boundary-value problems is studied. The transformation of a boundary-value problem into an initial-value problem is examined. The numerical solution process is also examined.
--	--

	Theoretical Foundations of the Finite Difference Method. The fundamental principles of the finite difference method are studied. Discretization of differential equations is examined. Construction of difference schemes is also studied. Concepts of Approximation, Stability and Convergence. The fundamental concepts of approximation and stability of numerical methods are examined. Convergence of numerical methods is also examined. Their importance in assessing the reliability and accuracy of computational algorithms is analyzed. Partial Differential Equations and Their Classification. Partial differential equations are examined. Their classification into elliptic, parabolic, and hyperbolic types is studied. The characteristics and applications of different equation classes are examined. Numerical Methods for First-Order Partial Differential Equations. Numerical methods for solving first-order partial differential equations are studied. Characteristic-based approaches are examined. Finite difference approaches are also examined. Numerical Methods for Solving the Heat Conduction Equation. Numerical methods for modeling heat conduction processes are studied. Difference schemes for these processes are analyzed. Their stability and convergence properties are also analyzed. Difference Schemes for Quasilinear Heat Conduction Equations. The construction of difference schemes for quasilinear heat conduction equations is studied. The application of these schemes is examined. The influence of nonlinear terms on numerical solutions is also examined. Explicit and Implicit Difference Schemes. Explicit and implicit finite difference schemes are examined. Their computational characteristics and stability conditions are comparatively analyzed. Their
--	--

	advantages and limitations are also comparatively analyzed. Two- and Three-Layer Difference Schemes. Two-layer difference schemes for time-dependent differential equations are studied. Three-layer difference schemes are also studied. Their construction, stability, and practical applications are analyzed. Numerical Solution of the Two-Dimensional Heat Conduction Equation. Numerical methods for solving two-dimensional heat conduction problems are studied. Computational grids and boundary conditions are examined. Temperature distributions are also examined. Linear Difference Schemes and Their Properties. The theoretical foundations of linear difference schemes are studied. Their properties are examined. Their approximation, stability, and convergence characteristics are analyzed. Difference Problems for the Wave Equation. The construction of difference problems for the wave equation is examined. Numerical modeling of wave propagation is studied. The stability of computational schemes is also studied. Numerical Solution of Hyperbolic Differential Equations. Numerical methods for solving hyperbolic differential equations are examined. Difference schemes for modeling wave processes are analyzed. Difference schemes for modeling propagation processes are also analyzed. Numerical Methods for Solving the Laplace Equation. Numerical methods for solving the Laplace equation are studied. Finite difference methods for this equation are examined. Iterative methods and their convergence properties are also examined. Numerical Methods for Solving the Poisson Equation. Numerical methods for solving the Poisson equation are examined. Discretization techniques and
--	---

	iterative algorithms are analyzed. The accuracy of numerical solutions is also analyzed. Numerical Modelling of Convection-Diffusion Problems. The mathematical modeling of convection-diffusion processes is studied. The numerical modeling of these processes is also studied. The interaction between transport and diffusion mechanisms is examined. Evaluation of the Accuracy and Efficiency of Computational Algorithms. The accuracy of numerical algorithms is evaluated. Their computational efficiency and performance are also evaluated. Different criteria for comparing computational methods are examined. Analysis of Computational Errors in Numerical Methods. Sources and types of computational errors in numerical methods are examined. Truncation and discretization errors are analyzed. Rounding and accumulated errors are also analyzed. Computer Modelling of Boundary-Value Problems. The principles of computer modeling of boundary-value problems are studied. Mathematical models are transformed into computational algorithms. Numerical results are analyzed. Solving Ordinary Differential Equations in the Python Environment. Solving ordinary differential equations using Python is studied. Numerical algorithms are implemented, and computational results are obtained. Solution behavior is visualized. Modelling Partial Differential Equations in the Python Environment. Numerical models of partial differential equations are developed using Python. Computational algorithms are implemented. The obtained results are visualized and analyzed. Implementation of Computational Methods in the MATLAB Environment. The implementation of numerical methods using MATLAB is examined. Computational algorithms and mathematical models
--	--

	are studied. Graphical analysis of numerical results is also studied. Visualisation and Analysis of Computational Results. Methods for visualizing numerical and computational results are studied. Tools for these visualizations are also studied. Graphical representations are used to analyze solution behavior and model dynamics. Mathematical Modelling of Engineering and Technical Problems. The formulation of mathematical models for engineering and technical processes is studied. The numerical solution of these models is examined. Computational methods are applied to analyze real-world systems. Preparation of a Project Based on Computational Methods and Modelling. A project based on computational methods and mathematical modeling is prepared. The project includes problem formulation, mathematical modeling, and numerical implementation. Result visualization, analysis, and presentation of conclusions are also included.
Teaching Methods	The module is delivered through advanced interactive lectures, problem-based and inquiry-based learning, practical computational exercises, numerical experiments, and research-oriented project activities. Interactive lectures introduce the theoretical foundations of computational methods and mathematical modelling, including differential equations, boundary-value problems, finite difference methods, approximation, stability, convergence, and numerical methods for elliptic, parabolic, and hyperbolic equations. Problem-based learning enables master's students to formulate mathematical models, select appropriate numerical methods, analyze computational problems, and justify the accuracy and efficiency of obtained solutions. Practical exercises focus on the numerical solution of ordinary and partial differential equations, boundary-value problems, heat conduction, Laplace, Poisson, wave, and convection-diffusion equations. Python

	and MATLAB environments are used to implement computational algorithms, perform numerical experiments, visualize computational results, and analyze approximation errors, stability, convergence, and algorithmic efficiency. Research-oriented project-based learning and independent study develop master's students' analytical, computational, research, and scientific skills through individual and group assignments, comparative studies of numerical methods, mathematical modelling projects, analytical reports, scientific presentations, and computational experiments, and students are encouraged to critically evaluate existing methods, formulate and investigate mathematical models of engineering and technical processes, and develop independent solutions using modern computational tools and programming environments.
Prerequisites	Mathematical Analysis Differential Equations Linear Algebra Numerical Methods Programming Fundamentals
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, practical performance, computer-based laboratory activities, seminar presentations, and the successful completion of independent study assignments, including the formulation of mathematical models, analysis of ordinary and partial differential equations, boundary-value problems, collocation and Galerkin methods, shooting methods, finite difference schemes, approximation, stability and convergence analysis, numerical solution of heat conduction, Laplace, Poisson, wave, and convection-diffusion equations, computational error analysis, and

	computer modelling using Python and MATLAB. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as a study of boundary-value problems for ordinary differential equations, an analysis of collocation and Galerkin methods, a comparative study of explicit and implicit difference schemes, an analysis of approximation, stability, and convergence, a numerical modelling project for heat conduction or wave equations, a study of numerical methods for Laplace and Poisson equations, an analysis of convection-diffusion problems, a comparative evaluation of computational algorithms, and a computer modelling project implemented using Python or MATLAB. The final assessment is a written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of differential equations, mathematical modelling, boundary-value problems, numerical methods, finite difference schemes, approximation, stability and convergence, numerical solutions of elliptic, parabolic, and hyperbolic equations, heat conduction, wave, Laplace, Poisson, and convection-diffusion problems, computational errors, and computer modelling using Python and MATLAB, as well as their ability to apply theoretical principles to practical and analytical problems. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Israilov M.I. Computational Methods. Part 1. Tashkent: O'qituvchi, 2003. 450 pp. 2. Israilov M.I. Computational Methods. Part 2. Tashkent: O'qituvchi, 2008. 340 pp. 3. S.S. Irisqulov, K.D. Ismanova, M. Olimov, A. Imomov. Numerical Methods and Algorithms. Namangan Publishing House, 2013. 244 pp. 4. Kalitkin N.N., Alshina E.A. Numerical Methods: in 2 books. Numerical Analysis: Textbook for Higher Professional Education Students. Moscow:

	Akademiya Publishing Centre, 2013. Book 1, 304 pp.; Book 2, 304 pp. 5. Richard L. Burden and J. Douglas Faires. Numerical Analysis. 9th Edition. Youngstown State University Press, 2011. 895 p. 6. L.Ridgway Scott. Numerical Analysis. Princeton University Press, 2011. 342 p. 7. Ismatullayev G.P., Pulatov S.I., Fayazov Q.S. A Guide to Numerical Methods. Tashkent: Universitet, 2006. 140 pp. 8. Bakhvalov N.S., Zhidkov N.P., Kobelkov G.M. Numerical Methods. Moscow: BINOM. Laboratory of Knowledge, 2011. 640 pp. 9. Samarskii A.A., Vabishchevich P.N., Samarskaya E.A. Problems and Exercises in Numerical Methods. Moscow: Librokom, 2009. 208 pp. 10. Samarskii A.A. Introduction to Numerical Methods. Moscow: Lan Publishing House, 2009. 288 pp. 11. Wikipedia. Nonlinear modelling. https://en.wikipedia.org/wiki/Nonlinear_modelling
Responsible Lecturer	Kholliev Fakhriddin Bokhodirovich

2.03. WEB APPLICATION DEVELOPMENT TECHNOLOGIES

Field	Description
Module Title	WEB APPLICATION DEVELOPMENT TECHNOLOGIES
Module Code	VIYAT204
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	VI semester
Module objective	The objective of the Module Web Application Development Technologies is to familiarise postgraduate students with the theoretical foundations and practical skills of web programming. Students are introduced to HTML and CSS. Students are prepared to create web pages and web applications. Core technologies such as JavaScript, PHP, and databases are used for this purpose. Students are developed to the stage of independent project creation. The module also shapes and develops the knowledge, skills, and competencies that a master's student must acquire and apply in their professional field.
Module Learning Outcomes	Knowledge: LO1. Can explain the basic concepts of the World Wide Web, web pages, web servers, client-server technology and web design. LO2. Can explain the basic capabilities of the HTML, CSS, JavaScript, PHP and SQL languages. LO3. Has a theoretical understanding of web security, cookies and session mechanisms. Skills: LO4. Can design simple and responsive web pages using HTML and CSS.

	LO5. Creates interactive elements (dynamic menus, forms, calculators and so on) using JavaScript. LO6. Can build a simple web application in PHP (login/registration, data storage and output). LO7. Performs CRUD operations on a MySQL database (creating tables, adding, modifying and deleting data). LO8. Creates a ready-made website using a CMS (Joomla). LO9. Can implement a simple CRUD project using a framework. Competencies: LO10. Formulates mathematical models of nonlinear processes and real-world systems; analyzes nonlinear functions, algebraic and differential equations, dynamic systems, and equilibrium points; and investigates the qualitative behavior of nonlinear systems. LO11. Applies numerical methods to solve nonlinear algebraic and differential equations; implements computational algorithms and mathematical models using Python and MATLAB; and evaluates the accuracy, stability, convergence, and computational efficiency of numerical solutions. LO12. Analyzes phase trajectories, equilibrium points, and local and global properties of nonlinear systems; applies Lyapunov methods to investigate system stability; and identifies and analyzes bifurcations and qualitative changes in the behavior of dynamic systems. LO13. Independently uses scientific literature, digital resources, and modern computational tools to investigate mathematical modelling problems; develops, implements, and analyzes mathematical models of scientific, engineering, and technical processes; and presents and defends research results and project outcomes individually and as part of a team.
--	---

Content	Lectures Introduction to Web Development. Key Concepts and Stages of WWW Development. The fundamental concepts of web development are introduced. The architecture of the World Wide Web is examined. The major stages of WWW development, from static web pages to dynamic, interactive, and modern web applications, are studied. Web Design. The principles of web design are studied, including page structure and usability. Visual hierarchy, layout, navigation, responsiveness, and user experience are also examined. The relationship between design and technical implementation is considered. Introduction to HTML. Structure of an HTML Document. Header Elements. HTML is introduced as the fundamental markup language of the web. The structure of an HTML document is examined. The use of head elements, metadata, titles, and other header components is also examined. Basic HTML Element Tags. Page Layout Using Tables. The basic HTML elements and tags used to structure web content are studied. The use of tables for organizing tabular information is examined. Creating structured page layouts using tables is also studied. HTML Frame Navigation, Forms and Objects. Navigation structures and HTML forms are studied. Embedded objects are also examined. Students examine how user input is collected and how different types of content can be integrated into web pages. HTML5 New Standards. Tags and Attributes. Working with Graphics, Audio and Video. Visual Effects. The main features of HTML5 are explored, including semantic elements and multimedia integration. Graphics, audio, video, and modern visual effects are also examined. The role of HTML5 in developing interactive web pages is analyzed.
-----------------------	--

	Introduction to CSS. Methods of Linking to HTML. Types of Selectors. Elements and Classes. Units of Measurement. CSS is introduced as a language for controlling the appearance and layout of web pages. Different methods of linking CSS to HTML are studied. Selectors, classes, elements, and measurement units are also studied. JavaScript Fundamentals. Variables, Data Types, Arrays, Operators and Programming Fundamentals. The basic principles of JavaScript programming are introduced. Variables, data types, arrays, and operators are examined. Expressions and fundamental programming concepts are studied as the basis for creating interactive web applications. PHP Basics. Variables, Constants, Data Types, Operators, Functions and Objects. The fundamentals of PHP server-side programming are introduced. Variables, constants, data types, and operators are studied. Functions and object-oriented programming concepts are also studied. PHP Cookies and Sessions. Variable Scope. File Handling and PHP Classes. The mechanisms for maintaining user state in web applications are examined through cookies and sessions. Variable scope and file operations are studied. The use of PHP classes is also studied. Databases and Database Management Systems. The concepts of databases and Database Management Systems are introduced. Database structures, data organization, storage, and management are examined. The role of databases in web applications is also examined. SQL: Structured Query Language for Databases. The fundamental principles of SQL are studied. Commands for creating and retrieving data in relational databases are examined. Commands for modifying and managing data in relational databases are also examined. Application of CMS Technology in Creating Websites. Content Management Systems and their
--	--

	application in website development are studied. The principles of content management and templates are examined. Plugins and website administration are also examined. Web Programming Using Frameworks. The role of modern frameworks in web application development is studied. Framework architecture and reusable components are examined. Routing, database integration, and efficient application development approaches are also examined. Web Application Editor Applications: The Case of Dreamweaver. Visual web development environments, particularly Dreamweaver, are examined for creating and editing web pages. The integration of visual design tools with HTML and CSS is explored. Integration with other web technologies is also explored. Practical lessons Introduction to the Web Application Development Environment. The basic tools used for web application development are studied. Software and environments used for web development are examined. The structure of a typical web development workflow is also examined. Designing the Structure of a Website or Web Application. The structure of a website or web application is planned. Pages and navigation are defined. Content organization and the basic layout of the planned project are also defined. Structure of an HTML Document. A basic HTML document is created. Its main structural components are examined, including the document type declaration. The html, head, and body elements are also examined. Formatting Text in an HTML Document. Textual content is formatted using HTML elements. Headings, paragraphs, and emphasis are applied. Line breaks and other basic text formatting techniques are also applied.
--	--

	Graphics and Multimedia on a Web Page. Images and multimedia content are added to web pages. The use of graphical elements is covered. The basic integration of visual and multimedia resources is also covered. Linking HTML Documents. Hyperlinks and navigation between HTML documents are examined. Internal links are created. External links are created, and navigation within a website is organized. Lists in HTML Documents. Ordered lists are created and formatted. Unordered and nested lists are also created and formatted. Different types of lists are used to organize and present structured information on web pages. Tables in HTML Documents. HTML tables are created and formatted. Structured data is organized using rows and columns. Headers and other table elements are also used to organize structured data. Frames. The principles of dividing a web page into separate frames are examined. Frame-based page organization is explored. Navigation between different content areas is also explored. Working with HTML5 Semantic Tags and Multimedia. Semantic HTML5 elements, including header, nav, section, and footer, are used. Video and audio are embedded into web pages. Simple graphics are created using the Canvas element. CSS Fundamentals and Basic Page Layout. Inline, internal, and external CSS are introduced. Selectors and classes are applied. Identifiers are applied to create and style a simple web page layout. JavaScript Basics and Interactive Web Elements. A simple calculator is developed using JavaScript. Form data validation is practiced. Dynamic effects, such as changing text and colours, are also implemented. PHP Basics and Form Data Processing. The use of PHP code within HTML documents is introduced. Variables and data types are used in PHP. Simple
--	--

	form data is processed by receiving and displaying user input. MySQL Database and PHP Connection. A database table is created in MySQL. Basic CRUD operations are performed using phpMyAdmin. A connection between the MySQL database and PHP application is also established. Working with Web Editors: Dreamweaver. The Dreamweaver workspace and its tools for web development are introduced. A simple web page is created. The completed website is uploaded to a server using FTP. Self-study HTML5 and CSS3 Technologies: Concepts and Capabilities. The fundamental concepts and capabilities of HTML5 and CSS3 are examined. Semantic elements, multimedia support, and advanced styling are analyzed. Responsive layouts, animations, and modern web design features are also analyzed. Browser Rendering Mechanisms. How web browsers process and display web pages is investigated. The stages of parsing HTML and CSS, and constructing the DOM and CSSOM, are examined. Creating the render tree, layout, and painting are also examined. Client-Server Architecture in Web Applications. The principles of client-server architecture are studied. The interaction between browsers, web servers, databases, and application logic is examined. The flow of requests and responses in web applications is also analyzed. JavaScript: History and Application Areas. The development of JavaScript is reviewed. Its evolution from a browser scripting language to a widely used programming technology is examined. Its applications in front-end development, back-end systems, mobile applications, and interactive web services are considered.
--	--

Using Open-Source Resources (W3Schools, MDN, GitHub). The effective use of open-source educational and development resources is studied. Documentation and code examples are explored. Repositories and collaborative tools for independent web development are also explored.

Responsive Design and Bootstrap. The principles of responsive web design are examined, including flexible layouts and media queries. Mobile-first development is also examined. Bootstrap components and grid systems are studied as tools for creating adaptable web interfaces.

Web Security (XSS, CSRF, SQL Injection). Common web application vulnerabilities are investigated, including Cross-Site Scripting and Cross-Site Request Forgery. SQL Injection is also investigated. Basic principles of secure web development and protection mechanisms are discussed.

Modern Front-End Frameworks (React, Angular, Vue). A comparative overview of modern front-end frameworks, including React, Angular, and Vue, is provided. Their component-based architectures and development approaches are analyzed. Their advantages, limitations, and typical application areas are also analyzed.

Agile Methodology in Web Application Development. The principles of Agile software development are examined. Its application to web projects is studied. Iterative development, Scrum, teamwork, task planning, continuous feedback, and incremental delivery are considered.

Development of a Personal Portfolio Web Page Using HTML and CSS. A personal portfolio web page is developed that presents information about skills, projects, education, and achievements. The project emphasizes semantic HTML structure. CSS styling, layout, and visual presentation are also emphasized.

JavaScript Interactivity and Form Validation. Interactive behavior is added to web pages using

	JavaScript. Form validation and event handling are implemented. Dynamic content updates and user interaction are also implemented and analyzed. Responsive Web Page Development Using Bootstrap. A responsive web page is created using the Bootstrap framework. Grid systems and responsive components are applied. Navigation elements and predefined interface components are applied to ensure compatibility with different screen sizes. Mini-Project: Online Book Catalogue (Front-End). A front-end prototype of an online book catalogue is developed. The project may include book categories, search and filtering functions, and book cards. Navigation and a responsive user interface are also included. Development of an Electronic Textbook Web Application and Deployment on GitHub. A web-based electronic textbook is created. Educational content is organized, and the interface is developed. The completed application is deployed using GitHub-based development and hosting tools. Deployment of a Web Application to a Hosting Environment. The process of deploying a web application to a hosting environment is examined. Domain and hosting concepts, and server configuration, are studied. File transfer, application testing, and basic deployment procedures are also studied.
Teaching Methods	The module is delivered through interactive lectures, problem-based learning, practical exercises, computer-based activities, and project-based learning. Lectures introduce the fundamental concepts of web development, including HTML, CSS, JavaScript, PHP, databases, SQL, web frameworks, client-server architecture, and web security. Practical exercises focus on developing web pages and applications, creating responsive interfaces, implementing JavaScript functionality, processing data with PHP, connecting applications to MySQL

	databases, and deploying web applications. Project-based learning and independent study develop students' analytical, technical, creative, and problem-solving skills through web development projects, presentations, reports, and mini-projects. Modern programming tools, open-source resources, frameworks, version-control platforms, and hosting services support independent learning and practical application of web technologies.
Prerequisites	Database Database Management Systems Programming 1,2 PHP Programming Language
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, practical performance, computer-based laboratory activities, seminar presentations, and the successful completion of independent study assignments, including HTML and CSS web page development, responsive design, JavaScript programming and interactivity, form validation, PHP programming, database creation and SQL operations, client-server architecture, web security, modern front-end frameworks, and web application deployment. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as an analysis of HTML5 and CSS3 technologies, a study of browser rendering mechanisms, a comparative analysis of modern front-end frameworks, a research project on web security vulnerabilities, a responsive web design project using Bootstrap, a personal portfolio website, a JavaScript interactivity project, or a mini-project for developing an online book catalogue or electronic textbook web application. The final assessment is a

	written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of web development fundamentals, HTML, CSS, JavaScript, PHP, databases, SQL, client-server architecture, web security, responsive design, CMS technologies, modern web frameworks, and web application deployment, as well as their ability to apply theoretical principles to practical and analytical problems. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Duckett J. HTML & CSS: Design and Build Websites. Indianapolis: John Wiley & Sons, Inc., 2011. 514 p. ISBN: 978-1-118-00818-8. 2. Jennifer Niederst Robbins. Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics. 5th Edition. O'Reilly Media, 2018. 810 p. 3. X.N. Zaynidinov, E.Sh. Nazirova, D.S. Yaxshibayev, S.U. Maxmudjanov. Creating Web Applications. Textbook. Tashkent: Aloqachi, 2020. 348 pp. 4. M. Aripov, S. Dottoyev, M. Fayziyeva. WEB Technologies: A Textbook. Tashkent: National Society of Philosophers of Uzbekistan Publishing House, 2013. 280 pp. 5. I.N. Vasilyeva, D.Yu. Fedorov. Web Technologies: A Textbook. St. Petersburg: SPbGUE Publishing House, 2014. 67 pp. 6. V.I. Sukhanov. Development of Web Applications on the Spring Platform: A Teaching and Methodological Manual. Yekaterinburg: Publishing House of Ural University, 2023. 180 pp. 7. A.V. Kirichenko, A.P. Nikolsky, E.V. Dubovik. Web in Practice: CSS, HTML, JavaScript, MySQL, PHP for Full-Stack Developers. St. Petersburg: Nauka i Tekhnika, 2021. 432 pp.

	8. D. Yatubovich, E.S. Yeropova, I.A. Yeropov. Fundamentals of Web Development: A Teaching and Methodological Manual for Conducting Laboratory Sessions. Vladimir State University. Vladimir: Sherlock-Press Publishing House, 2017. 102 pp. 9. O. Jakbarov, U. G'oyipov. Web Programming: A Textbook. Namangan: Iste'dod ziyo press, 2023. 200 pp. 10. Web Application Development: Key Stages. http://www.purrweb.com/ru/blog/kak-sozdat-veb-prilozhenie/ 11. ClevaDesk Designer. http://clevadesk.com/ru/clevadeskdesigner/ 12. https://tproger.ru/articles/10-luchshih-platform-dlya-sozdaniya-prilozhenij
Responsible Lecturer	Zaripova Mukaddas Djumayozovna

2.04. PROMPT ENGINEERING

Field	Description
Module Title	PROMPT ENGINEERING
Module Code	PRI204
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Prompt Engineering is to develop theoretical knowledge and practical skills in creating effective prompts for generative AI and large language models. Students learn prompt design. Testing and optimization of prompts are studied. Responsible application of prompts in research is developed. Prompts are also applied in writing and data analysis. Professional activities using prompt engineering are also covered.
Module Learning Outcomes	Knowledge: LO1. Explain key concepts, principles and contemporary development trends in prompt engineering. LO2. Describe working principles of generative AI and large language models (LLMs). LO3. Explain components of prompts and requirements for creating effective prompts. LO4. Differentiate between zero-shot, one-shot and few-shot prompting methods. LO5. Explain theoretical foundations of Chain-of-Thought, RAG and agent-based approaches. LO6. Describe safety, academic integrity and ethical standards in prompt engineering. Skills:

	LO7. Create clear, effective and logical prompts for various tasks and objectives. LO8. Apply zero-shot, one-shot, few-shot and chain-of-thought prompting methods. LO9. Develop prompts for scientific articles, annotations, reports and academic text editing. LO10. Create prompts for data analysis and interpretation of tables and charts. LO11. Use prompts to create, analyze and debug software code. LO12. Test, evaluate, optimize and refine prompt results. Competencies: LO13. Use generative AI tools responsibly in scientific and professional activities. LO14. Critically evaluate AI results and verify facts from AI-generated content. LO15. Apply prompt engineering in information systems and software projects. LO16. Independently explore new AI platforms and prompting technologies.
Content	Lectures Introduction to Prompt Engineering. The concept and definition of prompts, application areas, and benefits are introduced. Prompt engineering in modern AI systems and the importance of effective prompts are examined. The difference between good and poor prompts and their impact on AI model outputs are analyzed. Fundamentals of Generative AI and Large Language Models. Neural network foundations and transformer architecture basics are studied. How LLMs learn and generate text is examined, including token-level processing. Probability distributions in language generation and the limitations of language models are analyzed.

Prompt Structure and Principles for Creating Effective Prompts. Essential prompt components, including instructions, context, and examples, are studied. Prompt engineering best practices, clarity, and specificity requirements are examined. Avoiding ambiguity, structuring complex tasks, and testing prompt effectiveness are analyzed.

Types of Prompts and Their Practical Applications. Instructional, descriptive, and conversational prompts are studied. Creative versus factual prompts, context-setting prompts, and constrained prompts are examined. Prompts for different AI models and domain-specific prompt types are analyzed.

Role-Based Prompts and Providing Context. Role definition in prompts and expert persona prompts are studied. Adopting specific perspectives and context window management are examined. Persona consistency and building on established roles are analyzed.

Zero-Shot, One-Shot and Few-Shot Prompting. The zero-shot learning concept and one-shot and few-shot learning principles are studied. Example selection strategies and the quality of examples are examined. Task generalization and performance improvements with examples are analyzed.

Chain-of-Thought and Step-by-Step Prompts. Chain-of-thought reasoning is studied. Breaking complex problems into steps and intermediate reasoning steps are examined. Verification within prompts and debugging with step-by-step approaches are analyzed.

Using Prompts in Scientific Research. Literature review automation and research question formulation are studied. Hypothesis generation and methodology assistance are examined. Statistical analysis help and citation and reference management are analyzed.

Prompt Engineering in Academic Writing. Essay and paper structure, academic tone, and style are studied. Argument development assistance and

	literature integration are examined. Paraphrasing, summarization, and citation formatting help are analyzed. Preventing Plagiarism and Maintaining Academic Integrity. Plagiarism detection assistance and authentic paraphrasing are studied. Proper attribution in prompts and transparency about AI use are examined. Original thought encouragement and academic honesty principles are analyzed. Using Prompts in Data Analysis and Programming. Data interpretation tasks and statistical summary requests are studied. Code generation prompts and debugging assistance are examined. Algorithm explanation and documentation generation are analyzed. Prompt Engineering for Documents and RAG Approach. The Retrieval-Augmented Generation concept and document context integration are studied. Question-answering on documents and multi-document processing are examined. Source attribution and fact-grounding in prompts are analyzed. Working with Multimodal Prompts and Formats. Image analysis prompts and chart and graph interpretation are studied. Audio and video description and mixed modality handling are examined. Format conversion tasks and cross-modal understanding are analyzed. Methods for Evaluating, Testing and Improving Prompts. Prompt evaluation metrics and A/B testing prompts are studied. Consistency checking and performance benchmarking are examined. Iterative refinement strategies and feedback analysis are analyzed. Safety, Ethical Principles and Development Prospects. Bias in prompts and outputs and safety guidelines are studied. Responsible AI use and copyright and intellectual property considerations are
--	--

	examined. Misinformation risks and the future of prompt engineering are analyzed. Practical lessons Methods for Constructing and Designing Effective Prompts. Prompt planning and outlining are practiced. Component identification and structure building are implemented. Initial prompts are tested, and refinement techniques are applied. Creating Simple, Contextual and Role-Based Prompts. Basic prompts are written. Context information is added, and roles are defined. Multiple prompt types are combined, and effectiveness is verified. Using the Zero-Shot Prompting Method. Prompts are designed without examples using a direct instruction approach. Performance is evaluated. Modification strategies are applied, and domain application is practiced. Using the One-Shot Prompting Method. A single example is integrated into prompts. Example quality selection is practiced. The transition to few-shot prompting is explored, and consistency and output improvement are evaluated. Using the Few-Shot Prompting Method. Multiple examples are prepared. Example ordering strategies and diversity in examples are practiced. Pattern recognition and prompt optimization are applied. Crafting Chain-of-Thought and Step-by-Step Prompts. Problems are decomposed into steps. Step descriptions and intermediate outputs are prepared. Verification steps and debugging with reasoning paths are practiced. Using Prompts for Scientific Research Topic Formulation. Research questions are designed using prompts. Aims and objectives are clarified, and hypotheses are suggested. Literature gaps are identified, and significance is articulated.
--	--

	Using Prompts in Literature Reviews. Source recommendation prompts are practiced. Literature synthesis and thematic organization are applied. Gap analysis and critical evaluation prompts are used. Creating Prompts for Academic Writing and Editing. Essay outlines are generated using prompts. Paragraph structure improvement and argument strengthening are practiced. Writing clarity enhancement and citation integration are applied. Using Prompts to Prevent Plagiarism. Paraphrasing instruction prompts are practiced. Originality verification and citation formatting are applied. Acknowledgment generation and an academic integrity checklist are used. Crafting Prompts for Data Analysis. Data interpretation requests and statistical summary prompts are practiced. Trend identification and anomaly detection are applied. Pattern explanation prompts are also used. Using Prompts in Programming and Coding. Code generation prompts and debugging requests are practiced. Algorithm optimization is applied. Comment and documentation generation and code review assistance are used. Crafting Prompts Based on Documents and RAG. Document context integration is practiced. Fact-based question answering and information extraction are applied. Source attribution and accuracy verification are used. Multimodal Prompts: Images, Tables, Diagrams. Image description requests and graph interpretation are practiced. Chart analysis and diagram explanation are applied. Format conversion prompts are also used. Evaluating, Comparing and Improving Prompts. Output quality is assessed, and comparative evaluation is practiced. Performance metrics are applied. Iterative improvement, documentation, and reporting are completed.
--	--

	Self-study Theoretical Foundations and Modern Trends in Prompt Engineering. The history and development of prompt engineering are studied. The current state and research directions are examined. Key publications, breakthroughs, emerging applications, and future technologies are analyzed. Factors Affecting Prompt Quality with LLMs. Input characteristics and model-specific considerations are studied. Token limitations, temperature, and parameters are examined. Context window effects and instruction sensitivity are analyzed. Comparative Analysis of Prompting Methods. Zero-shot versus few-shot trade-offs are studied. Performance metrics comparison and efficiency analysis are examined. Application scenarios and method selection criteria are analyzed. Chain-of-Thought and Complex Problem Solving. Reasoning chain construction is studied. The importance of intermediate steps and error analysis in reasoning are examined. Verification mechanisms and task complexity assessment are analyzed. Role-Based and Expert Prompting Methodology. Persona development and expertise simulation are studied. Consistency maintenance and context preservation are examined. Advanced role combinations are analyzed. Scientific Research Applications of Prompts. Research design assistance and methodology selection are studied. Literature automation and data interpretation are examined. Publication preparation is also analyzed. Prompts for Literature Reviews and Sources. Systematic review prompts and citation management are studied. Source evaluation and synthesis strategies are examined. Quality assessment is also analyzed. Academic Writing Enhancement with AI. Quality improvement strategies and style consistency are
--	--

	studied. Argument development and clarity enhancement are examined. Structure optimization is also analyzed. Plagiarism Prevention and Academic Integrity. Ethical guidelines and original contribution emphasis are studied. Attribution practices and transparency requirements are examined. Integrity verification is also analyzed. RAG and Document-Based Prompting. Document preprocessing and context retrieval optimization are studied. Fact-grounding techniques and source tracking are examined. Accuracy assurance is also analyzed. Prompt Evaluation and Quality Metrics. Performance measurement and reliability assessment are studied. Consistency checking and bias identification are examined. Quality standards are also analyzed. Prompt Optimization and Refinement. Iterative improvement strategies and A/B testing approaches are studied. Performance analysis and parameter tuning are examined. Best practice development is also analyzed. Data Analysis and Statistical Prompting. Exploratory analysis requests and statistical inference are studied. Visualization requests and interpretation assistance are examined. Error identification is also analyzed. Programming and Code-Related Prompts. Code generation quality and debugging efficiency are studied. Documentation automation and optimization techniques are examined. Testing assistance is also analyzed. Safety, Ethics and Responsible AI. Bias mitigation and harmful content prevention are studied. Privacy protection and copyright compliance are examined. Responsible deployment is also analyzed.
Teaching Methods	The module is delivered through interactive lectures with real-world case studies and hands-on practical

	sessions with AI tools. Prompt creation and testing exercises are conducted. Peer review and collaboration are incorporated. Independent exploration of AI platforms and research-based learning activities are included. Critical analysis of AI outputs is practiced. Portfolio development with prompt samples is also required.
Prerequisites	Artificial Intelligence and Neural Networks Design of Information Systems Scientific Research Methodology
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation in practical sessions, prompt creation ability during hands-on work, and completion of teacher-guided independent work assignments, including prompt portfolios, analysis reports, academic writing samples, data analysis requests, code generation examples, and mini-projects on various prompting applications. The midterm assessment is a written test covering theoretical knowledge of prompt engineering concepts, LLM principles, prompting methods, the RAG approach, and evaluation criteria, together with independent work projects evaluating practical skills in prompt design, optimization, and application to real problems. The final examination is a written examination with problems requiring students to design appropriate prompts, evaluate outputs, improve prompt effectiveness, and explain design choices. The assessment measures theoretical understanding, practical prompting skills, and the ability to apply prompt engineering to complex tasks.
Recommended Literature	1. Phoenix James, Taylor Mike. Prompt Engineering for GenAI: Patterns for Reliable Prompts. Sprint Book, 2025.

	2. Hernández Gutiérrez J.A., et al. ChatGPT: Learning Prompt Engineering with 100+ Examples. Amazon KDP, 2024. 3. Berryman J., Ziegler A. Prompt Engineering for LLMs: The Art and Science. O'Reilly Media, 2025. 4. Sharma V., Thakur D.S. Prompt Engineering. MATS Centre, 2024.
Responsible Lecturer	Zaripova Mukaddas Djumayozovna, PhD, Associate Professor, Department of Computer and Software Engineering, Termez State University

2.04. MACHINE LEARNING

Field	Description
Module Title	MACHINE LEARNING
Module Code	MSHO'204
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Machine Learning is to develop theoretical knowledge and practical skills in machine learning for solving applied problems. Students learn data processing techniques. Supervised and unsupervised learning methods are studied. Neural networks and deep learning are covered. Model evaluation is performed using Python, Scikit-Learn, and TensorFlow/Keras. These methods are applied to real-world problems in economics, medicine, agriculture, and other domains.
Module Learning Outcomes	Knowledge: LO1. Explain fundamental concepts, developmental stages and contemporary trends in machine learning. LO2. Describe processes of training data preparation, features, feature engineering and model evaluation. LO3. Distinguish theoretical foundations of supervised, unsupervised and reinforcement learning algorithms. LO4. Describe principles of regression, classification, clustering, PCA, regularization and neural networks. LO5. Explain metrics for evaluating model performance and issues of overfitting and underfitting. LO6. Describe ethical issues, data security and model interpretability in machine learning.

	Skills: LO7. Create and test machine learning models using Python and Scikit-Learn libraries. LO8. Perform data cleaning, processing and visual analysis using Pandas, NumPy and Matplotlib. LO9. Apply linear regression, logistic regression, decision trees, Random Forest, SVM and K-means to problems. LO10. Build neural network, CNN and RNN/LSTM models using TensorFlow/Keras. LO11. Evaluate model performance using precision, recall, F1-score and other metrics. LO12. Create complete ML projects based on datasets, analyze results and prepare reports. Competencies: LO13. Use machine learning technologies responsibly in scientific and professional projects. LO14. Critically evaluate ML results, identify error sources and propose model improvements. LO15. Select appropriate ML approaches for problems in economics, medicine, agriculture and cybersecurity. LO16. Independently explore new ML platforms, AutoML and model interpretation technologies.
Content	Lectures Concept, History and Main Directions of Machine Learning. Machine learning is defined and distinguished from artificial intelligence. Historical development and key milestones are examined. Supervised, unsupervised, and reinforcement learning are introduced, along with current applications and future trends across industries. Key Concepts in ML: Training Data, Features and Feature Engineering. Training and test data concepts are studied. Feature definition, importance, and data quality and preprocessing are examined. Feature extraction, selection, dimensionality

	reduction, and their impact on model performance are analyzed. Statistical Approaches and Probability Foundations in ML. Probability theory basics for machine learning are studied. Bayesian methods, inference, and statistical distributions are examined. Maximum likelihood estimation, hypothesis testing, and uncertainty quantification are analyzed. Supervised Learning Key Concepts. The supervised learning paradigm and input-output mapping are studied. Loss functions, objectives, and training procedures are examined. Generalization, overfitting, and model validation strategies are analyzed. Linear and Logistic Regression Models. Linear regression principles, mathematics, and the normal equation are studied. Gradient descent and the interpretation of coefficients are examined. Logistic regression for classification, decision boundaries, and probabilistic interpretation are analyzed. Decision Trees and Ensembles. Tree structure, splitting criteria, pruning, and complexity control are studied. Information gain and the Gini index are examined. Random Forest, Gradient Boosting, bagging, and bootstrap aggregation are analyzed. Classification Methods: k-NN, Naive Bayes, SVM. The k-Nearest Neighbors algorithm and distance metrics are studied. The Naive Bayes probabilistic approach is examined. Support Vector Machines, hyperplane optimization, soft margins, and regularization are analyzed. Unsupervised Learning: Clustering and Dimensionality Reduction. Clustering concepts, objectives, and the K-means algorithm are studied. Hierarchical clustering methods and Principal Component Analysis are examined. Latent factor models and evaluation metrics for clustering are analyzed.
--	--

	Overfitting and Regularization Methods. Overfitting definition, causes, and the bias-variance tradeoff are studied. L1 and L2 regularization, early stopping, and cross-validation are examined. Dropout, data augmentation, and regularization parameter selection are analyzed. Fundamentals of Neural Networks and Deep Learning. The artificial neuron and activation functions are studied. Multilayer perceptrons, backpropagation, and gradient descent variants are examined. Optimization algorithms such as Adam and RMSprop, the universal approximation theorem, and computational complexity are analyzed. Convolutional Neural Networks and Computer Vision. CNN architecture, including convolution, pooling, and flattening, is studied. Feature map extraction and visualization are examined. Classical architectures such as LeNet, AlexNet, and ResNet, transfer learning, and object detection and segmentation basics are analyzed. Recurrent Neural Networks, LSTM and Time Series Modeling. RNN architecture, sequences, and the vanishing gradient problem are studied. LSTM cells, gates, and GRU simplifications are examined. Sequence-to-sequence models, attention mechanisms, and time series prediction applications are analyzed. Reinforcement Learning in Planning and Decision-Making. Markov decision processes and value and policy functions are studied. The Q-learning algorithm and policy gradient methods are examined. The exploration-exploitation tradeoff and applications in games and control are analyzed. Ethical Issues and Data Security in ML. Algorithmic bias and fairness are studied. Privacy preservation and differential privacy are examined. Model interpretability methods such as SHAP and LIME, data governance, and responsible AI principles are analyzed.
--	---

	Machine Learning in Practice: Economics, Medicine and Agriculture. Machine learning applications in finance, including credit risk and fraud detection, are studied. Medical diagnosis and treatment planning are examined. Agricultural yield and pest prediction, transportation optimization, and manufacturing quality control are analyzed. Practical lessons Introduction to Python and Scikit-Learn Library. Python setup and environment configuration are practiced. Jupyter notebooks for interactive coding are used. The Scikit-Learn library, basic data structures, and module import procedures are explored. Data Cleaning and Visualization. Data loading and inspection are practiced. Missing values are handled, and outliers are detected and treated. Data transformation, scaling, and exploratory data analysis with Matplotlib are performed. Data Partitioning and Model Evaluation. The train-test split procedure and cross-validation techniques are practiced. Performance metrics for regression and classification are applied. Confusion matrices, ROC curves, and hyperparameter tuning methods are used. Linear Regression Practice: Predicting House Prices. House price prediction is formulated as a problem, and data exploration is performed. Feature engineering for houses is practiced. Model training, evaluation, residual analysis, and prediction on new data are performed. Logistic Regression Application: Disease Diagnosis. A medical dataset is prepared for disease diagnosis. Class imbalance handling is practiced. Model training, prediction, performance evaluation, and clinical significance assessment are performed. Decision Trees for Data Classification. Decision trees are created and visualized. Hyperparameter selection and handling of imbalanced classes are
--	---

	practiced. Feature importance analysis and comparison with other methods are performed. Improving Results Using Random Forest. Random Forest ensembles are created and trained. Out-of-bag error estimation is practiced. Feature importance from trees, parameter optimization, and computational efficiency are analyzed. Classification Using SVM with Iris Dataset. SVM classification is performed on the Iris dataset. Linear, RBF, and polynomial kernels are selected and compared. SVM training, hyperplane visualization, and multi-class extension are practiced. Clustering: K-means and Hierarchical Methods. K-means initialization and convergence are practiced. The optimal number of clusters is selected. Dendrogram interpretation and agglomerative clustering approaches are examined. Dimensionality Reduction Using PCA. The covariance matrix is calculated for PCA. Eigenvalue decomposition and variance explained by components are practiced. Visualization in reduced space and reconstruction error analysis are performed. Creating a Recommendation System. A collaborative filtering approach is implemented for a recommendation system. Matrix factorization and user-item similarity computation are practiced. Recommendations are generated, and evaluation metrics are applied. Model Evaluation and Optimization. Comprehensive metrics are calculated for model evaluation. Parameter tuning strategies and ensemble methods combination are practiced. Performance analysis and bottleneck identification are performed. Working with Big Data. Large dataset handling techniques are practiced. Sampling and approximation methods are applied. Distributed
--	--

	computing concepts and memory and computation optimization are examined. Fundamentals of Speech Recognition. Audio signal processing is practiced. Feature extraction from speech and acoustic models are examined. Language model integration and end-to-end approaches are analyzed. Object Detection in Computer Vision. Bounding box prediction is practiced for object detection. Region-based methods, including R-CNN variants, are examined. Single-shot detectors such as YOLO and SSD, performance evaluation, and transfer learning applications are analyzed. Self-study Key Discoveries in ML History and ML vs AI Distinction. The historical timeline and milestones of machine learning are studied. Machine learning evolution and its distinction from traditional AI are examined. The current state of the art and future research directions are analyzed. Training Data Quality, Feature Engineering and Transfer Learning. Data collection and labeling strategies are studied. Feature importance analysis and domain adaptation approaches are examined. Pre-trained models and fine-tuning techniques are analyzed. Big Data and ML Integration and Ensemble Methods. Scalable ML algorithms and distributed computing frameworks are studied. Ensemble learning theory is examined. Boosting, bagging, stacking, and blending methods are analyzed. Clustering Methods and PCA/Dimensionality Reduction. Partitional and hierarchical clustering are studied. Density-based approaches and dimensionality reduction techniques are examined. Manifold learning and visualization methods are analyzed. Gradient Descent Algorithm Variants. Batch, stochastic, and mini-batch gradient descent are
--	--

	studied. Momentum and acceleration are examined. Adaptive learning rates, convergence analysis, and parameter tuning are analyzed. Activation Functions, Dropout and Batch Normalization. Sigmoid, ReLU, and other activation function variants are studied. Non-linearity in networks and dropout as regularization are examined. Batch normalization and layer normalization approaches are analyzed. CNN Architectures: LeNet, AlexNet, ResNet. The evolution of CNN design is studied. Skip connections, residual learning, and bottleneck blocks are examined. Architecture comparison, benchmarks, and modern improvements are analyzed. NLP Fundamentals and Sentiment Analysis Methods. Text preprocessing and tokenization are studied. Word embeddings such as Word2Vec and GloVe are examined. Sentiment classification approaches, attention for NLP, and transformer models are analyzed. Reinforcement Learning Algorithms: Q-learning and SARSA. Bellman equations are studied. Value iteration and policy iteration are examined. Temporal difference learning, the SARSA on-policy approach, and function approximation in reinforcement learning are analyzed. ML Applications in Medicine and Agriculture. Disease diagnosis and prognosis systems are studied. Treatment outcome prediction is examined. Crop yield forecasting, pest and disease detection, and resource optimization are analyzed. ML in Finance, Marketing and Energy Sectors. Credit risk assessment and customer churn prediction are studied. Fraud detection systems are examined. Demand forecasting and price optimization models are analyzed. Ethical Issues in Transport, Games and ML. Autonomous vehicle decision-making is studied. Game AI and player engagement are examined.
--	---

	Fairness in algorithmic decisions, privacy in personalization, and transparency and explainability are analyzed. Model Interpretation: SHAP and LIME Methods. SHAP value calculation and interpretation are studied. Local explanations with LIME are examined. Feature contribution analysis, model-agnostic explanations, and debugging of ML models are analyzed. AutoML, IoT, Cybersecurity and Edge Computing. Automated machine learning systems and neural architecture search are studied. ML deployment on edge devices is examined. Resource-constrained deployment and cybersecurity threat detection are analyzed. Future Trends in Machine Learning. Federated learning and privacy-preserving ML are studied. Continual learning and catastrophic forgetting are examined. Few-shot learning, zero-shot learning, and quantum machine learning prospects are analyzed.
Teaching Methods	The module is delivered through interactive lectures with case studies and real-world examples. Hands-on practical sessions with code implementation are conducted. Computational experiments, model training, code review, and optimization practice are included. Research-based learning with open datasets and independent project development with presentations are incorporated. Python ecosystem tools, including Jupyter, scikit-learn, and TensorFlow, are used. Collaborative problem-solving and peer feedback are also practiced.
Prerequisites	Data Structures and Algorithms Fundamentals of Databases Fundamentals of Artificial Intelligence
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and

	a final written examination (50%). The continuous assessment evaluates classroom participation in practical sessions, coding ability during hands-on ML projects, and completion of teacher-guided independent work assignments, including data analysis reports, model implementation code, experimental results, optimization studies, and mini-projects on ML applications. The midterm assessment is a written test covering theoretical knowledge of machine learning concepts, algorithms, data preprocessing, supervised and unsupervised learning, neural networks, and evaluation metrics, together with independent work projects evaluating practical skills in model development, performance analysis, and hyperparameter tuning. The final examination is a written examination with problems requiring students to design ML solutions, implement algorithms, analyze datasets, optimize models, and explain design decisions. The assessment measures theoretical understanding, algorithmic thinking, and the ability to apply machine learning to real-world problems.
Recommended Literature	1. Hastie T., Tibshirani R., James G. An Introduction to Statistical Learning. Springer, 2013. 2. Goodfellow I., Bengio Y., Courville A. Deep Learning. MIT Press, 2016. 3. Géron A. Hands-On Machine Learning with Scikit-Learn and TensorFlow. O'Reilly, 2019. 4. Murphy K.P. Probabilistic Machine Learning: An Introduction. MIT Press, 2022.
Responsible Lecturer	Jurayev Olim Albayevich, Department of Computer and Software Engineering, Termez State University

2.05. MATHEMATICAL MODELLING OF NONLINEAR PROCESSES

Field	Description
Module Title	MATHEMATICAL MODELLING OF NONLINEAR PROCESSES
Module Code	CHJMM204
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Mathematical Modelling of Nonlinear Processes is to cover the theoretical foundations of mathematical modelling of nonlinear processes, methods for constructing models, and issues of analysis and computer-based investigation. The course focuses on the mathematical representation of nonlinear processes encountered in real-world technical, economic, physical, and biological systems. The dynamic characteristics of these processes are studied, and they are modelled using modern software tools. Particular attention is paid to the application of nonlinear differential equations, stability theory, and bifurcation phenomena to the solution of practical problems. Optimisation and numerical methods are also applied to these problems. Master's students develop knowledge, skills, and competencies in creating mathematical models of complex dynamic systems, analysing them, and investigating them through computer simulation.
Module Learning Outcomes	Knowledge: LO1. Knowledge of the theoretical foundations, principles and application areas of mathematical modelling of nonlinear processes.

	LO2. Understands nonlinear differential equations, dynamic systems and the methods for constructing their mathematical models. LO3. Is familiar with the fundamental concepts of stability, equilibrium states, bifurcations and chaotic processes in nonlinear systems. LO4. Is familiar with the theoretical foundations of methods for analysing mathematical models, numerical solution and computer simulation. LO5. Is familiar with the capabilities and tools for modelling and analysing nonlinear processes in MATLAB, Python and other modern computational environments. Skills: LO6. They will acquire the skill to analyse real processes and systems, construct their nonlinear mathematical models and determine their parameters. LO7. They can solve ordinary differential equations and dynamical systems using analytical and numerical methods, and interpret the results. LO8. Can implement mathematical models on a computer and conduct simulation experiments using MATLAB, Python or other modern software tools. LO9. Has the skills to analyse and evaluate the stability, dynamic behaviour and control characteristics of nonlinear systems. LO10. Can visualise modelling results, analyse them, draw scientifically grounded conclusions and apply them to solving practical problems. Competencies: LO11. They will have the competence to mathematically model complex nonlinear processes and systems encountered in various fields and to develop appropriate models for them. LO12. They can carry out professional activities involving the analysis, forecasting and optimisation of
--	---

	the dynamic characteristics of processes based on mathematical models. LO13. Using modern computing technologies and software packages, they can independently organise modelling, simulation and numerical experiments. LO14. They will have the competence to solve practical problems arising in technical, economic, ecological and information systems using mathematical modelling methods. LO15. Can analyse the obtained results on a scientific basis, present them in graphical, tabular and report form, and apply them in decision-making processes. LO16. Using interdisciplinary knowledge, they can develop and implement modelling projects integrated with artificial intelligence, data analysis and digital technologies. LO17. They will have the competencies to conduct research in the field of mathematical modelling, explore new methods and apply innovative approaches in their professional practice.
Content	Lectures Introduction to the Mathematical Modelling of Nonlinear Processes. Application Areas in Computer Engineering. The fundamental concepts and objectives of mathematical modelling of nonlinear processes are introduced. The principles of this modelling are studied. Particular attention is given to the application of nonlinear models in computer engineering, software systems, computational technologies, and system performance analysis. Linear and Nonlinear Models. The Role of Mathematical Modelling in Software Systems. The differences between linear and nonlinear mathematical models are examined. Their structural properties and application areas are studied. The role of mathematical modelling in the analysis, design, optimization, and prediction of software systems is also discussed.

	Nonlinear Functions and Their Analysis Using Computer Graphics. Nonlinear functions are investigated using analytical and graphical methods. The behavior of functions, including their growth and extrema, is examined. Asymptotic properties and parameter dependence are also examined through computer-based visualization. Algorithm Complexity and Nonlinear Dependencies. The relationship between algorithmic complexity and nonlinear dependencies is studied. Mathematical models are used to analyze the growth of computational costs. The performance of algorithms under different input conditions is also analyzed. Nonlinear Algebraic Equations and Their Modelling. The formulation and analysis of nonlinear algebraic equations are studied as mathematical models of real-world and computational processes. The existence of solutions is examined. Nonlinear dependencies and modelling approaches are also examined. Numerical Algorithms for Solving Nonlinear Equations. Numerical methods for solving nonlinear equations are examined. Iterative approaches such as the bisection, Newton, and secant methods are studied. Their convergence, accuracy, and computational efficiency are compared. Modelling of Iterative Processes and Recursive Algorithms. The mathematical modelling of iterative and recursive computational processes is explored. Recurrence relations and convergence behavior are analyzed. Fixed-point processes and the computational characteristics of recursive algorithms are also analyzed. Nonlinear Systems of Equations and Their Computer Solution. Methods for formulating systems of nonlinear algebraic equations are studied. Methods for solving these systems are also studied. Numerical algorithms and computational approaches
--	--

	for obtaining approximate solutions are analyzed and implemented. Modelling Traffic and Load Processes in Computer Networks. Mathematical models of network traffic and computational load are studied. Data transmission rates and congestion are analyzed. Resource utilization and changing network conditions are analyzed using appropriate nonlinear models. Mathematical Models of Server Systems and Service Processes. The principles of mathematical modelling of server systems and service processes are examined. Models of request arrival and service time are studied. System load and resource utilization models are used to analyze the performance of computer systems. Modelling and Analysis of Data Flows. Data flows in computational and information systems are represented using mathematical models. Their structure and dynamics are analyzed. Their intensity and transformation are also analyzed to support system performance evaluation and optimization. Modelling Dynamic Systems Based on Differential Equations. The use of differential equations for modelling dynamic systems is introduced. The evolution of system states over time is examined. Initial and boundary conditions and numerical analysis of dynamic processes are also examined. Fundamentals of Mathematical Modelling in the Python Programming Language. The basic principles of implementing mathematical models in Python are introduced. Students learn to formulate computational problems and develop algorithms. Numerical calculations are performed, and modelling results are visualized. Computing Nonlinear Models Using the NumPy and SciPy Libraries. The computational implementation of nonlinear mathematical models using NumPy and SciPy is studied. Numerical methods and optimization tools are applied to
--	--

	nonlinear problems. Equation solvers and scientific computing techniques are also applied. Mathematical Modelling of the Performance of Computer Systems. Mathematical approaches to evaluating and predicting computer system performance are examined. Processor utilization and memory and storage performance are modelled. Network load, response time, and other system characteristics are also modelled and analyzed. Practical lessons Installing the Python Environment. Working with Jupyter Notebook and Mathematical Computing Packages. The Python programming environment is installed and configured. Jupyter Notebook is used for interactive computing. Essential mathematical and scientific computing packages used for numerical modelling and data analysis are introduced. Plotting and Analysing Linear and Nonlinear Functions Using Python. The graphical representation and analysis of linear and nonlinear functions are practiced. Function behavior and parameter changes are investigated. Extrema and other properties are examined using Python visualization tools. Experimental Analysis of the Time Complexity of Algorithms. The execution time and computational complexity of algorithms are experimentally investigated. The obtained results are compared with theoretical complexity estimates. The results are represented graphically for further analysis. Graphical Solution of Nonlinear Algebraic Equations. The graphical interpretation of nonlinear algebraic equations is examined. Functions are plotted, and approximate roots are identified. The graphical behavior of a function is analyzed to determine the existence and location of solutions. Implementing the Bisection and Interval Methods for Dividing a Segment in Two in Python. Interval-based numerical methods for solving nonlinear
--	---

	equations are implemented in Python. The bisection method is applied to successively reduce the search interval. The required root is approximated with a specified accuracy. Programming the Simple Iteration and Newton Methods. Iterative methods for nonlinear equations are implemented. The simple iteration and Newton methods are programmed. Their convergence, accuracy, and computational efficiency are investigated and compared. Solving a System of Nonlinear Equations Using Numerical Methods. Computational procedures for solving systems of nonlinear equations are developed. Different numerical approaches are applied. The obtained solutions are evaluated according to accuracy and convergence criteria. Creating and Analysing a Simple Mathematical Model of Network Traffic. The modelling of network traffic using mathematical relationships is introduced. A simplified model is constructed, and changes in traffic intensity are simulated. The influence of network load on system behavior is analyzed. Building and Software Implementation of a Server Queue System Model. A mathematical model of a server queue system is developed and implemented in software. Request arrival and service processes are simulated. Waiting time and server workload are also simulated and analyzed. Modelling and Visualising Changes in Data Flow. A mathematical model describing the dynamics of data flow is created. Computational experiments are used to investigate changes in flow intensity and distribution. Graphical visualization is used to examine system behavior over time. Solving Differential Equations Using the Euler Method and Analysing Them by Means of Graphs. The numerical solution of differential equations using the Euler method is practiced. Approximate solutions are computed, and the results are visualized. The
--	--

	influence of model parameters and numerical accuracy is analyzed. Computing Nonlinear Models Using the NumPy and SciPy Libraries. Nonlinear mathematical models are implemented using the NumPy and SciPy libraries. Numerical solvers and optimization tools are applied. Scientific computing functions are also applied to analyze nonlinear processes. Creating a Mathematical Model of a Computer Network, Server or IT Service Process and Its Software Analysis. A computer network, server, or IT service process is selected, and an appropriate mathematical model is formulated. The model is implemented in software. Computational experiments are performed, and the obtained results are analyzed. Nonlinear Models in Artificial Intelligence and Machine Learning. The application of nonlinear models in Artificial Intelligence and Machine Learning is examined. Nonlinear activation functions and prediction models are explored. Other computational approaches used in intelligent systems are also explored. Modern Issues and Prospects of Mathematical Modelling in the Field of IT Services. Current challenges and future prospects of mathematical modelling in IT services are studied. A selected practical problem is analyzed. Possible modelling, computational, and software-based solutions are discussed. Self-study Subject and Objectives of the Science of Mathematical Modelling of Nonlinear Processes. The subject, objectives, and fundamental principles of mathematical modelling of nonlinear processes are introduced. The role of nonlinear models in scientific research and engineering is examined. Their role in computer science and the analysis of complex systems is also examined.
--	--

The Concept of a Mathematical Model and the Stages of Modelling. The concept of a mathematical model is studied. The main stages of the modelling process are examined. Problem formulation, abstraction, mathematical representation, computational implementation, verification, validation, and interpretation of results are considered.

Linear and Nonlinear Models: Similarities and Differences. A comparative analysis of linear and nonlinear models is conducted. Their mathematical properties and structural differences are examined. Their superposition characteristics, solution behavior, and application areas are also examined.

Nonlinear Functions and Their Main Properties. The main properties of nonlinear functions, including continuity and monotonicity, are examined. Extrema, asymptotic behavior, and sensitivity to parameters are also examined. Their role in mathematical modelling is analyzed.

Analysis of Nonlinear Functions Using Computer Graphics. Nonlinear functions are investigated through graphical visualization and computational experiments. The influence of parameters on function behavior is analyzed. Intersections, extrema, and other qualitative characteristics are also analyzed.

Parametric Functions and Their Modelling. The representation of parametric functions is studied. Analysis methods for parametric functions are examined. The influence of parameters on the behavior and geometric properties of mathematical models is also examined.

Fundamentals of Mathematical Model Identification. The identification of mathematical models based on experimental or observed data is studied. Parameter estimation and model selection are examined. The relationship between data and model structure is also examined.

Errors in Modelling and Assessment of Model Adequacy. The main sources of modelling errors are analyzed, including measurement and approximation

errors. Numerical and structural errors are also analyzed. Methods for evaluating model adequacy and the correspondence between model predictions and real data are studied.

Methods for the Mathematical Representation of Real Processes. Different approaches to representing real-world processes mathematically are examined. Physical laws and empirical relationships are considered. Differential equations, statistical models, and computational methods are also considered.

Modern Trends in Computer Modelling. Modern trends in computer modelling are reviewed, including digital twins and data-driven modelling. Artificial intelligence and high-performance computing trends are also reviewed. Simulation-based analysis is examined.

Capabilities of the MATLAB and Python Software Environments for Modelling. The modelling capabilities of MATLAB and Python are explored. Their libraries and computational tools are compared. Their visualization capabilities and applications in numerical simulation and nonlinear modelling are also compared.

The Concept of Nonlinear Differential Equations and Their Classification. The concept of nonlinear differential equations is introduced. Their main properties are examined. Their classification according to order, type, degree, and dependence on variables is also examined.

First-Order Ordinary Differential Equations. First-order ordinary differential equations and their mathematical models are studied. Analytical approaches to solving selected classes of nonlinear equations are examined. Numerical approaches to these equations are also studied.

Higher-Order Nonlinear Differential Equations. Higher-order nonlinear differential equations are examined as mathematical models of complex dynamic processes. Their formulation and

	transformation are discussed. Methods of analysis for these equations are also discussed. Numerical Methods for Solving Nonlinear Differential Equations. Numerical approaches for solving nonlinear differential equations are studied. Approximation methods and iterative procedures are analyzed. Discretization techniques and the accuracy of computational solutions are also analyzed. Dynamic Systems and Their Mathematical Models. The fundamental concepts of dynamic systems are examined. Their mathematical representation is studied. The evolution of system states over time and the role of differential equations in dynamic modelling are analyzed. State Space and Spatial Trajectories. State-space representations of dynamic systems are introduced. State variables and phase space are examined. Trajectories are used to analyze the behavior and evolution of nonlinear systems. Equilibrium Points and Methods for Their Determination. Equilibrium or stationary points of nonlinear dynamic systems are studied. Methods for determining equilibrium states are examined. Their role in system behavior is also examined. Local and Global Properties of Nonlinear Systems. The local and global behavior of nonlinear systems is investigated. Stability and trajectories are analyzed. Invariant regions and long-term system dynamics are also analyzed from different mathematical perspectives. Autonomous and Non-Autonomous Dynamic Systems. The characteristics of autonomous dynamic systems are studied. The characteristics of non-autonomous dynamic systems are also studied and compared. The influence of time-dependent parameters and external forces on system behavior is examined. Mathematical Models of Oscillatory Processes. Mathematical models describing oscillatory and
--	--

	periodic processes are studied. The formation of oscillations is analyzed. Their amplitude, frequency, and the influence of nonlinearities are also analyzed. Predator-Prey (Lotka-Volterra) Model. The Lotka-Volterra model is examined as a classical nonlinear model of population interaction. The relationships between predator and prey populations are analyzed. Equilibrium states and dynamic behavior are also analyzed. Analysis of the SIR Epidemiological Model. The SIR model for describing the spread of infectious diseases is examined. The dynamics of susceptible, infected, and recovered populations are analyzed. The influence of model parameters is also analyzed. The Concept of Stability and Its Types. The concept of stability in dynamic systems is introduced. Its main types are examined. The differences between stability, asymptotic stability, and other forms of system behavior are discussed. Lyapunov's Stability Theory. Lyapunov's approach to the analysis of stability in nonlinear systems is studied. Lyapunov functions are examined. Their role in determining the behavior of equilibrium points is also examined. Methods for Constructing Lyapunov Functions. Different approaches to constructing Lyapunov functions are analyzed. The selection of suitable functions is considered. Their application to the investigation of stability in nonlinear systems is also considered. Asymptotic Stability of Linear Systems. The concept of asymptotic stability in linear dynamic systems is examined. Eigenvalue-based methods for determining system stability are studied. Other analytical approaches are also studied. The Concept of Structural Stability. Structural stability is investigated. Its significance in nonlinear dynamic systems is studied. The persistence of
--	---

	qualitative system behavior under small changes in parameters or model structure is analyzed. Basic Concepts of Bifurcation Theory. The fundamental concepts of bifurcation theory are introduced. Changes in the qualitative behavior of dynamic systems are examined. These changes are studied as they are caused by variations in parameters. Saddle-Node Bifurcation. The saddle-node bifurcation is studied as one of the fundamental types of local bifurcation. The creation of equilibrium points as a control parameter changes is analyzed. The disappearance of equilibrium points is also analyzed using mathematical and graphical models.
Teaching Methods	The module is delivered through interactive lectures, problem-based learning, practical computational exercises, computer simulations, and project-based activities. Lectures introduce nonlinear mathematical models, nonlinear equations, dynamic systems, equilibrium points, stability, and bifurcation theory. Problem-based learning enables students to formulate mathematical models, apply analytical and numerical methods, and interpret computational results. Practical exercises use Python and MATLAB to solve nonlinear equations and differential equations, model dynamic systems, visualize phase trajectories, and analyze stability and bifurcations. Project-based learning and independent study develop students' mathematical, analytical, computational, and research skills through modelling projects, comparative studies, presentations, and computational experiments.
Prerequisites	Calculus 1,2 Differential Equations Linear Algebra Programming 1,2
Exams and Assessment	Type of examination (written, oral, project) and its weight

	Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, practical performance, computer-based laboratory activities, seminar presentations, and the successful completion of independent study assignments, including the analysis of linear and nonlinear models, nonlinear functions, mathematical model identification, modelling errors and model adequacy, nonlinear algebraic and differential equations, numerical solution methods, dynamic systems, equilibrium points, phase trajectories, stability, Lyapunov functions, bifurcation theory, and mathematical models of real-world processes. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as a comparative study of linear and nonlinear models, an analysis of nonlinear functions using computer graphics, a mathematical model identification project, an assessment of modelling errors and model adequacy, a comparative study of numerical methods for nonlinear equations, a dynamic system modelling project, an analysis of equilibrium points and stability, a study of Lyapunov methods, and a research project on bifurcation theory or nonlinear mathematical models. The final assessment is a written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of mathematical modelling of nonlinear processes, nonlinear functions, model identification, modelling errors, nonlinear algebraic and differential equations, numerical methods, dynamic systems, equilibrium points, stability theory, Lyapunov methods, structural stability, and bifurcation theory, as well as their ability to apply theoretical principles to practical and analytical problems. Successful completion of all independent study assignments is required as an essential component of the module assessment.
--	--

Recommended Literature	1. Isralov M.I. Computational Methods. Part 1. Tashkent: O'qituvchi, 2003. 450 pp. 2. Israyilov M.I. Computational Methods. Part 2. Tashkent: O'qituvchi, 2008. 340 pp. 3. S.S. Irisqulov, K.D. Ismanova, M. Olimov, A. Imomov. Numerical Methods and Algorithms. Namangan Publishing House, 2013. 244 pp. 4. Kalitkin N.N., Alshina E.A. Numerical Methods: in 2 vols. Numerical Analysis: A Textbook for Students of Higher Professional Education. Moscow: Publishing Centre Akademiya, 2013. Vol. 1, 304 pp.; Vol. 2, 304 pp. 5. Giordano F.R., Weir M.D., Fox W.P. A First Course in Mathematical Modelling. 5th Edition. Boston: Cengage Learning, 2014. 6. Strogatz S.H. Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry and Engineering. 2nd Edition. Boulder: Westview Press, 2015. 7. Hirsch M.W., Smale S., Devaney R.L. Differential Equations, Dynamical Systems and an Introduction to Chaos. 3rd Edition. Academic Press, 2013. 8. Chapra S.C., Canale R.P. Numerical Methods for Engineers. 8th Edition. McGraw-Hill Education, 2020. 9. Richard L. Burden and J. Douglas Faires. Numerical Analysis. 9th Edition. Youngstown State University Press, 2011. 895 p. 10. L. Ridgway Scott. Numerical Analysis. Princeton University Press, 2011. 342 p. 11. Ismatullayev G.P., Pulatov S.I., Fayazov Q.S. Handbook of Numerical Methods. Tashkent: University, 2006. 140 pp. 12. Bakvalov N.S., Zhidkov N.P., Kobelkov G.M. Numerical Methods. Moscow: Bynom Publishing House. Laboratory of Knowledge, 2011. 640 pp.
-------------------------------	---

	13. Samarsky A.A., Vabishevich P.N., Samarskaya E.A. Problems and Exercises on Numerical Methods. Moscow: Libronom Publishing House, 2009. 208 pp. 14. Samarsky A.A. Introduction to Numerical Methods. Moscow: Lan' Publishing House, 2009. 288 pp. 15. Toth J., Krisztin T. Computational Methods for Mathematical Modelling. Springer, 2021. 16. Downey A.B. Modeling and Simulation in Python. No Starch Press, 2023. 17. Kreyszig E. Advanced Engineering Mathematics. 10th Edition. Wiley, 2011. 18. https://devos.uz/article.php?id%3D1816 19. https://en.wikipedia.org/wiki/Nonlinear_modelling 20. https://www.sciencedirect.com/special-issue/10DNT4NRBGH
Responsible Lecturer	Kholliiev Fakhriddin Bokhodirovich

2.05. MATHEMATICAL FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

Field	Description
Module Title	MATHEMATICAL FOUNDATIONS OF ARTIFICIAL INTELLIGENCE
Module Code	SIMA204
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Mathematical Foundations of Artificial Intelligence is to provide students with knowledge of the theoretical foundations of artificial intelligence and neural networks. Modern mathematical methods of image understanding are studied. These methods are applied to solving applied problems with high quality and accuracy. Students are provided with knowledge on the creation of modern information systems. These information systems are used to interpret decision-making in difficult, non-formalised problems. Such problems arise in various applied domains.
Module Learning Outcomes	Knowledge: LO1. To develop the necessary knowledge and skills in artificial intelligence. LO2. Develop practical skills in creating programmes using modern programming languages. LO3. Have basic knowledge of the risks posed by artificial intelligence and how to protect against them. LO4. Develop practical skills in working with databases and Big Data. LO5. Formulate theoretical knowledge of the structure and development principles of modern information systems.

	LO6. To possess the necessary knowledge and skills regarding network topologies and their operational functions. LO7. Software development and testing. LO8. Familiarisation with methods for ensuring information security in network data processing environments. Skills: LO9. Familiarisation with software related to artificial intelligence technologies. LO10. Studies the theoretical foundations of using personal computers in management processes and the methods for applying them to various branches and sectors of the national economy. LO11. Studies database management systems and working with Big Data, to acquire the necessary knowledge and skills related to artificial intelligence platforms. LO12. Develop the necessary knowledge and skills in the fundamentals of artificial intelligence. LO13. Possess the necessary knowledge and skills in programming languages and their technologies. LO14. They will familiarise themselves with methods for ensuring information security when handling data and with its functional capabilities. Competencies: LO15. Students will gain the skills to apply artificial intelligence technologies across various fields and find practical solutions to real-world problems. LO16. Students acquire the skills to develop, test and optimise artificial intelligence algorithms using modern programming languages. LO17. Students will acquire the skills to collect, store, process and effectively utilise large-scale data (Big Data).
Content	Lectures

Artificial Intelligence: History of development and current state. The concept of Artificial Intelligence is introduced. Its historical development, major milestones, current capabilities, and contemporary trends are examined. The role of AI in science, industry, business, and everyday applications is also discussed.

Data Analysis. The fundamental principles of data analysis are studied. Data collection, preprocessing, and exploration are examined. Visualization, statistical analysis, and the preparation of data for machine learning are also examined.

Biological Neuron and Artificial Neural Network Model. The structure and functioning of biological neurons are examined. Their mathematical representation in artificial neural networks is studied. The basic concepts of artificial neurons, weights, activation functions, and network architecture are also studied.

Machine Learning Models: Linear Regression, Logistic Regression, Decision Tree. Fundamental supervised machine learning models are introduced. The principles and mathematical foundations of linear regression, logistic regression, and decision trees are examined. Their training processes and applications are also examined.

Unsupervised Learning: K-Means Clustering and PCA. Unsupervised learning methods for discovering patterns and structures in unlabeled data are studied. K-means clustering is studied for clustering unlabeled data. Principal Component Analysis (PCA) is studied for dimensionality reduction.

Deep Learning: Neural Network Structures. The fundamental principles of deep learning and modern neural network architectures are introduced. Multi-layer networks, activation functions, and loss functions are examined. Optimization methods and the training process are also examined.

Convolutional Neural Networks (CNN) and Computer Vision. The architecture and principles of

	Convolutional Neural Networks are examined. Convolution, pooling, and feature extraction are studied. The application of CNNs to image classification and computer vision tasks is also studied. Recurrent Neural Networks (RNN) and Natural Language Processing. Neural network architectures designed to process sequential data are studied. Recurrent Neural Networks are examined. Their applications in Natural Language Processing, text analysis, and sequence prediction are also examined. Reinforcement Learning: Q-Learning, Deep Q-Network. The fundamental concepts of reinforcement learning are introduced. Agents, environments, states, actions, and rewards are studied. Q-learning and Deep Q-Networks are studied for solving sequential decision-making problems. Model Evaluation and Validation Methods. Methods for evaluating and validating machine learning models are examined. Training and testing datasets and cross-validation are analyzed. Accuracy, precision, recall, F1-score, ROC-AUC, and overfitting are also analyzed. Big Data and Cloud Technologies. The fundamental concepts of Big Data and cloud computing technologies are introduced. Large-scale data processing and distributed computing are examined. Cloud infrastructure and its role in artificial intelligence applications are also examined. Recommendation Systems: Collaborative Filtering, Content-Based Filtering. The principles of recommendation systems are examined. Collaborative filtering methods are studied for generating personalized recommendations. Content-based filtering methods based on user behavior and item characteristics are also studied. Understanding Information Security and Attack Vectors in Artificial Intelligence Programmes. Information security risks and attack vectors in artificial intelligence systems are studied. Data
--	---

	security, adversarial attacks, and data poisoning are examined. Model vulnerabilities and methods for protecting AI applications are also examined. Getting Started with Libraries Like TensorFlow and PyTorch. Popular deep learning frameworks, including TensorFlow and PyTorch, are introduced. Their basic architecture and data processing capabilities are studied. Model construction, training, and application in AI projects are also studied. Examples of Using Artificial Intelligence in Entrepreneurial and Practical Projects. Practical applications of artificial intelligence in entrepreneurship and business are examined. Applications in science, industry, agriculture, healthcare, and other professional fields are also examined. Examples of AI-based projects and solutions are analyzed. Practical lessons Artificial Intelligence and the Technologies Used in It. The fundamental concepts of Artificial Intelligence are introduced. The technologies used to develop AI-based systems are introduced. Basic AI tools, application areas, and the general workflow of AI projects are explored. Data Pre-processing Using Python (Pandas, NumPy, Matplotlib). Basic data preprocessing is performed using Python libraries such as Pandas, NumPy, and Matplotlib. Data cleaning and transformation techniques are applied. Statistical analysis and visualization techniques are also applied. Building a Simple Regression Model. Constructing and training a simple regression model is practiced. Data is prepared, and the model is trained. Predictions are generated, and the model's performance is evaluated using appropriate metrics. Multi-Layer Perceptron (MLP) and Feedforward Architecture. A basic Multi-Layer Perceptron is implemented. The structure of feedforward neural
--	---

	networks is explored. The roles of input, hidden, and output layers in processing data are examined. Main Activation Functions (Sigmoid, Tanh, ReLU, etc.). Commonly used activation functions, including Sigmoid, Tanh, and ReLU, are implemented. These activation functions are compared. Their influence on neural network learning and model performance is analyzed. Binary Classification Practice. A machine learning model for binary classification is developed and trained. Data preparation and model training are performed. Prediction and evaluation using classification metrics are also performed. Clustering Using K-Means. The K-means clustering algorithm is applied to unlabeled data. Clusters are determined, and cluster centers are analyzed. The resulting groups are visualized. RNN Architecture. The structure and operating principles of Recurrent Neural Networks are explored. A basic RNN model is implemented for processing sequential data. Temporal dependencies are analyzed using the implemented model. Introduction to Convolutional Neural Networks (CNN). The basic architecture of CNNs is introduced. Convolution and pooling operations are applied. A simple model for image classification is developed. Regularisation Methods: L1, L2 and Early Stopping. L1 and L2 regularization techniques are applied to reduce overfitting. Early stopping is also applied to reduce overfitting. The influence of regularization on model complexity and generalization performance is analyzed. Creating a Recommendation System. Developing a basic recommendation system is practiced. Collaborative filtering methods are applied to generate personalized recommendations. Content-based filtering methods are also applied to generate personalized recommendations.
--	---

	Model Evaluation and Optimisation. Machine learning models are evaluated using appropriate performance metrics. Optimization techniques are applied to these models. Hyperparameter tuning and model improvement strategies are also applied. Working with Big Data. Techniques for processing large datasets are introduced. Techniques for analyzing large datasets are also introduced. Data storage, processing, and analysis methods suitable for Big Data applications are explored. Fundamentals of Speech Recognition. The fundamental principles of speech recognition systems are explored. Audio data processing and feature extraction are examined. Basic methods for converting speech into text are also examined. Object Detection in Computer Vision. The fundamental principles of object detection are studied. Computer vision techniques are applied to identify and locate objects in images. Deep learning techniques are also applied to identify and locate objects in images. Self-study The History and Modern State of Artificial Intelligence. The historical development of Artificial Intelligence is reviewed, from its early theoretical foundations to modern generative, autonomous, and intelligent systems. Current achievements and challenges are considered. Future development trends are also considered. Data Analysis: Data Cleaning and Visualisation. Techniques for cleaning, transforming, and visualising datasets are investigated. Missing data handling and outlier detection are studied. Data consistency analysis and graphical interpretation of analytical results are also studied. Mathematical Foundations of Artificial Neural and Perceptron Models. The mathematical basis of artificial neurons and perceptrons is explored. Weighted sums, activation functions, bias terms, and
--	---

	decision boundaries are studied. The relationship between biological neurons and their computational models is also examined. Comparative Analysis of Regression and Decision Tree Models. A comparative study of regression and decision tree models is conducted. Their learning principles, model structure, and interpretability are examined. Their strengths, limitations, and suitability for different machine learning tasks are also examined. K-Means Clustering and PCA Method. Two important unsupervised learning techniques are studied. K-means clustering is studied for grouping similar data points. Principal Component Analysis is studied for reducing data dimensionality while preserving significant information. MLP, Feedforward Architecture and Activation Functions. The structure of Multi-Layer Perceptrons and feedforward neural networks is analyzed. Their learning process is also analyzed. Particular attention is given to the role of Sigmoid, Tanh, ReLU, and other activation functions in nonlinear data transformation. Model Evaluation and Validation Metrics. Different approaches to assessing the quality of machine learning models are examined. Their generalization ability is also examined. Accuracy, precision, recall, F1-score, ROC-AUC, cross-validation, and overfitting detection are considered. Big Data and Cloud Technologies. The technologies used to store, process, and analyze large-scale datasets are reviewed. Big Data platforms and cloud computing resources are studied. Their integration with Artificial Intelligence applications is discussed. CNN Blocks: Pooling, Padding, Stride, Filter. The main computational components of Convolutional Neural Networks are studied. How filters extract features is examined. How padding, stride, and
--	--

pooling affect the spatial dimensions and performance of CNN models is also examined.

Advanced CNN Architectures such as DenseNet and Inception. The architectural innovations of advanced CNN models, including DenseNet and Inception, are investigated. Their approaches to feature reuse and multi-scale feature extraction are compared. Their computational efficiency and image recognition performance are also compared.

RNN, LSTM and GRU Architectures. Recurrent neural architectures for sequential data processing are explored. Conventional RNNs are compared with LSTM and GRU models. This comparison focuses on memory mechanisms and their ability to address long-term dependency problems.

Recommendation Systems. The principles of personalized recommendation are examined. Collaborative filtering, content-based approaches, and hybrid methods are studied. Applications in e-commerce, media platforms, and information services are also considered.

Information Security and Attack Vectors in AI Applications. Security vulnerabilities specific to Artificial Intelligence systems are analyzed, including adversarial examples and data poisoning. Model manipulation, privacy threats, and other attack vectors are also analyzed. Basic strategies for improving AI system security are discussed.

Fundamentals of Speech Recognition and Computer Vision. An integrated overview of two important AI application areas is provided. Speech recognition involves the processing and interpretation of audio signals. Computer vision focuses on extracting meaningful information from images and video.

Final Mini-Project: Developing a Machine Learning Model for a Selected Problem. The final independent study is completed as a practical mini-project. A relevant problem is selected, and data is prepared and analyzed. An appropriate machine

	learning model is chosen, trained, and evaluated, and the results are visualized and presented.
Teaching Methods	The module is delivered through interactive lectures, problem-based and inquiry-based learning, practical exercises, computer simulations, and project-based activities. Interactive lectures introduce the mathematical foundations of Artificial Intelligence, including linear algebra, probability and statistics, optimization, data analysis, artificial neurons, perceptrons, machine learning models, neural networks, and model evaluation methods. Practical exercises focus on data preprocessing, statistical analysis, linear and logistic regression, decision trees, K-means clustering, PCA, perceptrons, activation functions, neural network training, model evaluation, and basic deep learning applications. Python libraries and computational environments are used to implement mathematical models, perform numerical experiments, visualize data, train machine learning models, and analyze their results. Project-based learning and independent study develop students' analytical, mathematical, computational, research, and problem-solving skills through individual and group assignments, comparative studies, presentations, analytical reports, and mini-projects. Digital educational resources, programming libraries, visualization tools, and modern AI frameworks support independent learning and enable students to apply mathematical foundations of Artificial Intelligence to practical, scientific, engineering, and professional problems.
Prerequisites	Fundamentals of Informatics Programming Fundamentals Digital Logic and Electronics Operating Systems
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and

	a final written examination (50%). The continuous assessment evaluates classroom participation, practical performance, computer-based laboratory activities, seminar presentations, and the successful completion of independent study assignments, including data cleaning and visualisation, mathematical analysis of artificial neurons and perceptrons, regression and decision tree models, K-means clustering, PCA, MLP and feedforward architectures, activation functions, model evaluation and validation metrics, Big Data and cloud technologies, CNN architectures, RNN/LSTM/GRU models, recommendation systems, information security in AI applications, speech recognition, computer vision, and the development of a machine learning model for a selected problem. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as an analysis of the mathematical foundations of artificial neural networks, a comparative study of regression and decision tree models, a study of K-means clustering and PCA, an analysis of MLP architectures and activation functions, a comparative evaluation of machine learning validation metrics, a study of CNN architectures and their main components, an analysis of RNN, LSTM, and GRU models, a research project on recommendation systems or information security in AI applications, and a mini-project involving the development and evaluation of a machine learning model. The final assessment is a written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of the mathematical foundations of Artificial Intelligence, data analysis, artificial neurons and perceptrons, regression and classification models, clustering and dimensionality reduction, neural network architectures, activation functions, model evaluation and validation, deep learning fundamentals, Big Data and cloud technologies, recommendation systems, and security issues in AI applications, as well as their ability to apply
--	--

	mathematical and theoretical principles to practical and analytical problems. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Potapov A.S. Technologies of Artificial Intelligence. St. Petersburg: St. Petersburg State University ITMO, 2010. 218 pp. 2. Intelligent Information Systems and Technologies: A Textbook / Y.Y. Gromov, O.G. Ivanova, V.V. Alekseev et al. Tambov: Publishing House of FGBOU Higher Education Institution "TSTU", 2013. 244 pp. 3. Ignatyev N.A., Usmanov R.N., Madraximov Sh.F. Intellectual Analysis of Given Data. Tashkent, 2018. 144 pp. 4. Stamp M. Information Security: Principles and Practice. John Wiley & Sons, 2011. 606 pp. 5. Markov A. S., Barabanov A. V., Dorofeev A. V., Tsirlov V. L. Seven Safe Information Technologies / ed. by A. S. Markov. Moscow: DMK Press, 2017. 224 pp. 6. Gavrilova T.A., Khoreshevsky V.F. Knowledge Bases of Intelligent Systems. Textbook. St. Petersburg: Piter, 2001. 384 pp. 7. Joshi, Pratik. Artificial Intelligence with Examples in Python. Translated from English. St. Petersburg: OOO "Dialektika", 2019. 448 pp. 8. Nikolaev A.B., Fominykh I.B. Intellectual Analysis and Processing of Data. Textbook for the course Intelligent Systems (Part 1). Moscow, 2003. 117 pp. 9. https://www.cs.ox.ac.uk/teaching/courses/2025-2026/ai/ 10. https://ziyonet.uz - an information and educational portal.
Responsible Lecturer	Jo'rayev Olim Albayevich

2.06. ELECTRONIC GOVERNMENT

Field	Description
Module Title	ELECTRONIC GOVERNMENT
Module Code	EKH204
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Electronic Government is to develop in postgraduate students the knowledge, skills, and professional competencies concerning the theoretical foundations of the e-government system. The legal and regulatory framework of e-government is also studied. Its architecture, interactive public services, and open data are examined. Electronic document circulation and digital identification mechanisms of the e-government system are also examined. Students are prepared for the effective use of e-government services. They are also prepared for the analysis of these services and their application in professional practice.
Module Learning Outcomes	Knowledge: LO1. Can explain the concept, objectives, tasks and key principles of e-government. LO2. Can explain the main regulatory and legal documents in the field of e-government and the powers of the regulatory bodies. LO3. Describe the architecture of e-government, central databases, inter-agency electronic collaboration and the content of government information systems.

	LO4. Can explain the differences and practical applications of the G2G, G2B and G2C relationship models. LO5. Understands the nature of electronic digital signatures, electronic document circulation, and authentication and identification mechanisms. LO6. They will have knowledge of the requirements for information security, confidentiality and the handling of open data in an e-government system. Skills: LO7. Can make effective use of the Government Portal, the Single Interactive Government Services Portal and other official electronic platforms. LO8. Seeks, analyses and utilises information from the Lex.uz, regulation.gov.uz, my.gov.uz, gov.uz, data.egov.uz and openbudget.uz resources in practical assignments. LO9. Practically analyses electronic public services, applications, electronic payments, open data and open budget processes. LO10. Compares the functional capabilities of the official websites, interactive services and open data portals of government bodies. LO11. Analyses practical scenarios related to electronic document circulation, electronic digital signature and information security requirements. LO12. Prepares a presentation, report, table, analytical summary and mini-project on e-government projects. Competencies: LO13. Assesses the role of digital technologies and e-government mechanisms in public administration from a professional perspective. LO14. They will have the competence to identify issues and develop proposals for improving digital public services.
--	---

	LO15. Applies the principles of transparency, openness, convenience and citizen-centric service delivery in e-government projects. LO16. Utilises open data and e-government services in scientific research, analytical and professional activities. LO17. Studies the experience of foreign e-government systems and conducts a comparative analysis with the national system. LO18. Approaches the requirements for information security and personal data protection responsibly. LO19. Takes into account legal, technical and ethical requirements when using e-government services. LO20. Will be competent to independently study and implement new digital platforms and services in the e-government system.
Content	Lectures The Concept of 'E-Government', Its Definitions, Main Objectives and Principles. The concept, purpose, functions, and fundamental principles of e-government are studied. Its role in improving public administration and increasing transparency is explained. Its role in simplifying public services and strengthening communication between government institutions and citizens is also explained. Normative Legal Documents Regulating the Field of E-Government. The principal laws, regulations, governmental decisions, and technical standards governing e-government are examined. Their role in regulating electronic public services, information exchange, and digital identification is discussed. Their role in data protection and inter-agency cooperation is also discussed. State Bodies Regulating the E-Government Sector and Their Powers. The functions and responsibilities of public authorities involved in the development and coordination of e-government systems are studied. Their responsibilities in the regulation and monitoring

of these systems are also studied. Their powers in managing digital infrastructure, information systems, public services, and cybersecurity are examined.

Rights of Natural and Legal Persons to Use the E-Government System and the Procedure for Considering Requests. The rights of individuals and organisations to access electronic public services and government information resources are studied. Procedures for submitting and registering electronic applications, requests, complaints, and proposals are explained. Procedures for reviewing and responding to these are also explained.

E-Government Architecture, Central Databases and Inter-Agency Electronic Collaboration. The structure and main components of e-government architecture are examined. Central databases and government information systems are studied. Integration platforms and mechanisms for secure electronic data exchange between public institutions are also studied.

Official Websites of Government Bodies and the Main E-Government Projects in the Republic of Uzbekistan. The structure, functions, and requirements of official government websites are studied. Major national e-government projects are analyzed. Their role in providing information, delivering interactive services, and supporting communication with citizens and businesses is also analyzed.

Key Projects for Creating E-Government Databases and System Management Mechanisms. The development and operation of central government databases and integrated information systems are examined. Data collection, updating, and interoperability mechanisms are discussed. Access control, system administration, monitoring, and information security mechanisms are also discussed.

Forms of Interaction in E-Government: the Content of the G2G, G2B and G2C Models. The

	principal interaction models of e-government are studied. Government-to-Government, Government-to-Business, and Government-to-Citizen services are compared. This comparison considers their participants, objectives, information flows, and practical applications. Discussion Portal for Draft Regulatory and Legal Documents and Working with Legislative Information. The use of electronic platforms for publishing and discussing draft regulatory documents is examined. Methods for searching legal information and reviewing draft legislation are studied. Methods for submitting proposals and participating in public consultation processes are also studied. Ensuring the Confidentiality of Information Resources in the E-Government System, Electronic Document Circulation and Electronic Digital Signature. Information confidentiality, integrity, authentication, and access control within e-government systems are studied. Electronic document management and secure data exchange are examined. Digital signatures and protection against unauthorized access are also examined. The Government Portal of the Republic of Uzbekistan and the Unified Interactive State Services Portal: Objectives, Tasks and Opportunities. The purposes, functions, structure, and service capabilities of the principal government portals are studied. Access to public information, electronic applications, and online services is examined. Service tracking, user identification, and communication with government institutions are also examined. The Open Data Portal and Open Budget Portal of the Republic of Uzbekistan: Openness, Transparency and Public Oversight. The role of open government data and budget-information platforms in promoting transparency and accountability is studied. Methods for accessing and analyzing public datasets and financial information
--	---

are examined. Methods for using this information for research, decision-making, and public oversight are also examined.

Utilities and Housing Stock Portal, Electronic Payment Systems and Use of Interactive Services.

Digital platforms used for housing, communal services, utility management, and electronic payments are studied. Online account management and service requests are examined. Payment processing, information exchange, and user interaction mechanisms are also examined.

E-Government as a Factor in Modernising the National Economy and Improving the Quality of Public Services.

The influence of e-government on economic development and administrative efficiency is analyzed. Its influence on investment conditions and public-service quality is also analyzed. Its role in reducing bureaucracy, service costs, processing time, and corruption risks is considered.

Foreign E-Government Systems, International Rankings and Prospects for the Development of E-Government in Uzbekistan.

International e-government models, best practices, and evaluation indicators are studied. Foreign experiences are compared. Opportunities for improving digital public services, institutional integration, accessibility, cybersecurity, and citizen participation in Uzbekistan are analyzed.

Practical lessons

Working with the Internet, Its Software and Hardware, and Search Engines.

Internet infrastructure and network equipment are examined. Browsers, communication software, and search engines are also examined. Effective methods for finding, filtering, evaluating, and using online information are practiced.

Internet Communication Services and the Culture of Official Electronic Communication.

Email, messaging, video conferencing, and other Internet communication services are studied. Professional

	correspondence and official communication etiquette are practiced. Information security and responsible digital behaviour are also practiced. Analysis of the Structure of Official Websites of Government Bodies. The design, navigation, and content of official government websites are examined. Their accessibility, feedback mechanisms, and interactive services are also examined. Compliance with transparency and user-oriented service requirements is analyzed. Analysis of G2G, G2C and G2B Models Based on Practical Examples. Government-to-Government, Government-to-Citizen, and Government-to-Business interaction models are compared through real service scenarios. Their participants and information flows are identified. Their objectives and practical benefits are also identified. Analysis of Methods for Ensuring and Protecting the Confidentiality of Information Resources. Access control and authentication methods are studied. Encryption, data backup, and secure communication measures are also studied. These organisational measures are used to protect government information resources from unauthorized access and disclosure. Normative and Legal Foundations of E-Government: Working with Lex.uz and Other Official Sources. Legislation and regulatory documents related to e-government are searched and reviewed using official legal-information resources. These documents are also analyzed using these resources. Skills in identifying current documents and interpreting their main provisions are developed. Developing an E-Government Architecture and an Organisational Scheme for Interactive Services. A simplified e-government architecture is designed, showing users, government bodies, and databases. Integration systems, authentication tools, and electronic services are also shown. Organisational
--	--

	and information-exchange processes are presented schematically. Analysis of the Procedure for Electronic Document Exchange and Handling of Electronic Digital Signatures. The stages of creating, signing, and sending electronic documents are examined. The stages of receiving, verifying, registering, and storing these documents are also examined. The practical role of electronic digital signatures in ensuring authenticity and integrity is analyzed. Practical Study of the Features of the Government Portal of the Republic of Uzbekistan (gov.uz). The portal's structure, official information, and government-body pages are explored. Its public announcements, search tools, and feedback services are also explored. Its role as a central source of government information is assessed. Working with Services via the Single Interactive Government Services Portal (my.gov.uz). User registration and digital identification are examined through the unified portal. Service search, electronic application submission, and document attachment are also examined. Payment and application-status monitoring through the portal are examined as well. Analysis of the Capabilities of the Open Data Portal of the Republic of Uzbekistan (data.egov.uz). Public datasets on the Open Data Portal are searched. Their structure and formats are reviewed. Their relevance, accessibility, and potential use in research, analysis, public oversight, and application development are assessed. Working with the Regulation.gov.uz Portal for the Discussion of Draft Regulatory and Legal Documents. Draft legal documents on the portal are located. Explanatory materials are reviewed, and proposed regulations are analyzed. Mechanisms for submitting public comments and suggestions are also studied. Analysis of the Functions of the Republic of Uzbekistan's "Open Budget" Portal
--	---

(openbudget.uz). Budget information and public participation tools on the Open Budget portal are examined. Initiative-based budgeting and project information are also examined. Voting mechanisms and financial transparency features available through the portal are examined as well.

Practical Application of Utility Services, Housing Stock Portals and Electronic Payment Systems.

Online services for utility payments and account management are studied. Meter readings, housing-related requests, and electronic transactions are also studied. Service accessibility, usability, and payment security are evaluated.

Preparation and Presentation of a Final Mini-Project on E-Government Interactive Services. A mini-project proposing or modelling an interactive public service is developed and presented. The project includes service objectives, target users, and workflow. It also includes required information systems, security measures, and expected public benefits.

Self-study

The Essence of E-Government, Its Stages of Development and Its Main Principles. The concept, objectives, and development stages of e-government are studied. Its fundamental principles are also studied. Its role in improving public administration, transparency, accessibility, efficiency, and citizen participation is examined.

Normative Legal Documents in the Field of E-Government. The principal laws and regulations governing electronic public services are examined. Official documents governing digital identification and information exchange are also examined. Documents governing data protection and inter-agency cooperation are examined as well.

Competences of State Bodies in the Field of E-Government. The powers of government institutions involved in regulating e-government systems are studied. Their responsibilities in coordinating and

developing these systems are also studied. Their role in monitoring e-government systems is studied as well.

Rights of Natural and Legal Persons to Use Electronic Government Services. The rights of individuals and organisations to access public information are studied. Their rights to submit electronic applications and receive interactive services are also studied. Their rights to protect their personal and organisational data are studied as well.

E-Government Architecture and Inter-Agency Electronic Collaboration. The components of e-government architecture are examined. Information systems and central databases are studied as part of this architecture. Integration platforms and secure data-exchange mechanisms are also examined.

Official Websites of Government Bodies and E-Government Projects. The structure and content of official government websites are analysed. Their accessibility, transparency, and interactive capabilities are also analysed. Major national e-government projects are analysed as well.

Practical Application of G2G, G2B and G2C Models. Government-to-Government, Government-to-Business, and Government-to-Citizen models are studied through practical examples. Their participants and service processes are compared. Their information flows and benefits are also compared.

Information Security and Confidentiality in the E-Government System. Authentication and authorisation measures used to protect government information resources are examined. Encryption, access control, and data backup measures are also examined. Monitoring and other protective measures are examined as well.

Working with the Portal for Discussing Draft Regulatory and Legal Documents. Methods for searching and reviewing draft legal documents are studied. Methods for analysing these documents are also studied. Methods for participating in public

discussions by submitting comments and proposals are studied as well.

Using Services via the Single Interactive Government Services Portal. Service search and digital identification through the portal are examined. Electronic application submission and document attachment are also examined. Payment procedures and application-status monitoring are examined as well.

Analysis of the Open Data Portal and Open Data Sets. The structure and formats of open government datasets are analysed. Their accessibility and reliability are also analysed. Their practical use for research, decision-making, application development, and public oversight is analysed as well.

Open Budget Portal and Public Oversight Mechanisms. The role of open-budget platforms in promoting financial transparency is studied. Their role in promoting citizen participation is also studied. Budget information, initiative-based projects, voting, and public-control mechanisms are analysed.

Electronic Digital Signature, Authentication and Electronic Document Circulation. The processes of creating, signing, and transmitting electronic documents are studied. The processes of verifying, registering, and storing these documents are also studied. Digital signatures and authentication mechanisms are examined as tools for ensuring authenticity and integrity.

Innovative Technologies and Electronic Payment Systems in E-Government. The application of cloud technologies and mobile services in public services is examined. The application of artificial intelligence and digital identification is also examined. The application of electronic payments and other innovative solutions is examined as well.

Foreign E-Government Systems and Proposals for Uzbekistan. International e-government practices, service models, and development approaches are compared. Based on this analysis,

	proposals for improving accessibility, integration, and security in Uzbekistan are developed. Proposals for improving transparency and service quality are also developed.
Teaching Methods	The module is delivered through lectures, practical sessions, case-based and problem-based learning, legal and comparative analysis, demonstrations, group discussions, presentations, and project-based assignments. Students work with official e-government platforms, including gov.uz, my.gov.uz, lex.uz, regulation.gov.uz, data.egov.uz, and openbudget.uz. Students analyse G2G, G2B, and G2C interaction models. They also complete practical tasks involving electronic public services, open data, electronic documents, digital signatures, information security, and electronic payments. Teacher-guided and student independent learning, official legal sources, scientific literature, analytical reports, comparative tables, presentations, and mini-projects further develop students' digital, legal, analytical, research, and public-service design competencies.
Prerequisites	Information Technology Fundamentals of Information Security Web Technologies and Internet Services Foundations of Public Administration and Legal Information
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and final assessment (50%). The continuous assessment evaluates participation in 15 practical sessions and completion of six Teacher-Guided Independent Work assignments. A maximum of 8 points is awarded for practical participation, while 12 points are allocated to independent assignments, reports, analytical work, presentations, and mini-projects. The midterm assessment consists of a theoretical component worth 12 points and independent work worth 18

	points. The theoretical component may be conducted as a test or a written examination consisting of three theoretical and two practical questions. Independent work is prepared as a report, analytical paper, project, or presentation and submitted through the HEMIS system. The final assessment may be conducted as a written examination or a test. The written examination consists of five questions worth 10 points each, while the test consists of 25 questions worth 2 points each. All required independent study assignments must be submitted before the final assessment.
Recommended Literature	1. Zaynidinov X.N., Yakubov M.S., Qoraboyev J. Electronic Government. 2nd revised edition. Tashkent: Akademiya, 2014. 273 pp. https://cyberleninka.ru/article/n/o-zbekiston-elektron-hukumat-tizimida-axborot-kommunikatsiya-texnologiyalari-xizmatlaridan-foydalanish-mexanizmlari-va-ularning 2. Aripov A.N., Mirzahidov X.M. and others. Information and Communication Technologies in Public Administration. Tashkent, 2005. 204 pp. https://akbt.tkti.uz/uz/books/15597 3. United Nations. E-Government Survey 2024: Accelerating Digital Transformation for Sustainable Development. New York: United Nations, 2024. https://desapublications.un.org/publications/un-e-government-survey-2024 4. Information Security and Privacy. Korea Internet & Security Agency. UN-APCICT/ESCAP, 2011. Module 6. Second Edition. https://www.unapcict.org/sites/default/files/2019-08/Academy%202.0%20Module%206_1.pdf 5. E-Government Applications. Nag Yeon Lee and Kwangsok Oh. UN-APCICT/ESCAP, 2011. Module 3. Second Edition. https://www.unapcict.org/sites/default/files/2019-08/Academy%202.0%20Module%203.pdf 6. ICT for Development Policy, Process and Governance. Emmanuel C. Lallana. UN-

	APCICT/ESCAP, 2011. Module 2. Second Edition. https://www.unapcict.org/sites/default/files/2019-08/Academy%202.0%20Module%202_1.pdf 7. Stamp M. Information Security: Principles and Practice. John Wiley & Sons, 2011. 606 pp.
Responsible Lecturer	Zaripova Mukaddas Djumayozovna, Doctor of Philosophy (PhD) in Economics, Associate Professor ad interim

2.06. FUNDAMENTALS OF THE DJANGO FRAMEWORK

Field	Description
Module Title	FUNDAMENTALS OF THE DJANGO FRAMEWORK
Module Code	PDTM204
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Fundamentals of the Django Framework is to develop master's students' theoretical knowledge and practical skills related to the Django framework. Students learn to create web applications based on Django. They also learn to manage and test these applications. The applications addressed are modern, secure, and scalable. They are also database-integrated. These skills are developed and applied through practical projects.
Module Learning Outcomes	Knowledge: LO1. Can explain the purpose, operating principles, and role of the Django framework in developing web applications. LO2. Can describe the MVT (Model-Template-View) architecture and the concepts of project and app. LO3. Knows the functions of the settings.py, urls.py, views.py, models.py, admin.py, and forms.py files. LO4. Understands the fundamentals of Django ORM, migrations, model fields, and database operations. LO5. Knows the principles of organizing the user interface through templates, static files, and media files.

	LO6. Understands the fundamentals of authentication, authorization, sessions, permissions, and web security in Django. Skills: LO7. Can create a Django project, configure a virtual environment, add an application, and run the server. LO8. Can create dynamic web pages using URL routes, view functions, and templates. LO9. Will acquire skills in creating models, performing migrations, and managing data through the Django Admin panel. LO10. Can receive and validate user data using Forms and ModelForms. LO11. Can add CRUD, search, filter, pagination, file upload, and email-sending functions to a web application. Competencies: LO12. Will have the competence to create a simple API using Django REST Framework, test endpoints, and document them. LO13. Will acquire the ability to manage a Django project based on security requirements, user roles, and permissions. LO14. Knows how to deploy a Django project, manage environment settings, and prepare the project for a production environment. LO15. Will have the competence to independently develop a Django-based web project, analyze project code, fix errors, and present the result professionally.
Content	Lectures Introduction to the Django Framework: What is Django and How Does It Work?. The architecture, features, and working principles of the Django framework are introduced. Its role in modern web development and the Model-View-Template (MVT) architecture are explained. The request-response

	cycle and advantages for rapid, secure, and scalable application development are also explained. Stages of Creating a Django Project and Application (Project/App). The procedures for creating Django projects and applications are studied. Project structure and application organisation are learned. Configuration and best practices for developing modular web applications are also learned. Django Settings: settings.py, Main Parameters, and URL Routing. The configuration of Django projects through the settings.py file is examined. Core settings, installed applications, and middleware are studied. Database configuration, static files, and URL routing mechanisms are also studied. Working with View Functions, the Template System, and Static Files. View functions and template rendering are studied. Template inheritance and dynamic content generation are also studied. The management of static resources such as CSS, JavaScript, and images is studied as well. Creating Models in Django and Connecting to a Database. The principles of database modelling in Django are examined. Data models are created, and relationships between tables are established. Databases are configured, and object-oriented data management is implemented. Migrations and the Fundamentals of Django ORM (Object-Relational Mapping). The migration process and Django ORM are studied. Database schemas are created, modified, and synchronized. Database operations are performed through Python objects instead of SQL queries. Django Admin Panel: Registering and Managing Models. The Django administration interface is explored. Models are registered, and the admin panel is customised. Application data is managed, and administrative operations are performed efficiently.
--	--

	Django Forms and ModelForms: Data Input and Validation. The creation of forms using Django Forms and ModelForms is studied. Methods for data validation and error handling are examined. Methods for secure user input processing and form customisation are also examined. CRUD Operations: Create, Read, Update, Delete, Search, and Filters. The implementation of CRUD functionality is studied using Django views and ORM. Applications are developed that perform data creation, retrieval, updating, and deletion. These applications also perform searching, filtering, and sorting. User Authentication and Registration Mechanisms. Django's authentication framework is examined. User registration and authentication are implemented. Account management, password handling, and secure access to application resources are also implemented. User Permissions, Roles, Login, Logout, and Permissions. The concepts of authorization and user roles are studied. Permission management, login, logout, and access control are also studied. Role-based access management is implemented in Django applications. Security in Django Projects: CSRF, Sessions, Passwords, and Privacy. Security mechanisms provided by Django are examined, including CSRF protection and session management. Password hashing and authentication security are also examined. Privacy protection and common web security practices are examined as well. Fundamentals of Django REST Framework and API Development Capabilities. The architecture and components of Django REST Framework are introduced. RESTful APIs are developed, and data is serialized. Requests and responses are processed, and backend services are integrated with external applications.
--	--

	Integration with AJAX, JavaScript, Bootstrap, Pagination, and Media Files. The integration of Django applications with frontend technologies is studied. Dynamic page updates and responsive interfaces are examined. Pagination, media file management, and interactive web application development are also examined. Deploying a Django Project and the Final Project Concept. The procedures for deploying Django applications to production environments are studied. Project deployment, server configuration, and application optimisation are learned. The planning, implementation, and presentation of a complete web application project are also learned. Practical lessons Creating a Virtual Environment, Installing Django, and Launching a New Project. A Python virtual environment is created, and the Django framework is installed. A new project is initialized. Its operation is verified using the built-in development server. Creating a Django Application (App) and Adding It to the Project. A Django application is created and registered within the project configuration. The application structure is organized. Communication between project components is established. Working with settings.py and urls.py Files and Creating Routes. Project settings are configured, and installed applications are managed. Databases, static files, and URL routing are also managed. Navigation between application pages is implemented. Creating a Dynamic Page Using Views and Templates and Connecting Static Files. Dynamic web pages are developed using Django views and templates. Template inheritance is implemented. CSS, JavaScript, and image resources are integrated. Creating a Model, Performing Migrations, and Connecting to a Database. Database models are designed, and relationships between entities are
--	--

	defined. Database migrations are executed. Database connectivity is established within a Django project. Adding, Reading, Updating, and Deleting Data Through Django ORM. Create, Read, Update, and Delete (CRUD) operations are performed using Django ORM. Efficient database management is practiced. This database management is performed without writing SQL queries. Registering a Model and Managing Data in the Django Admin Panel. Models are registered in the Django Admin interface. Administrative features are customized. Data management, editing, searching, and filtering operations are performed. Creating Forms Using Django Forms and ModelForms. Forms for user input are designed, and ModelForms are implemented. Data validation is performed, and errors are handled. Submitted information is securely processed. Creating CRUD Functions and Connecting Them to the User Interface. Complete CRUD functionality is implemented. Database operations are integrated with templates. Interactive user interfaces for managing application data are provided. Creating User Registration, Login, and Logout Functions. User authentication features are implemented, including account registration and login. Logout and password management features are also implemented. Secure access to protected resources is implemented as well. Restricting Permissions Based on Permissions, Groups, and Roles. Authorization mechanisms are configured by assigning permissions. User groups and roles are created. Access to application resources is controlled according to security requirements. Checking Security Settings, CSRF, and Password Policy in a Django Project. Django security mechanisms, including CSRF protection, are
--	---

	examined and configured. Secure password policies and session management are also examined and configured. Authentication settings and application security best practices are examined as well. Creating and Testing a Simple API Using Django REST Framework. RESTful API endpoints are developed using Django REST Framework. Serializers are implemented. HTTP requests and responses are processed, and API functionality is tested. Search, Filter, Pagination, File/Image Upload, and Email-Sending Functions. Advanced application features, including search and filtering, are implemented. Pagination, file and image uploading, and email services are also implemented. Efficient data presentation is implemented as well. Bootstrap/AJAX Integration, Deployment, and Preparation of the Final Mini-Project Report. Bootstrap and AJAX are integrated to create responsive and interactive web interfaces. The completed Django application is deployed to a hosting environment. The documentation and presentation of the final mini-project are prepared. Self-study History of the Development of the Django Framework and Comparison with Other Frameworks. The evolution of the Django framework is studied. Its architecture, features, and areas of application are also studied. Django is compared with other popular web frameworks in terms of performance, scalability, security, and development efficiency. Detailed Analysis of the Django MVT (Model-View-Template) Architecture. The structure and components of the Model-View-Template architecture are examined. The interaction between models, views, and templates in Django applications is analysed. URL routing and the request-response cycle are also analysed.
--	--

	The Process and Importance of Database Migration in Django. The purpose of migrations and schema version control are studied. Database synchronization is also studied. Best practices for managing structural changes in Django projects are studied as well. The Django Template System and Its Capabilities. Template inheritance, template tags, and filters are examined. Reusable components and dynamic content generation are also examined. The efficient separation of presentation logic from application logic is examined as well. Methods for Ensuring Security in a Django Project. Django's built-in security mechanisms, including CSRF protection and authentication, are analysed. Session management, password hashing, and input validation are also analysed. Protection against common web vulnerabilities is analysed as well. Advantages and Challenges of Working with Forms in Django. Django Forms and ModelForms are studied. Data validation techniques and error handling are also studied. Form customization and methods for securely processing user input are studied as well. File and Image Upload Process in Django. File and media management are examined. Image uploading and storage configuration are also examined. Validation and secure handling of uploaded files within Django applications are examined as well. Email Sending Functionality in Django. Django's email framework and SMTP configuration are studied. Automated notifications and email verification are also studied. Password recovery mechanisms and secure email communication are studied as well. Managing User Roles and Permissions in Django. Django's authentication and authorization systems are analysed, including user roles and groups.
--	---

	Permissions and access control are also analysed. Role-based security management is analysed as well. Methods of Integrating Django with AJAX and JavaScript. Techniques for integrating Django applications with AJAX and JavaScript are examined. These techniques are used to create responsive interfaces. They are also used to create asynchronous communication and dynamic user experiences. Django and Bootstrap: Using Them Together to Improve Design. The integration of Bootstrap with Django templates is studied. This integration is used to develop responsive and user-friendly web applications. It is also used to develop visually consistent web applications. Fundamentals of Django REST Framework and Its Application Opportunities. The architecture of Django REST Framework and API development principles are examined. Serializers and viewsets are also examined. Authentication mechanisms and integration with frontend and third-party applications are examined as well. Creating Pagination and Search Functionality in Django. Methods for implementing efficient search and filtering are studied. Sorting and pagination methods are also studied. These methods are used to improve application usability and data management. Deploying Django Projects via Heroku or PythonAnywhere. The procedures for preparing and configuring Django applications on cloud hosting platforms are examined. Procedures for deploying and maintaining these applications are also examined. Environment configuration and production deployment are examined as well. Project Proposal for Creating a Django-Based Learning Platform or Testing System. A project proposal for a web application based on the Django framework is developed. This proposal defines system objectives, functional requirements, and
--	---

	database structure. It also defines user roles, security features, and implementation methodology.
Teaching Methods	The module is delivered through lectures, practical laboratory sessions, inquiry-based and problem-based learning, coding exercises, database modelling, project-based learning, group discussions, presentations, and software development activities. Students develop Django web applications by working with the MVT architecture, models, ORM, templates, forms, authentication, REST APIs, and deployment techniques. Practical activities include creating dynamic websites, implementing CRUD operations, managing databases, developing secure web applications, integrating frontend technologies, and deploying completed projects. Teacher-guided and student independent learning, official Django documentation, scientific literature, electronic educational resources, reports, presentations, and mini-projects further develop students' programming, analytical, software development, database management, problem-solving, and web application deployment skills.
Prerequisites	Python Programming Object-Oriented Programming Database Systems Web Technologies Internet Programming
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and final assessment (50%). The continuous assessment evaluates students' participation in 15 practical sessions and completion of six Teacher-Guided Independent Work assignments. A maximum of 8 points is awarded for practical activities, while 12 points are allocated to independent assignments, coding exercises, reports, presentations, web application development tasks, and mini-projects. The

	midterm assessment consists of a theoretical component worth 12 points and independent work worth 18 points. The theoretical component may be conducted as a test or a written examination consisting of three theoretical questions and two practical programming tasks. Independent work is submitted through the HEMIS system in the form of reports, analytical papers, software projects, presentations, or programming assignments. The final assessment may be conducted as a written examination or a test. The written examination consists of five questions worth 10 points each, while the test consists of 25 questions worth 2 points each. The final assessment evaluates students' theoretical knowledge, programming skills, database management abilities, API development competencies, web application security knowledge, and practical experience in developing and deploying Django-based web applications. Successful completion of all independent study assignments is required before the final assessment.
Recommended Literature	1. Django Software Foundation. Django Documentation. Version 5.2. 2. Vincent W.S. Django for Beginners: Build Websites with Python and Django. 5th Edition. Still River Press, 2024. 3. Vincent W.S. Django for APIs: Build Web APIs with Python and Django. Still River Press. 4. Mele A. Django 5 By Example: Build Powerful and Reliable Python Web Applications from Scratch. Packt Publishing, 2024. 5. Greenfeld D.R., Greenfeld A.R. Two Scoops of Django. Two Scoops Press. 6. Mozilla Developer Network. Django Web Framework (Python). 7. Django REST Framework. Official Documentation. 8. GeeksforGeeks. Django Tutorial.

	9. Python Software Foundation. Python Documentation.
Responsible Lecturer	Norqobilov Sobir Hamza o'g'li, Lecturer, Department of Computer and Software Engineering

3.01. Scientific-Pedagogical Work and Qualification Practice

Field	Description
Module Title	SCIENTIFIC-PEDAGOGICAL WORK AND QUALIFICATION PRACTICE
Module Code	3.01
Credit Points (ECTS)	20 Credits
Workload	Total workload: 600 hours Scientific-pedagogical work: 480 hours (120 hours in each of Semesters I, II, III, IV) Professional practice: 120 hours (Semester IV)
Frequency/Duration	Semester I, II, III, IV
Module objective	In accordance with the Regulation on the Master's Degree and the State Educational Standard, a master's student must be comprehensively knowledgeable, capable of scientific thinking, proficient in modern information technologies, and ready for the analysis of scientific data, scientific-research, and scientific-pedagogical activities. Based on the results of scientific-pedagogical work and professional practice, the master's student must: master new pedagogical and information technologies and methods of activating the educational process and assessing students' knowledge through the rating control system; master the organizational and educational-methodological support of independent study and distance-learning technologies; know the principles of creating educational-methodological manuals and be able to use them; be able to draw up lesson plans for various types of classes; gain experience in the organizational and educational-methodological support of educational-upbringing work; understand the integration of education with

	science and production and become acquainted with innovative scientific-technical solutions; improve specialty-related knowledge and skills and conduct experimental-testing work on the topic of the master's dissertation; and develop competencies related to scientific-research work.
Module Learning Outcomes	Knowledge: LO1. Will possess knowledge of modern pedagogical and information technologies, methods of activating the educational process, and the rating-based system for assessing students' knowledge. LO2. Will possess knowledge of the organizational and educational-methodological support of independent study and distance-learning technologies. LO3. Will possess knowledge of the principles of designing educational-methodological manuals, lesson plans, working syllabi, and calendar-thematic plans for various types of classes. LO4. Will possess knowledge of the regulatory and legal framework governing higher education, master's degree programs, and the organization of professional practice/internships in the Republic of Uzbekistan. LO5. Will possess knowledge of the integration of education with science and production, and of innovative scientific-technical solutions relevant to the computer systems and software specialty. Skills: LO6. Will have the skill of delivering lectures and conducting practical/seminar classes, and of preparing lecture texts, presentation slides, and methodological guidelines for an assigned subject. LO7. Will have the skill of compiling assessment materials (tests and examination questions) and applying the rating control system to evaluate students' knowledge. LO8. Will have the skill of organizing extracurricular and educational-upbringing activities, such as subject clubs and student events, at the department. LO9. Will have the skill of conducting experimental-testing work related to the specialty and to the topic of the master's dissertation during professional practice at a host organization.

	LO10. Will have the skill of preparing and defending structured reports on scientific-pedagogical work and professional practice at department meetings. Competence: LO11. Will have the competence to independently plan, organize, and conduct pedagogical activity at a higher education, secondary specialized, or vocational education institution. LO12. Will have the competence to integrate theoretical knowledge, scientific research, and production/practice experience within a professional educational or research environment. LO13. Will have the competence to reflect on and continuously improve their own teaching practice and specialty-related knowledge and skills.
Content	Year 1, Semester I – Scientific-Pedagogical Work Attending classes of the department's leading professors and instructors (12 sessions) (24h) Preparation of lecture texts and delivery of lectures (5 lectures) (8h) Preparing a lesson plan and lesson slides for teaching a specific topic using new pedagogical and information technologies (8h) Preparing methodological guidelines for a specific subject on conducting students' independent study hours (10h) Developing skills in compiling questions and tests for interim and final assessments in the assigned subject, and compiling 200 tests and 200 questions for the assigned subject (20h) Studying the State Educational Standard of Higher Education and preparing a brief report on it (8h) Studying the qualification requirements for a designated field of education and preparing a brief report on it (8h) Studying and developing a subject syllabus for the assigned subject based on the scientific supervisor's guidance (8h) Studying and developing a subject's working syllabus for the assigned subject based on the scientific supervisor's guidance (8h)

	Studying and developing a calendar-thematic plan based on the scientific supervisor's guidance (8h) Preparing a report on the results of scientific-pedagogical work in the 1st semester and defending it at the department meeting (10h) Year 1, Semester II – Scientific-Pedagogical Work Attending classes of the department's leading professors and instructors (12 lessons). (24h) Preparation of lecture texts and delivery of lectures (5 lectures). (10h) Preparing a lecture text for the assigned subject using foreign literature. (20h) Preparing slides for each lecture of the assigned subject. (10h) Studying and developing the digital-methodological support of the assigned subject based on the scientific supervisor's guidance (20h) Studying and developing the program of a subject club for the assigned subject based on the scientific supervisor's guidance (10h) Participating with lectures in faculty scientific and subject clubs (3 times). (6h) Organizing evening events in student dormitories (6h) Preparing methodological guidelines for a specific subject on conducting students' independent study hours (8h) Preparing a report on the results of scientific-pedagogical work in the 2nd semester and defending it at the department meeting (6h) Year 2, Semester III – Scientific-Pedagogical Work Attending classes of the department's leading professors and instructors (12 lessons). (24h) Preparation of lecture texts and delivery of lectures (5 lectures). (10h) Studying how to work with the instructor profile on the Hemis platform and preparing a report on the results of the study. (10h) Studying the regulatory documents on organizing the activities of the faculty dean's office and preparing a report on the results of the study. (10h)
--	---

	Studying the regulatory documents on organizing the department's activities and preparing a report on the results of the study. (20h) Preparing methodological guidelines for a specific subject on conducting students' independent study hours (10h) Participating with lectures in faculty scientific and subject clubs (3 times). (6h) Organizing evening events in student dormitories (10h) Becoming familiar with the regulatory documents on organizing the teaching process of a higher education institution's professor and preparing a report on the results of the study. (10h) Preparing a report on the results of scientific-pedagogical work in the 3rd semester and defending it at the department meeting (10h) Year 2, Semester IV – Scientific-Pedagogical Work Attending classes of the department's leading professors and instructors (12 lessons). (24h) Preparation of lecture texts and delivery of lectures (5 lectures). (10h) Studying how to work with the instructor profile on the Hemis platform and preparing a report on the results of the study. (10h) Studying the regulatory documents on organizing the activities of the faculty dean's office and preparing a report on the results of the study. (10h) Studying the regulatory documents on organizing the department's activities and preparing a report on the results of the study. (20h) Preparing methodological guidelines for a specific subject on conducting students' independent study hours (10h) Participating with lectures in faculty scientific and subject clubs (3 times). (6h) Organizing evening events in student dormitories (10h) Becoming familiar with the regulatory documents on organizing the teaching process of a higher education institution's professor and preparing a report on the results of the study. (10h)
--	--

	Preparing a report on the results of scientific-pedagogical work in the 4th semester and defending it at the department meeting (10h) Year 2, Semester IV – Professional Practice Preparatory stage Occupational safety, internal code of conduct, requirements and conditions for completing professional practice at the organization, the organization's structure, tasks and functions, areas of activity; becoming familiar with the host organization's regulatory and legal documents; engaging in scientific-research work and expanding theoretical knowledge, practical skills and competencies (5h) Participating in scientific-research projects, scientific-practical conferences, seminars, and studying the work of journals at the educational institution. (5h) Studying specialty-related virtual educational technologies and interactive educational technologies as well as modern technical tools; mastering professional and scientific-research competencies. (5h) Collecting materials related to the dissertation topic (15h) Main stage Improving specialty-related knowledge, skills and competencies. Performing tasks related to experimental and scientific-research activities for the master's dissertation (30h) Conducting consultations with leading specialists and scholars of the host organization on the master's dissertation (20h) Final stage Compiling, generalizing, analyzing and correcting the materials obtained on the research topic (20h) Keeping a diary and analyzing the work performed (10h) Preparing the internship documentation file and report. Defense at the department and the master's degree office (10h)
Teaching Methods	The module is delivered as guided independent activity under the supervision of the department and an assigned academic/scientific supervisor, rather than

	through scheduled lecture or seminar contact hours. Students complete the semester-by-semester activity plan set out in the Content field above — including, as applicable, classroom observation and teaching practice, preparation of methodological materials, professional practice at a host organization, and/or research, drafting and revision of the master's dissertation — and report on progress to the Department of Computer and software at the end of each semester.
Prerequisites	Completion of the first-year compulsory and elective theoretical modules of the Computer systems and software engineering master's program.
Exams and Assessment	Form of assessment: Rating-based assessment of semester reports, defended at the Department of Computer and software engineering (final control, 100-point scale) Assessment is conducted on a final-control basis on the overall 100-point scale: a report on scientific-pedagogical work is submitted and assessed at the end of each of Semesters I–IV, while the professional-practice report is submitted and assessed at the end of Semester IV. The record sheet is completed by the supervising instructor of the scientific-pedagogical work and professional practice and submitted to the Master's Degree Office. Grading scale: 5 – "Excellent" (90–100 points) – timely and full completion of the assignments designated for the semester, with a well-substantiated defense of the report at the department meeting and clear development of pedagogical ability and skill; 4 – "Good" (70–89.9 points) – timely and full completion of the assignments, with a substantiated defense of the report; 3 – "Satisfactory" (60–69.9 points) – timely but only partial completion of the assignments, with a satisfactory defense of the report; 2 – "Unsatisfactory" (0–59.9 points) – failure to complete the assignments designated for the semester. The entire module is organized as guided independent scientific-pedagogical and practical work (attending and delivering lectures, preparing methodological materials, and completing a period of professional

	practice at a host organization) rather than lecture/seminar contact hours; there is therefore no separate Independent Study (IS) component distinct from the module's own content – the semester activity plan listed above constitutes the assessed independent work, and its full and timely completion, together with a successful defense of the semester report at the department, is required for a passing score.
Recommended Literature	Main literature: 1. O‘zbekiston Respublikasining “Ta’lim to‘g‘risida”gi Qonuni, T.;2020-y. 2. Magistratura to‘g‘risida Nizom. O‘zbekiston Respublikasi Vazirlar Mahkamasining Magistratura to‘g‘risidagi nizomni tasdiqlash haqidagi 2015-yil 2-martdagi 36-son qarori. 3. Sh.M. Mirziyoyev Erkin va farovon, demokratik O‘zbekiston davlatini birgalikda barpo etamiz. - Toshkent, “O‘zbekiston”, 2016-yil. 4. Sh.M. Mirziyoyev Tanqidiy tahlil, qat’iy tartib - intizom va shaxsiy javobgarlik –har bir rahbar faoliyatining kundalik qoidasi bo‘lishi kerak. Toshkent, “O‘zbekiston”, 2017-yil. 5. Xodjayev B.X., Choriyev A., Saliyeva Z.T. Pedagogik tadqiqotlar metodologiyasi. – T.: IQTISODIYOT DUNYOSI, 2018. 6. Choriyev A., Choriyev I. Pedagogikaning fanlararo bog‘liqlik metodologiyasi. – T.: Fan, 2011. 7. Choriyev A., Choriyev I. Pedagogika tarixi metodologiyasi. – T.: Fan, 2011. 8. To‘raqulov X.A., Hasanboyev J.J., Usmanov N.O‘., Alqarov I.Sh., To‘raqulov O.X. Ilmiy tadqiqot asoslari. – T.: “Fan va texnologiya” nashriyoti, 2011. 9. Краевский, В.В. Методология педагогического исследования: Пособие для педагога-исследователя. – Самара: Изд-во СамГПИ, 1994. Supplementary literature: 10. Барсков, А.Г. Научный метод: возможности и иллюзии. – М.,1994.

	11. Безрукова, В.С. Педагогика. Проективная педагогика: Учеб. пособие для пед. ин-тов и индустр.-пед. техникумов – Екатеринбург, 1996. 12. Брызгалова, С.И. Научно-педагогическое исследование. – Калининград, 1996. 13. Гершунский, Б.С. Философия образования для ХХИ века. – М., 1998. 14. Аристер Н.И., Загузов Н.И. Протсекура подготовки и защиты диссертатсий. - М.: АОЗТ "ИКАР", 1995. 15. 6. Кузин Ф.А. Магистерская диссертация. Методика написания и правила оформления. - М.: МАМАРМЕН, 1993.
Responsible Lecturer	KHatamov Ochildi Kurbonovch, Doctor of Economics, Professor. Zaripova Mukaddas Djumayozovna, PhD in Economics, Associate Professor Juraev Olim Albaevich, PhD in Economics, Head of department. KHolliyev Faxriddin Bokhodirovich, PhD in Physics and Mathematics, Associate Professor

3.02. Research Work, Preparation and Defense of Master's Thesis

Field	Description
Module Title	RESEARCH WORK, PREPARATION AND DEFENSE OF MASTER'S THESIS
Module Code	3.02
Credit Points (ECTS)	40 Credits
Workload	Total workload: 1200 hours (independent research study) Semester I: 60 hours; Semester II: 180 hours; Semester III: 300 hours; Semester IV: 660 hours
Frequency/Duration	Semester I, II, III, IV
Module objective	Scientific research work covers the study of standard models, objects, and methods of investigation of processes in the master's specialization; the study of the scientific and technical support of experimental research; the conduct of experimental research; and the processing, analysis, and formulation of conclusions and recommendations from the results obtained. The master's student must conduct a thorough study of existing scientific research in the chosen field, identify its shortcomings, analyze its achievements, continue the investigation, and make an original contribution to the field, culminating in the preparation and defense of a master's dissertation of 70–80 pages comprising an introduction, at least three main chapters, a conclusion, a list of references, and appendices (if any).
Module Learning Outcomes	Knowledge: LO1. Will possess knowledge of standard models, objects, and methods of investigation used in scientific research within the master's specialization. LO2. Will possess knowledge of the scientific and technical support — test benches, modern apparatus, and other equipment — required for conducting experimental research. LO3. Will possess knowledge of the structure, content, and formatting requirements of a master's dissertation

	(introduction, main chapters, conclusion, references, and appendices). LO4. Will possess knowledge of the stages of scientific research work — scientific-research work, dissertation preparation, and scientific-pedagogical work — as defined by the State Educational Standard. Skills: LO5. Will have the skill of critically reviewing existing scientific literature in the chosen field, identifying its shortcomings, and formulating an original research problem. LO6. Will have the skill of designing and carrying out experimental or theoretical research, including the assembly and use of relevant measuring instruments and equipment. LO7. Will have the skill of processing, analyzing, and interpreting research results, and of drawing well-founded conclusions and recommendations. LO8. Will have the skill of drafting, structuring, and revising the chapters of a master's dissertation, and of preparing scientific articles and conference theses based on the research. LO9. Will have the skill of presenting and defending research progress and results at department meetings and at the preliminary and official defense of the dissertation. Competence: LO10. Will have the competence to independently plan and manage a multi-semester research project, from topic selection through to the successful defense of the master's dissertation. LO11. Will have the competence to make an original scientific contribution to the field of physics, integrating theoretical, experimental, and computational approaches as appropriate. LO12. Will have the competence to communicate research findings effectively, in both written (dissertation, articles) and oral (departmental report, defense) form, in accordance with academic standards.
Content	Year 1, Semester I – Research Work / Dissertation Preparation

	Selection of the master's dissertation topic Study of resolutions and regulatory documents in the field of the discipline Study of newly completed and current scientific research works in the field of the discipline (8h) Study of the retrieval of materials related to the master's dissertation topic from databases such as Web of Science, Scopus, Springer Nature, and others (8h) Compilation of a catalogue of the most important scientific literature and websites Familiarisation with research being conducted in Uzbekistan in the field of the discipline (8h) Preparation of a scientific thesis on the collected literature and its analysis on the dissertation topic and submission for publication (4h) Study of the preparation of standard models or devices on the master's dissertation topic Preparation of the dissertation annotation (4h) Preparation of the dissertation table of contents (4h) Preparation of the introduction of the dissertation (4h) Study of the selected research object on the basis of the master's dissertation topic Preparation of a thesis based on the results of the study of the dissertation research object and submission for publication (6h) Presentation at the departmental scientific seminar on the results of scientific research work carried out during Semester 1 of the academic year Submission of a report on the scientific research work carried out during Semester 1 of the academic year Year 1, Semester II – Research Work / Dissertation Preparation Receiving instructions from the academic supervisor on the preparation of Chapter I of the dissertation (6h) Drafting of Section 1.1 of Chapter I (14h) Drafting of Section 1.2 of Chapter I (14h) Drafting of Section 1.3 of Chapter I (14h) Preparation of conclusions on Chapter I and submission to the academic supervisor (12h) Development of algorithms, methods, and software solutions for the ongoing scientific research work (12h)
--	---

	Acquiring skills in the assembly of measuring instruments, measurement methods, and in conducting experimental research and processing its results (12h) Receiving instructions from the academic supervisor on the preparation of Chapter II of the dissertation (6h) Drafting of Section 2.1 of Chapter II (16h) Drafting of Section 2.2 of Chapter II (16h) Drafting of Section 2.3 of Chapter II (16h) Preparation of conclusions on Chapter II and submission to the academic supervisor (16h) Preparation of a scientific article on Chapter II of the dissertation topic and submission for publication (22h) Presentation at the departmental scientific seminar on the results of scientific research work carried out during Semester 2 of the academic year Submission of a report on the scientific research work carried out during Semester 2 of the academic year Year 2, Semester III – Research Work / Dissertation Preparation Conducting experimental trials for the dissertation research (44h) Presentation of the results of the conducted experimental trials to the academic supervisor in graphical and report form (44h) Preparation of a scientific article on the basis of the results of the conducted experimental trials and submission for publication (40h) Receiving instructions from the academic supervisor on the preparation of Chapter III of the dissertation (6h) Drafting of Section 3.1 of Chapter III (44h) Drafting of Section 3.2 of Chapter III (44h) Drafting of Section 3.3 of Chapter III (44h) Preparation of conclusions on Chapter III and submission to the academic supervisor (30h) Presentation at the departmental scientific seminar on the results of scientific research work carried out during Semester 3 Submission of a report on the scientific research work carried out during Semester 3 Year 2, Semester IV – Research Work / Dissertation Preparation
--	--

	Formulation of the general conclusions of the dissertation: a) explanation of the existence of the problem posed and its solution; b) development of recommendations on the practical significance of the results and the utilisation of the results obtained (60h) Revision of the introductory section of the dissertation on the basis of the ideas set forth in the conclusions (60h) Formal preparation of the dissertation in accordance with the requirements of the Regulation (60h) Printing of the preliminary draft of the dissertation text and submission to the academic supervisor (60h) Review and revision of the work taking into account the academic supervisor's amendments and comments, and introduction of relevant additions and changes to the work (60h) Obtaining a written review of the dissertation from the academic supervisor (60h) Submission of the dissertation text and the articles and theses published on the dissertation to the designated internal and external reviewers, and obtaining their reviews (60h) Preparation of a 20-minute presentation and presentation materials and clearance with the academic supervisor (60h) Conduct of the preliminary defense of the dissertation work at the department (or at a production enterprise or organisation) (60h) Incorporation of the necessary suggestions and recommendations received during the preliminary defense into the dissertation work (60h) Preparation of an annotation — indicating the master's student's surname, first name, and patronymic, as well as the academic supervisor, specialisation, relevance of the topic, methodological foundations, aim and objectives of the research, object and subject, and a brief description of the research stages — for publication on the official university website, clearance with the academic supervisor, and submission to the head of department (60h)
Teaching Methods	The module is delivered as guided independent activity under the supervision of the department and an

	assigned academic/scientific supervisor, rather than through scheduled lecture or seminar contact hours. Students complete the semester-by-semester activity plan set out in the Content field above — including, as applicable, classroom observation and teaching practice, preparation of methodological materials, professional practice at a host organization, and/or research, drafting and revision of the master's dissertation — and report on progress to the Department of Computer and software engineering at the end of each semester.
Prerequisites	Completion of Methodology of Scientific Research and Modern Research Methods in Computer systems and software, and approval of a master's dissertation topic by the Department of Computer and software engineering.
Exams and Assessment	Form of assessment: Rating-based assessment of semester research reports and the preliminary/official defense of the master's dissertation (final control, 100-point scale) At the end of each semester, a report on the scientific research work and master's dissertation preparation completed during that semester is submitted and assessed as a final-control activity on the overall 100-point scale; the grade record is completed by the supervising academic advisor and submitted to the Master's Division. Grading scale: 5 – "Excellent" (90–100 points) – timely and complete fulfilment of the semester's tasks, substantiated presentation of reports at departmental meetings, dissertation progress in accordance with the plan, and publication of scientific articles and theses with conference participation; 4 – "Good" (70–89.9 points) – timely and complete fulfilment of tasks and substantiated reports, with publication of theses; 3 – "Satisfactory" (60–69.9 points) – timely and complete fulfilment of tasks and substantiated reports, without publication requirements met; 2 – "Unsatisfactory" (0–59.9 points) – failure to fulfil the semester's tasks, to present substantiated reports, or to progress the dissertation according to plan.

	The module consists entirely of guided independent research work – selecting and justifying the dissertation topic, literature review, experimental/theoretical investigation, drafting and revising dissertation chapters, publishing articles and theses, and preliminary and official defense – detailed in the semester-by-semester plan above (60 hours in Semester I, 180 in Semester II, 300 in Semester III, and 660 in Semester IV). There is no separate Independent Study (IS) component beyond this work: successful completion of every stage of the plan, confirmed by the department at the end of each semester, is required for a passing score, culminating in the preliminary and official defense of the master's dissertation.
Recommended Literature	Main literature: 1. Shermuhamedova N.A. Ilmiy tadqiqot metodologiyasi: darslik. – Toshkent, 2021. 2. Kubayeva Sh. Ilmiy tadqiqot metodologiyasi: o'quv qo'llanma. – Toshkent, 2022. 3. Aytbayev M., Kimsanbayeva Sh., Umarova F. Ilmiy tadqiqot metodologiyasi: o'quv qo'llanma. – Toshkent, 2023. 4. Новиков А.М., Новиков Д.А. Методология научного исследования. – Москва: Либроком, 2007. URL: 5. Creswell J.W., Creswell J.D. Research Design: Qualitative, Quantitative and Mixed Methods Approaches. – 6th ed. – SAGE Publications, 2023. Supplementary literature: 6. Kothari C.R. Research Methodology: Methods and Techniques. – 3rd ed. – New Age International Publishers, 2019. 7. Kumar R. Research Methodology: A Step-by-Step Guide for Beginners. – 5th ed. – SAGE Publications, 2023. 8. Booth W.C., Colomb G.G., Williams J.M. The Craft of Research. – 5th ed. – University of Chicago Press, 2024. 9. Кузнецов И.Н. Научное исследование: методика проведения и оформление. –

	Москва: Издательско-торговая корпорация «Дашков и К°», 2020.
Responsible Lecturer	KHatamov Ochildi Kurbonovch , Doctor of Economics, Professor. Zaripova Mukaddas Djumayozovna , PhD in Economics, Associate Professor Juraev Olim Albaevich , PhD in Economics, Head of department. KHolliyev Faxriddin Bokhodirovich , PhD in Physics and Mathematics, Associate Professor